

Final Report

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Executive Summary

Phase 3 extends the Group 5 monitoring infrastructure deployed in Phases 1 and 2 by implementing all eighteen CIS Top 18 Controls across the segmented lab environment, configuring automated active response workflows, and integrating a SOAR platform for orchestrated incident response. The Wazuh Manager at 172.16.23.34 serves as the central SIEM, with seven enrolled agents spanning Windows Server, Ubuntu Linux, and container hosts. Two fully operational automation workflows are demonstrated: an EternalBlue active response that automatically blocks attacker IPs via iptables when rule 100004 fires, and a SOAR pipeline in which Wazuh forwards SSH brute force alerts to n8n, which extracts the attacker IP and executes a remote iptables block on the VPN server. All eighteen CIS controls are addressed with technical implementation evidence; where lab constraints prevent full implementation, gaps are explicitly documented.

Deployment Topology

The G5 environment is built on a single hypervisor host running VMware, segmented across five primary zones by a CoreRouter (172.16.23.33) running Ubuntu 24.04.2 LTS. The Management zone (172.16.23.32/28) hosts the Wazuh Manager, n8n SOAR, and Security Onion. The AD zone (172.16.33.0/26) hosts GS-AD running Windows Server 2019 with Active Directory, Sysmon64, and DHCP. The DMZ (172.16.48.x/29 series) hosts jumpbox-ssh, jumpbox-rdp, OpenVPN, a reverse proxy, WAF, and the container host running CorpWeb, IntWeb, FileSharing, and E-Site as Docker containers across VLANs 30 through 34. The Internal zone (172.16.23.0/28) hosts internal user and admin workstations.

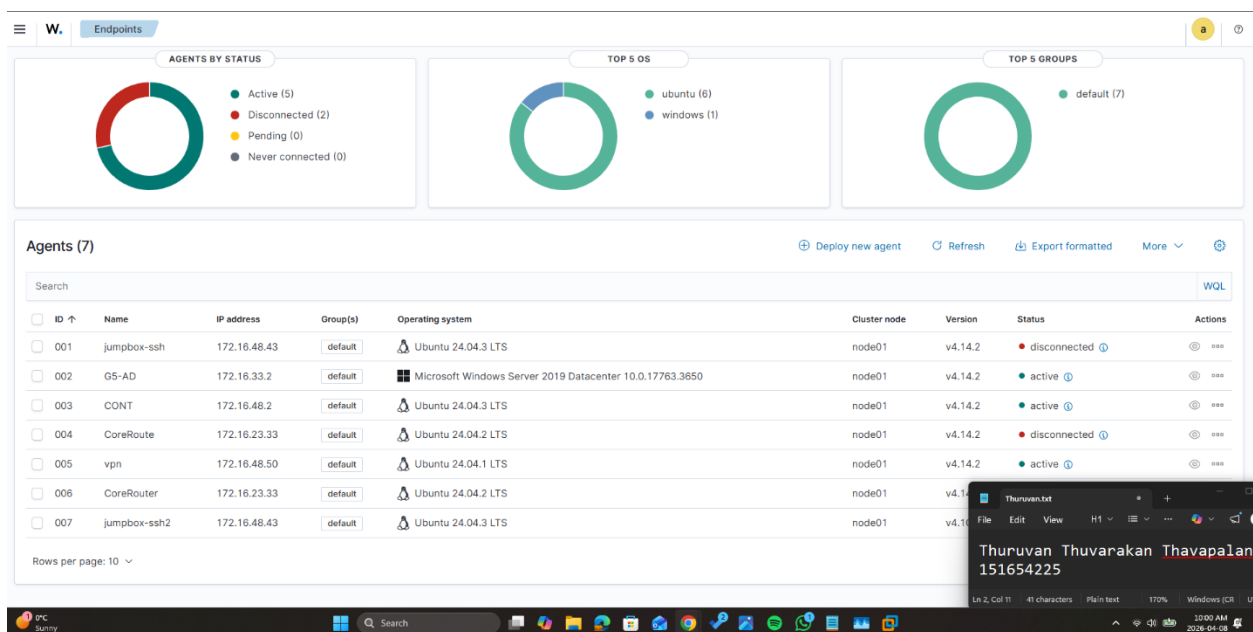
Seven Wazuh agents are enrolled: GS-AD (002), mgmt (003/004), vpn (005), CoreRouter (006), jumpbox-ssh2 (007), and CONT (009). The Wazuh dashboard and OpenSearch are accessible at 172.16.23.34, and the n8n SOAR instance is deployed at 172.16.23.35.

CIS Top 18 Controls Implementation

Control 1 — Inventory of Enterprise Assets

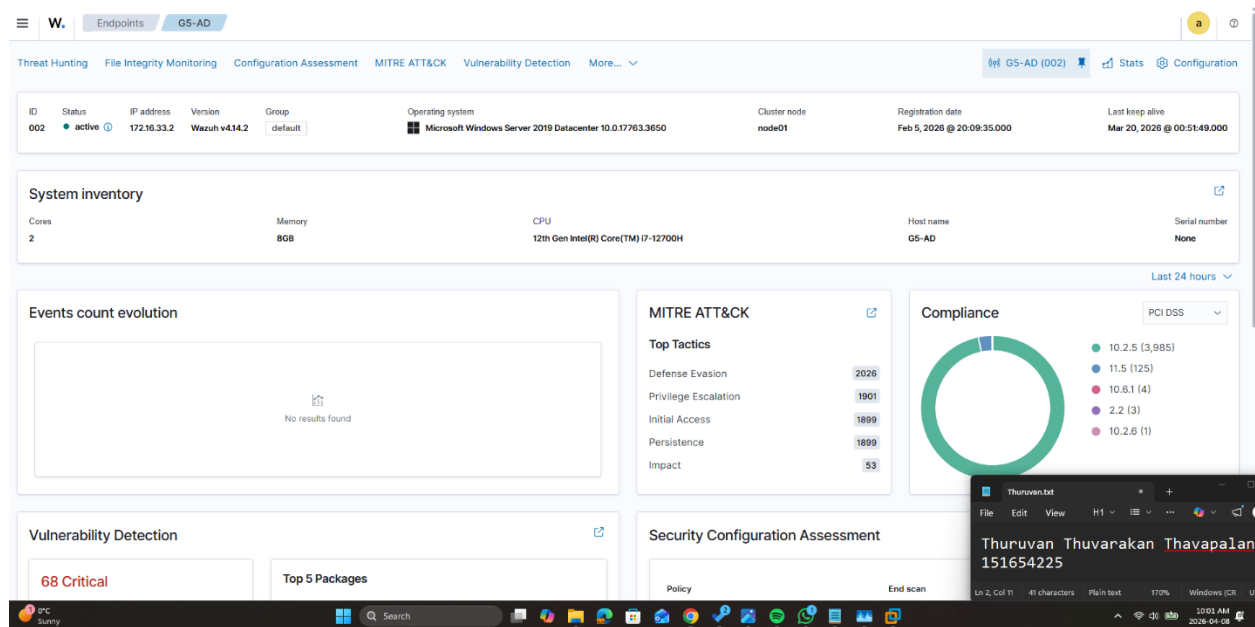
Active and continuous asset discovery is implemented through three complementary mechanisms. Wazuh's syscollector module provides real-time visibility into all monitored hosts, collecting hardware, network interface, and OS data from all seven enrolled agents automatically every 24 hours. An active nmap-based discovery script runs on the core router daily at 06:00 via cron, scanning all nine subnets and comparing discovered hosts against an approved baseline, with unknown IPs triggering a syslog alert forwarded to Wazuh. A passive tcpdump-based listener also runs on the core router as a systemd service, capturing traffic on all interfaces for five minutes every hour without transmitting any discovery packets. DHCP audit logging is enabled on GS-AD with logs stored at C:\Windows\System32\dhcp\, recording all IP lease events including hostname, MAC address, and timestamp.

Wazuh Endpoints dashboard displaying all seven enrolled enterprise assets with agent ID, hostname, IP address, operating system, version, and active status, satisfying CIS Control 1.1 asset inventory requirements.



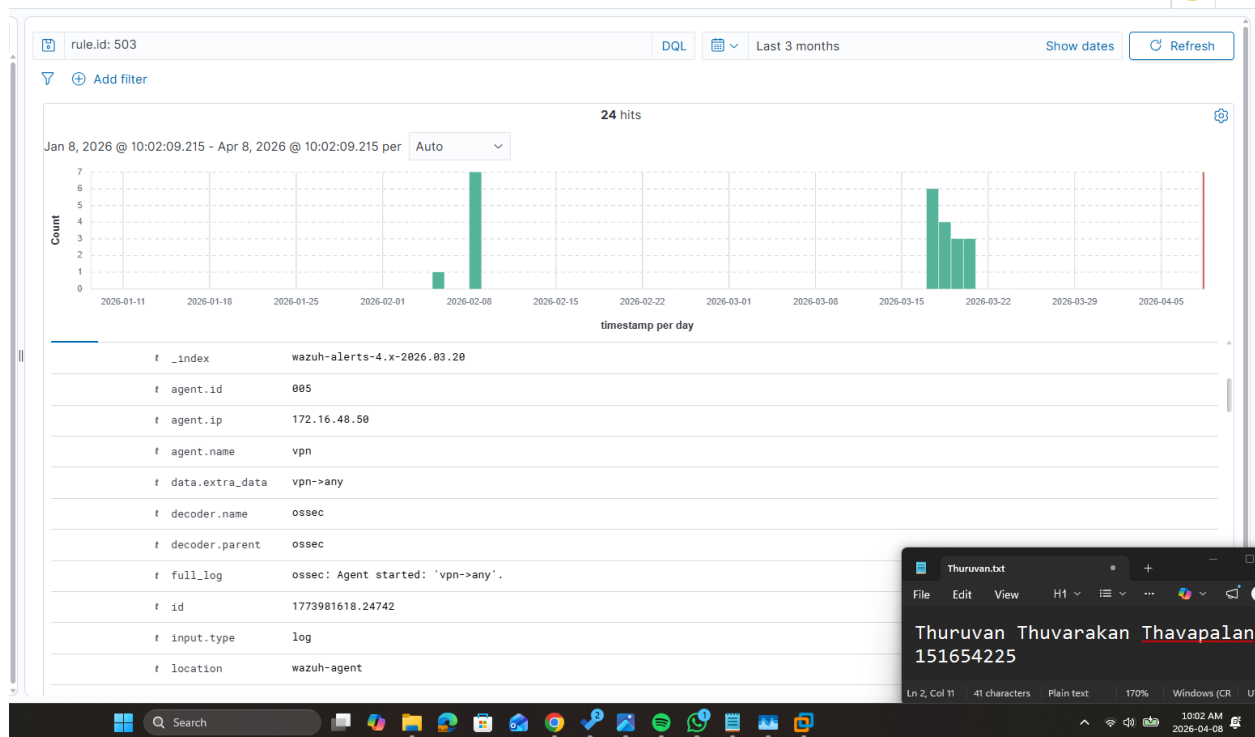
1.1a - Wazuh Agents Summary Dashboard

G5-AD agent detail page showing system inventory including CPU model, memory (8GB), hostname, and compliance mapping, confirming hardware-level asset data collection by Wazuh syscollector.



1.1b - Wazuh Agent Info with System Inventory

Wazuh Security Events filtered by rule.id 503 showing agent enrollment events, with the most recent event from the VPN agent (172.16.48.50), demonstrating automated detection of new assets connecting to the network.



1.2 - Wazuh Agent Enrollment Alert

Active asset discovery script executing on the core router, scanning all nine network subnets sequentially and displaying live discovered hosts per subnet.

```

SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 ~ 10:07:24 $ sudo /usr/local/bin/asset_discovery.sh
[sudo] password for ttthavapalan:
=====
Asset Discovery Scan - 2026-04-08_10:37:05
=====

--- Scanning 172.16.33.0/26 ---
172.16.33.2
172.16.33.1

--- Scanning 172.16.23.0/28 ---
172.16.23.1

--- Scanning 172.16.23.16/28 ---
172.16.23.17

--- Scanning 172.16.23.32/28 ---
172.16.23.34
172.16.23.33

--- Scanning 172.16.48.0/29 ---
172.16.48.2
172.16.48.1

--- Scanning 172.16.48.8/29 ---

```

1.3a - Asset Discovery Script nmap Output

Root crontab entry on the core router scheduling the asset discovery script to execute daily at 06:00, satisfying the CIS Control 1.3 requirement for daily active discovery execution.

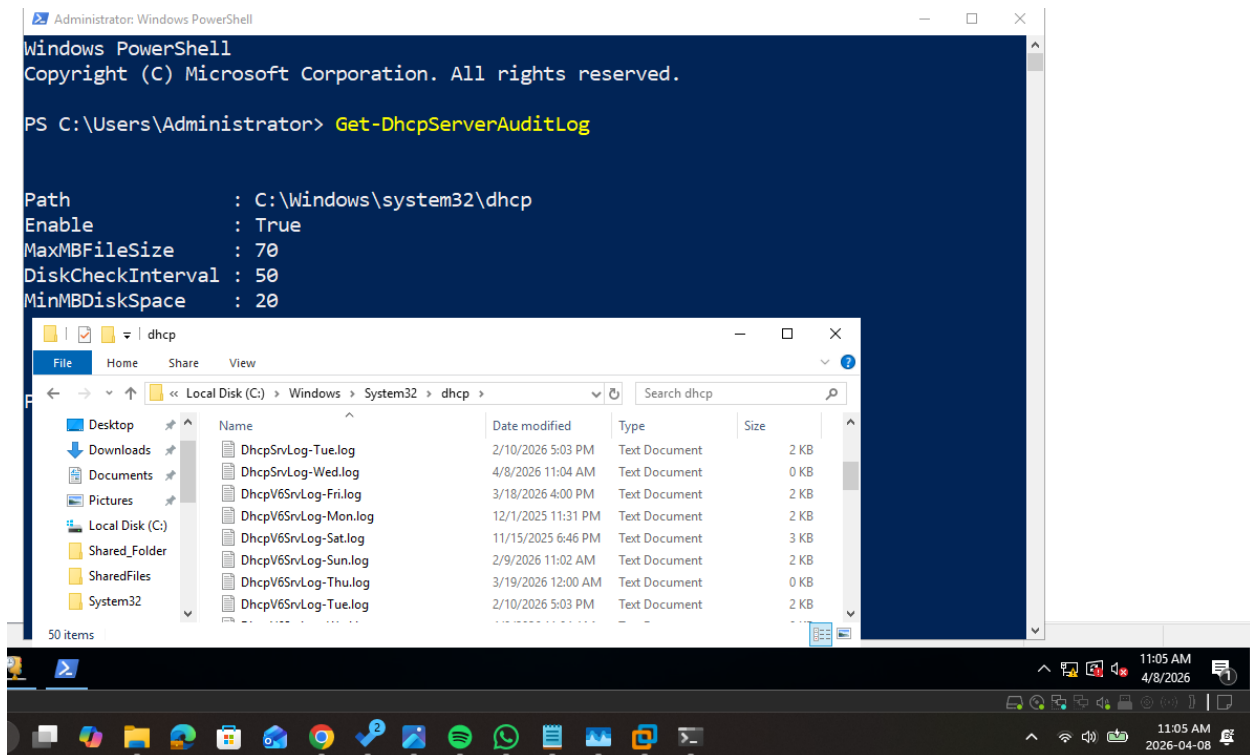
```

# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow co
0 6 * * * /usr/local/bin/asset_discovery.sh >> /var/log/asset_discovery/cron.log 2>&1

```

1.3b - Cron Job for Asset Discovery

Windows PowerShell output of Get-DhcpServerAuditLog on GS-AD confirming DHCP audit logging is enabled with log path C:\Windows\System32\dhcp



1.4a - DHCP Server Audit Log Enabled

Contents of the DHCP server audit log file showing IP lease renewal events for jumpbox-ssh with MAC addresses and timestamps, demonstrating DHCP-based asset inventory updates.

The screenshot shows a Windows desktop environment. In the background, a Notepad window titled 'DhcpSrvLog-Thu.log' displays DHCP server audit log entries. The log contains various events such as 'Database Cleanup Begin', 'DNS Update Request', 'leases expired and leases deleted', and 'Renew' events for the IP address 172.16.48.43, associated with the hostname 'jumpbox-ssh'. The log entries include timestamps and MAC addresses. In the foreground, a text editor window titled 'Thuruva.txt' is open, displaying the name 'Thuruva Thuvakaran Thavapalan' and the phone number '151654225'. The Windows taskbar at the bottom shows the time as 11:26 AM on 2026-04-08.

```

24,03/19/26,15:07:56,Database Cleanup Begin,,,,,0,6,,,,,,0
30,03/19/26,15:07:56,DNS Update Request,172.16.23.3,user-g5.G5.LOCAL,,,0,6,,,,,,0
25,03/19/26,15:07:56,0 leases expired and 0 leases deleted,,,,,0,6,,,,,,0
25,03/19/26,15:07:56,0 leases expired and 0 leases deleted,,,,,0,6,,,,,,0
31,03/19/26,15:08:06,DNS Update Failed,172.16.23.3,user-g5.G5.LOCAL,,,0,6,,,,,,9002
30,03/19/26,15:40:50,DNS Update Request,172.16.48.43,jumpbox-ssh.G5.LOCAL,,,0,6,,,,,,0
11,03/19/26,15:40:50,Renew,172.16.48.43,jumpbox-ssh.G5.LOCAL,2B9434C100020000AB11B55589BCE74D20EC,,4219687;
31,03/19/26,15:41:00,DNS Update Failed,172.16.48.43,jumpbox-ssh.G5.LOCAL,,,0,6,,,,,,9005
24,03/19/26,16:07:56,Database Cleanup Begin,,,,,0,6,,,,,,0
30,03/19/26,16:07:56,DNS Update Request,172.16.23.3,user-g5.G5.LOCAL,,,0,6,,,,,,0
25,03/19/26,16:07:56,0 leases expired and 0 leases deleted,,,,,0,6,,,,,,0
25,03/19/26,16:07:56,0 leases expired and 0 leases deleted,,,,,0,6,,,,,,0
31,03/19/26,16:08:07,DNS Update Failed,172.16.23.3,user-g5.G5.LOCAL,,,0,6,,,,,,9002
30,03/19/26,16:18:44,DNS Update Request,172.16.48.43,jumpbox-ssh.G5.LOCAL,,,0,6,,,,,,0
12,03/19/26,16:18:44,Release,172.16.48.43,jumpbox-ssh.G5.LOCAL,2B9434C100020000AB11B55589BCE74D20EC,,421968
12,03/19/26,16:18:44,Release,172.16.48.43,jumpbox-ssh.G5.LOCAL,2B9434C100020000AB11B55589BCE74D20EC,,421968
12,03/19/26,16:18:44,Release,172.16.48.43,jumpbox-ssh.G5.LOCAL,2B9434C100020000AB11B55589BCE74D20EC,,421968
30,03/19/26,16:18:54,DNS Update Request,172.16.48.43,jumpbox-ssh.G5.LOCAL,,,0,6,,,,,,0
30,03/19/26,16:18:54,DNS Update Request,172.16.48.43,jumpbox-ssh2.G5.LOCAL,,,0,6,,,,,,0
11,03/19/26,16:18:54,Renew,172.16.48.43,jumpbox-ssh2.G5.LOCAL,,,0,6,,,,,,0
31,03/19/26,16:18:54,DNS Update Failed,172.16.48.43,jumpbox-ssh2.G5.LOCAL,,,0,6,,,,,,0
31,03/19/26,16:18:54,DNS Update Failed,172.16.48.43,jumpbox-ssh2.G5.LOCAL,,,0,6,,,,,,0
31,03/19/26,16:19:04,DNS Update Failed,172.16.48.43,jumpbox-ssh2.G5.LOCAL,,,0,6,,,,,,0
30,03/19/26,16:21:54,DNS Update Request,172.16.48.43,jumpbox-ssh2.G5.LOCAL,,,0,6,,,,,,0
11,03/19/26,16:21:54,Renew,172.16.48.43,jumpbox-ssh2.G5.LOCAL,,,0,6,,,,,,0

```

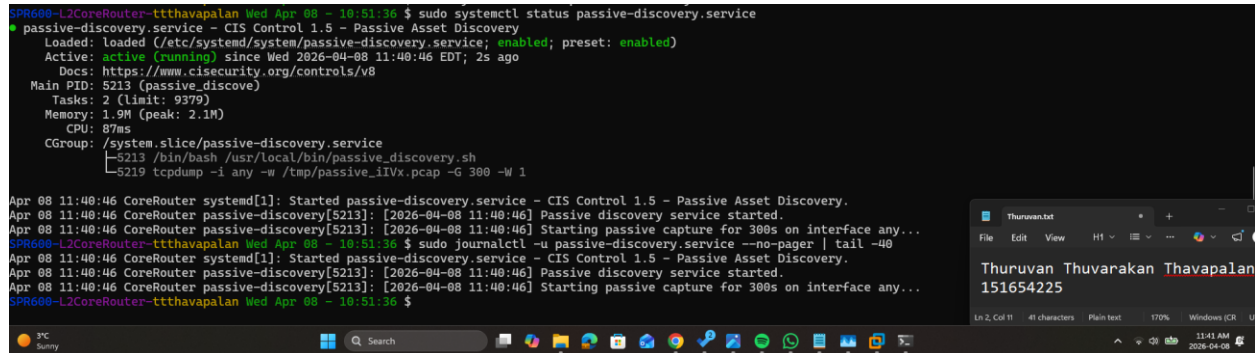
Thuruva Thuvakaran Thavapalan
151654225

Ln 2, Col 11 41 characters Plain text 170% Windows (CR) UT

11:26 AM 2026-04-08

1.4b - DHCP Log IP Lease Entries

Systemd service status of passive-discovery.service on the core router showing the service active and running, with journal logs confirming passive tcpdump capture without transmitting any discovery packets.



```

SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 10:51:36 $ sudo systemctl status passive-discovery.service
● passive-discovery.service - CIS Control 1.5 - Passive Asset Discovery
   Loaded: loaded (/etc/systemd/system/passive-discovery.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-08 11:48:46 EDT; 2s ago
     Docs: https://www.cisecurity.org/controls/v8
   Main PID: 5213 (passive_discover)
      Tasks: 2 (limit: 9379)
     Memory: 1.9M (peak: 2.1M)
        CPU: 87ms
    CGroup: /system.slice/passive-discovery.service
            └─5213 /bin/bash /usr/local/bin/passive_discovery.sh
               └─5219 tcpdump -i any -w /tmp/passive_iIVx.pcap -G 300 -W 1

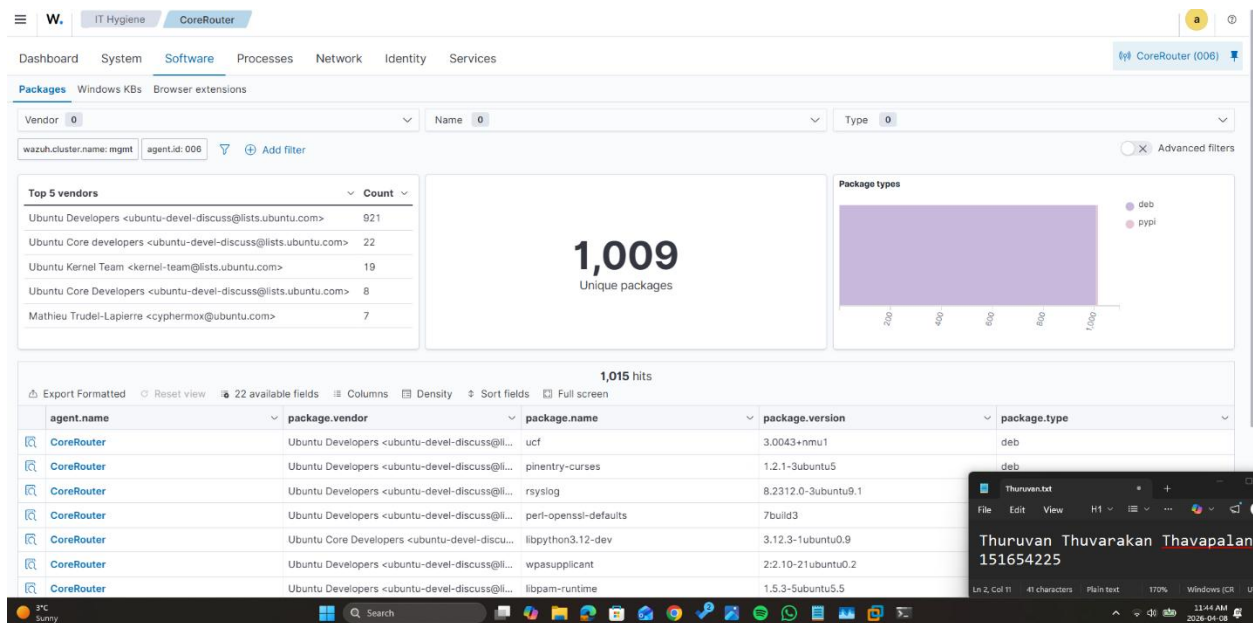
Apr 08 11:48:46 CoreRouter systemd[1]: Started passive-discovery.service - CIS Control 1.5 - Passive Asset Discovery.
Apr 08 11:48:46 CoreRouter passive-discovery[5213]: [2026-04-08 11:48:46] Passive discovery service started.
Apr 08 11:48:46 CoreRouter passive-discovery[5213]: [2026-04-08 11:48:46] Starting passive capture for 300s on interface any...
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 10:51:36 $ sudo journalctl -u passive-discovery.service --no-pager | tail -40
Apr 08 11:48:46 CoreRouter systemd[1]: Started passive-discovery.service - CIS Control 1.5 - Passive Asset Discovery.
Apr 08 11:48:46 CoreRouter passive-discovery[5213]: [2026-04-08 11:48:46] Passive discovery service started.
Apr 08 11:48:46 CoreRouter passive-discovery[5213]: [2026-04-08 11:48:46] Starting passive capture for 300s on interface any...
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 10:51:36 $
  
```

1.5 - Passive Discovery Service Running

Control 2 — Inventory of Software Assets

Software asset inventory is maintained through Wazuh's syscollector module, which automatically collects installed package data from all enrolled agents. For Linux agents, dpkg package data is collected; for Windows agents, installed programs from the registry are enumerated. Unauthorized software detection is handled by a custom baseline comparison script deployed as a systemd timer service running hourly on the Wazuh Manager and jumpbox-ssh. Application allowlisting is enforced on jumpbox-ssh through AppArmor with 30 profiles in enforce mode.

Wazuh IT Hygiene software inventory for CoreRouter showing 1,009 unique packages with package name, vendor, version, and type columns, demonstrating comprehensive Linux software inventory collection.



2.1a - Linux Software Inventory CoreRouter

Wazuh IT Hygiene software inventory for G5-AD showing 12 Windows packages including Wazuh Agent, Wireshark, and Nmap, confirming Windows software inventory collection via syscollector.

The screenshot displays the Wazuh IT Hygiene interface for the G5-AD agent. The 'Software' tab is active, showing a list of 12 unique packages. The interface includes a navigation bar with 'Dashboard', 'System', 'Software', 'Processes', 'Network', 'Identity', and 'Services'. The 'Software' section has filters for 'Vendor', 'Name', and 'Type'. A 'Top 5 vendors' table lists Microsoft Corporation (6), Mozilla (2), Broadcom Inc. (1), Nmap Project (1), and The Wireshark developer community (1). A central graphic shows '12 Unique packages'. A 'Package types' chart shows a single bar for 'win'. Below, a table lists 12 hits with columns for agent.name, package.vendor, package.name, package.version, and package.type. A text editor window is open in the foreground showing the name 'Thuruvan Thavarakan Thavapalan' and the ID '151654225'.

agent.name	package.vendor	package.name	package.version	package.type
G5-AD	Microsoft Corporation	Microsoft Visual C++ 2022 X64 Additional Run...	14.40.33816	win
G5-AD	Wazuh, Inc.	Wazuh Agent	4.14.2	win
G5-AD	Microsoft Corporation	Microsoft Visual C++ 2022 X64 Minimum Runti...	14.40.33816	win
G5-AD	The Wireshark developer community, https://...	Wireshark 4.4.5 x64	4.4.5	win
G5-AD	Mozilla	Mozilla Maintenance Service	125.0.1	win
G5-AD	Nmap Project	Npcap	1.79	win
G5-AD	Microsoft Corporation	Microsoft Visual C++ 2015-2022 Redistributab...	14.40.33816.0	win

2.1b - Windows AD Software Inventory

Wazuh Vulnerability Detection inventory showing 22,877 evaluated vulnerability hits across all agents, with CVE IDs, affected package names, versions, and severity ratings.

agent.name	package.name	package.version	vulnerability.description	vulnerability.severity	vulnerability.id
CONT	gpgsm	2.4.4-2ubuntu17.3	GnuPG can be made to spin on a relati...	Low	CVE-2022-3219
CONT	gpgsm	2.4.4-2ubuntu17.3	In GnuPG before 2.4.9, armor_filter in g...	High	CVE-2025-68973
CONT	gpgsm	2.4.4-2ubuntu17.3	In GnuPG through 2.4.8, if a signed me...	Medium	CVE-2025-68972
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68766
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68765
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68763
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68758
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68757
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68756
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68753
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68752
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68751
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68750
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68749
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68748
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68747
CONT	linux-image-6.8.0-87-generic	6.8.0-87.88	In the Linux kernel, the following vulner...	-	CVE-2025-68746

2.2 - Vulnerability Detection Inventory

Systemd timer status of software-audit.timer on jumpbox-ssh showing the timer active and waiting, confirming hourly automated software audit execution.

```

tthavapalan@jumpbox-ssh2:~$ sudo systemctl status software-audit.timer
• software-audit.timer - CIS Control 2.3 - Run Software Audit Hourly
   Loaded: loaded (/etc/systemd/system/software-audit.timer; enabled; preset: enabled)
   Active: active (waiting) since Wed 2026-04-08 11:54:19 EDT; 21s ago
     Trigger: Wed 2026-04-08 12:54:19 EDT; 59min left
   Triggers: • software-audit.service

Apr 08 11:54:19 jumpbox-ssh2 systemd[1]: Started software-audit.timer - CIS Control 2.3
tthavapalan@jumpbox-ssh2:~$
  
```

2.3a - Software Audit Timer on Jumpbox

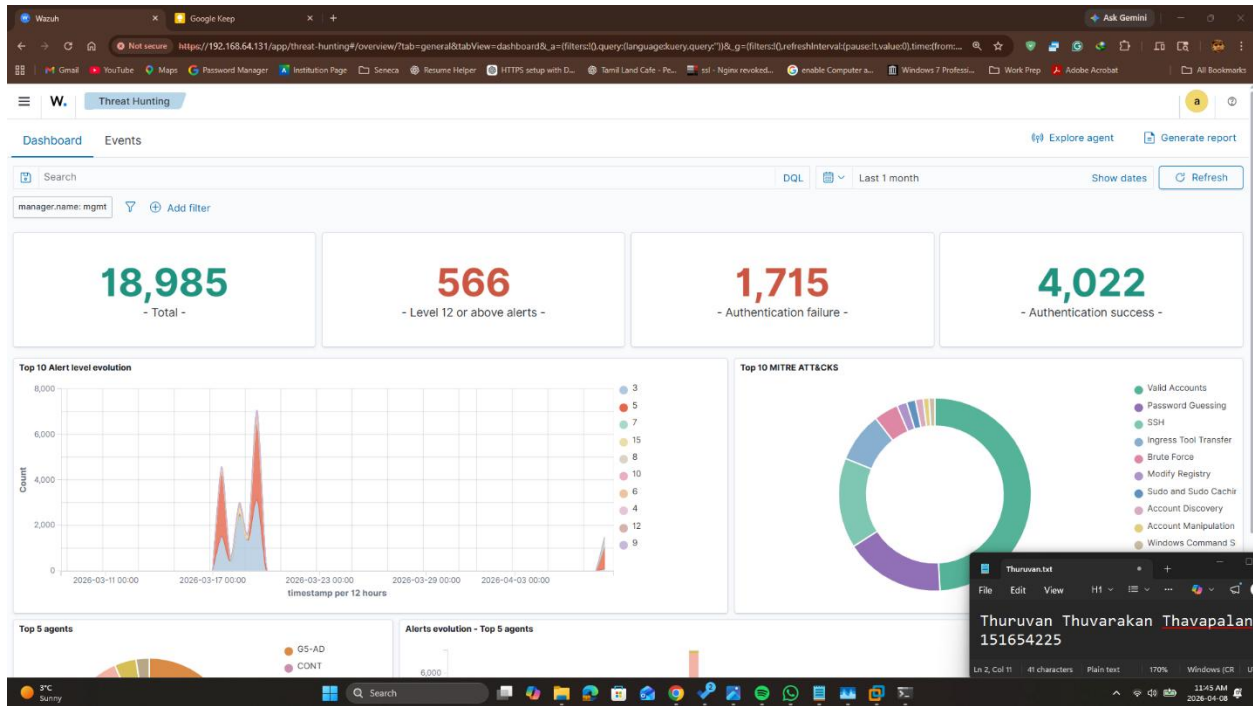
Wazuh Manager journal output showing the software audit service detecting cowsay and libtext-charwidth-perl as unauthorized packages with an [ALERT] flag, confirming the unauthorized software detection mechanism functions correctly.

```
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 11:35:54 $ sudo systemctl status software-audit.service
o software-audit.service - CIS Control 2.3 - Unauthorized Software Detection
   Loaded: loaded (/etc/systemd/system/software-audit.service; disabled; preset: enabled)
   Active: inactive (dead) since Wed 2026-04-08 11:49:58 EDT; 3s ago
     Duration: 70ms
   TriggeredBy: ● software-audit.timer
   Process: 5282 ExecStart=/usr/local/bin/software_baseline_check.sh (code=exited, status=0/SUCCESS)
   Main PID: 5282 (code=exited, status=0/SUCCESS)
     CPU: 88ms

Apr 08 11:49:58 mgmt software-audit[5286]: =====
Apr 08 11:49:58 mgmt software-audit[5288]: Software Audit - 2026-04-08 11:49:58
Apr 08 11:49:58 mgmt software-audit[5298]: =====
Apr 08 11:49:58 mgmt software-audit[5295]: --- Unauthorized Software Check ---
Apr 08 11:49:58 mgmt software-audit[5303]: [ALERT] New unauthorized packages detected:
Apr 08 11:49:58 mgmt software-audit[5305]: cowsay
Apr 08 11:49:58 mgmt software-audit[5305]: libtext-charwidth-perl
Apr 08 11:49:58 mgmt software-audit[5308]: =====
Apr 08 11:49:58 mgmt software-audit[5310]: Audit complete. Log: /var/log/software_audit/software_check_2026-04-08_11:49:58.log
Apr 08 11:49:58 mgmt systemd[1]: software-audit.service: Deactivated successfully.
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 11:35:54 $ cat /var/log/software_audit/cat /var/log/software_audit/software_check_$(ls -t /var
cat: '/var/log/software_audit/software_check_2026-04-08_11:49:58.log': No such file or directory
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 11:35:54 $
```

2.3b - Unauthorized Package Detected

Wazuh Threat Hunting dashboard showing 18,985 total events and top MITRE ATT&CK techniques across all agents managed by the Wazuh Manager.



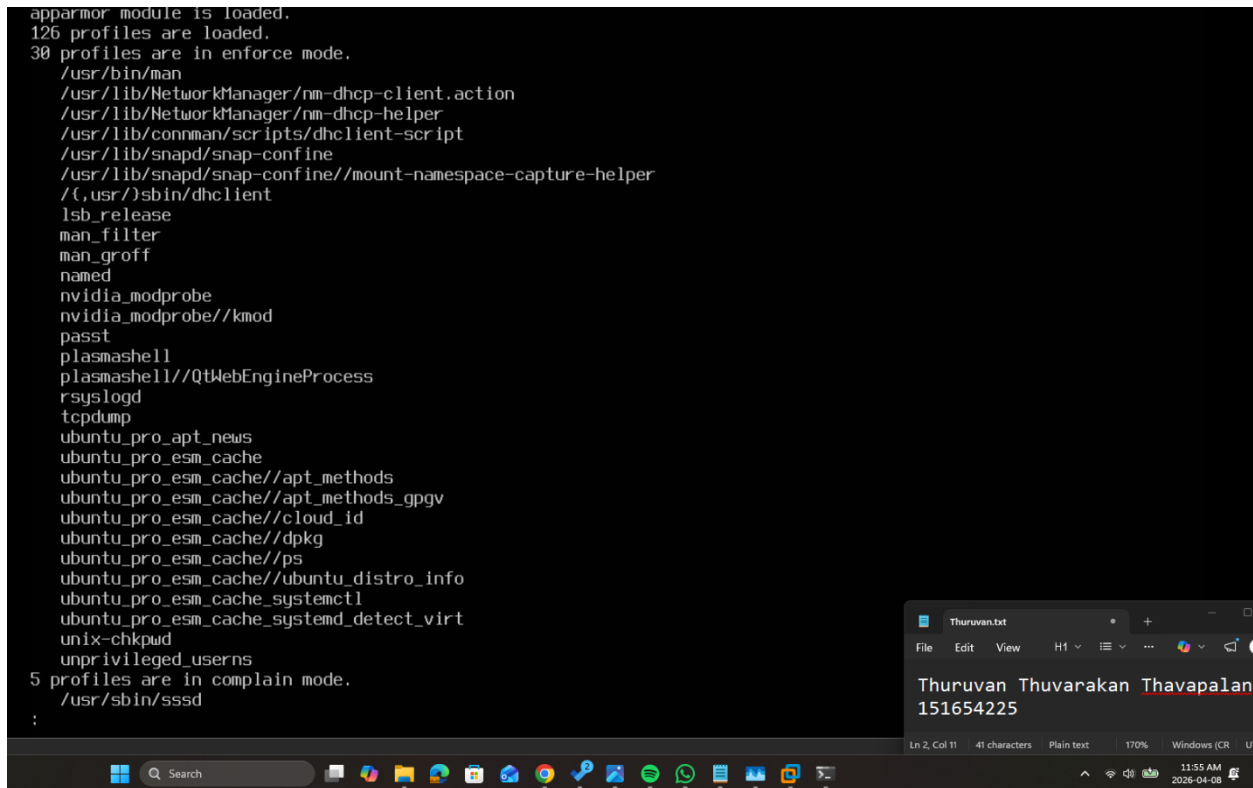
2.4 - Threat Hunting Dashboard

AppArmor status on jumpbox-ssh showing the module loaded with 126 profiles, 30 in enforce mode including tcpdump, rsyslogd, named, and sssd, implementing application allowlisting per CIS Control 2.5 through 2.7.

```

apparmor module is loaded.
126 profiles are loaded.
30 profiles are in enforce mode.
  /usr/bin/man
  /usr/lib/NetworkManager/nm-dhcp-client.action
  /usr/lib/NetworkManager/nm-dhcp-helper
  /usr/lib/connman/scripts/dhclient-script
  /usr/lib/snapd/snap-confine
  /usr/lib/snapd/snap-confine/mount-namespace-capture-helper
  /usr/sbin/dhclient
  ls_release
  man_filter
  man_groff
  named
  nvidia_modprobe
  nvidia_modprobe//kmod
  passt
  plasmashell
  plasmashell//QtWebEngineProcess
  rsyslogd
  tcpdump
  ubuntu_pro_apt_news
  ubuntu_pro_esm_cache
  ubuntu_pro_esm_cache//apt_methods
  ubuntu_pro_esm_cache//apt_methods_gpgv
  ubuntu_pro_esm_cache//cloud_id
  ubuntu_pro_esm_cache//dpkg
  ubuntu_pro_esm_cache//ps
  ubuntu_pro_esm_cache//ubuntu_distro_info
  ubuntu_pro_esm_cache_systemctl
  ubuntu_pro_esm_cache_systemd_detect_virt
  unix_chkpwd
  unprivileged_usersns
5 profiles are in complain mode.
  /usr/sbin/sssd
:

```

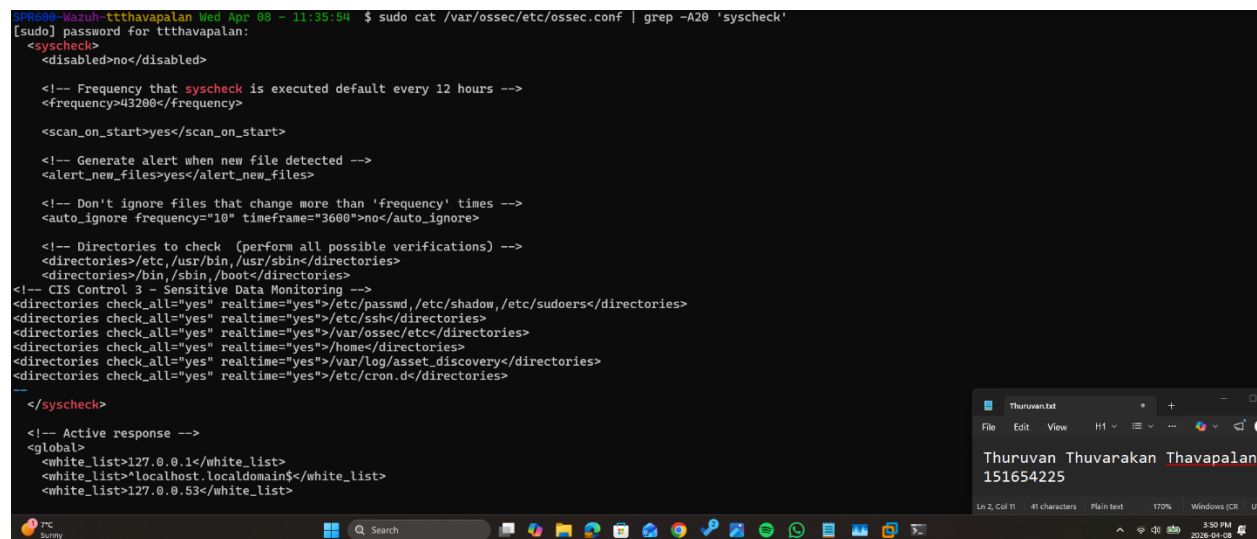


2.5 - AppArmor Enforced Profiles

Control 3 — Data Protection

Data protection is implemented through Wazuh File Integrity Monitoring, network segmentation by data sensitivity, encryption of data in transit, and strict file permission controls. FIM is configured with real-time monitoring on sensitive paths including credential files, SSH configuration, Wazuh security configuration, home directories, and cron paths. All inter-agent communications use TLS on port 1514, and remote access is encrypted using OpenVPN with AES-256-GCM.

Wazuh Manager ossec.conf syscheck configuration showing FIM enabled with 12-hour scan frequency, scan_on_start, alert_new_files, and six sensitive path entries with check_all="yes" and realtime="yes" attributes, confirming the data management process.



```
SPR600-Wazuh-tthavapalan Wed Apr 08 ~ 11:35:54 $ sudo cat /var/ossec/etc/ossec.conf | grep -A20 'syscheck'
[sudo] password for tthavapalan:
<syscheck>
<disabled>no</disabled>

<!-- Frequency that syscheck is executed default every 12 hours -->
<frequency>43200</frequency>

<scan_on_start>yes</scan_on_start>

<!-- Generate alert when new file detected -->
<alert_new_files>yes</alert_new_files>

<!-- Don't ignore files that change more than 'frequency' times -->
<auto_ignore frequency="10" timeframe="3600">no</auto_ignore>

<!-- Directories to check (perform all possible verifications) -->
<directories>/etc,/usr/bin,/usr/sbin</directories>
<directories>/bin,/sbin,/boot</directories>

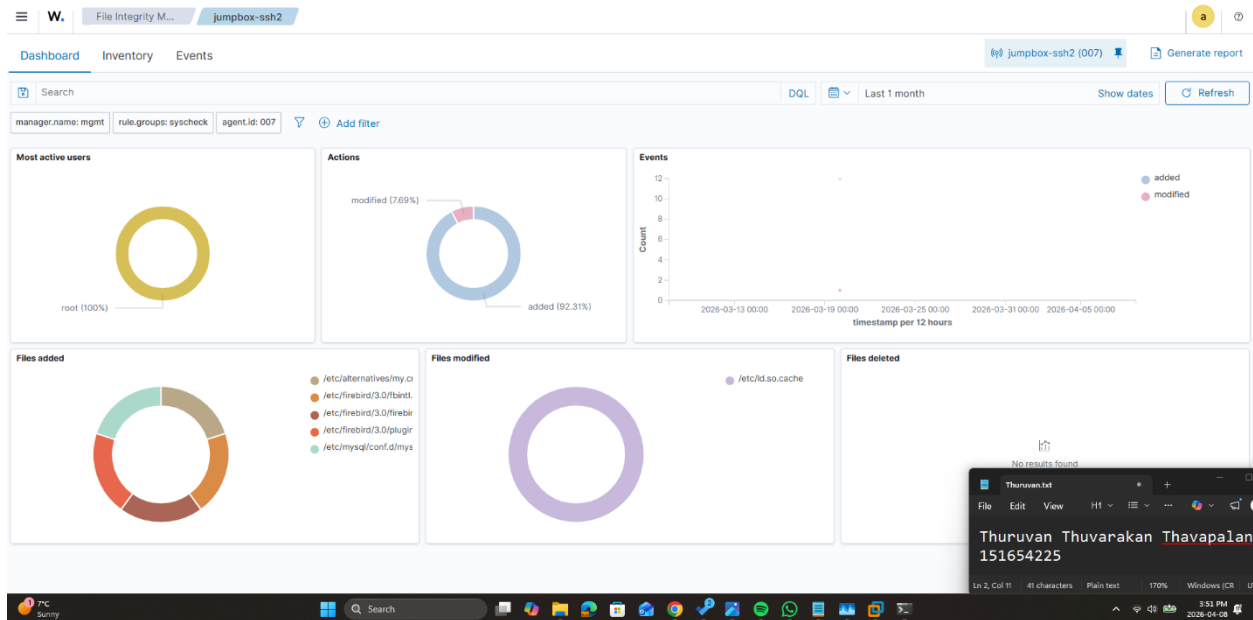
<!-- CIS Control 3 - Sensitive Data Monitoring -->
<directories check_all="yes" realtime="yes">/etc/passwd,/etc/shadow,/etc/sudoers</directories>
<directories check_all="yes" realtime="yes">/etc/ssh</directories>
<directories check_all="yes" realtime="yes">/var/ossec/etc</directories>
<directories check_all="yes" realtime="yes">/home</directories>
<directories check_all="yes" realtime="yes">/var/log/asset_discovery</directories>
<directories check_all="yes" realtime="yes">/etc/cron.d</directories>

</syscheck>

<!-- Active response -->
<global>
<white_list>127.0.0.1</white_list>
<white_list>localhost.localdomain</white_list>
<white_list>127.0.0.53</white_list>
```

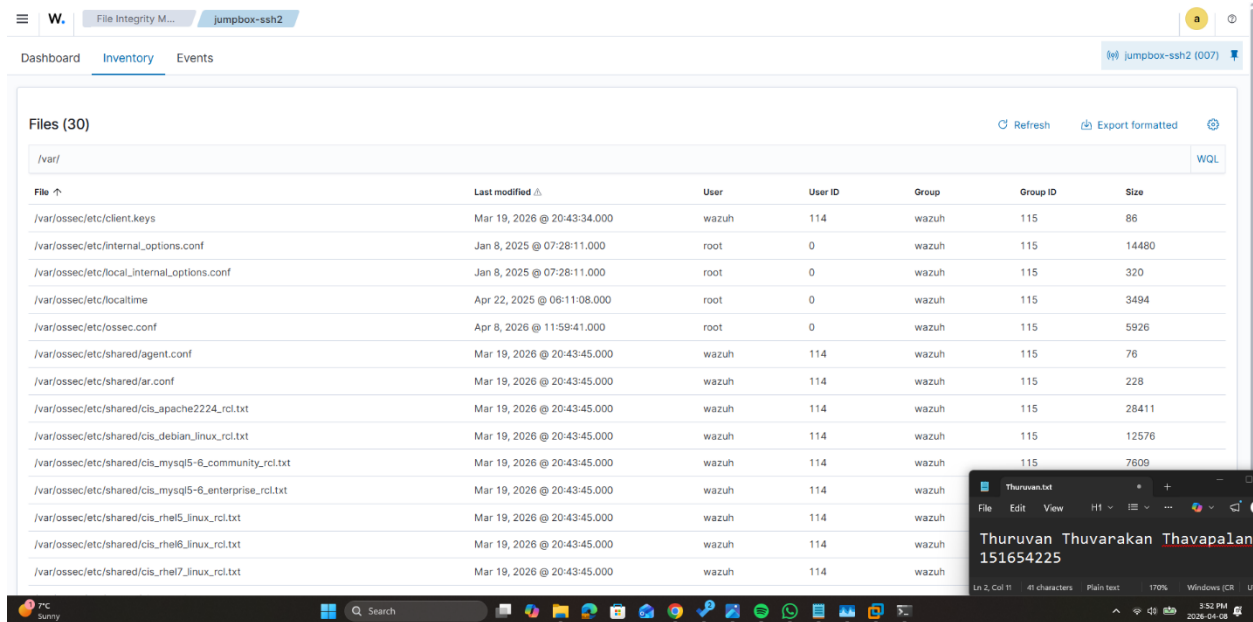
3.1 - FIM Configuration in ossec.conf

Wazuh File Integrity Monitoring dashboard for jumpbox-ssh2 showing files added and modified breakdown, confirming active FIM data inventory collection.



3.2a - FIM Dashboard

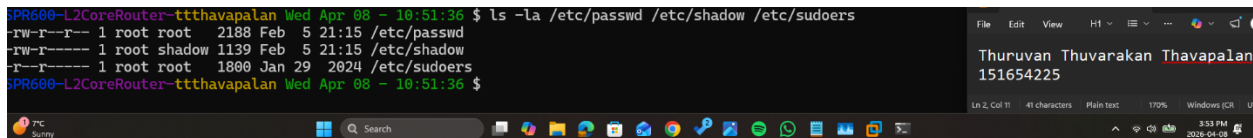
Wazuh FIM Inventory tab for jumpbox-ssh2 showing 30 monitored files under /var/ossec/etc/ with last modified timestamp, owner, user ID, group ID, and file size, constituting the sensitive data inventory.



File	Last modified	User	User ID	Group	Group ID	Size
/var/ossec/etc/client.keys	Mar 19, 2026 @ 20:43:34.000	wazuh	114	wazuh	115	86
/var/ossec/etc/internal_options.conf	Jan 8, 2025 @ 07:28:11.000	root	0	wazuh	115	14480
/var/ossec/etc/local_internal_options.conf	Jan 8, 2025 @ 07:28:11.000	root	0	wazuh	115	320
/var/ossec/etc/localtime	Apr 22, 2025 @ 06:11:08.000	root	0	wazuh	115	3494
/var/ossec/etc/ossec.conf	Apr 8, 2026 @ 11:59:41.000	root	0	wazuh	115	5926
/var/ossec/etc/shared/agent.conf	Mar 19, 2026 @ 20:43:45.000	wazuh	114	wazuh	115	76
/var/ossec/etc/shared/ar.conf	Mar 19, 2026 @ 20:43:45.000	wazuh	114	wazuh	115	228
/var/ossec/etc/shared/cis_apache2224_rcl.txt	Mar 19, 2026 @ 20:43:45.000	wazuh	114	wazuh	115	28411
/var/ossec/etc/shared/cis_debian_linux_rcl.txt	Mar 19, 2026 @ 20:43:45.000	wazuh	114	wazuh	115	12576
/var/ossec/etc/shared/cis_mysql5-6_community_rcl.txt	Mar 19, 2026 @ 20:43:45.000	wazuh	114	wazuh	115	7609
/var/ossec/etc/shared/cis_mysql5-6_enterprise_rcl.txt	Mar 19, 2026 @ 20:43:45.000	wazuh	114	wazuh	115	7609
/var/ossec/etc/shared/cis_rhel5_linux_rcl.txt	Mar 19, 2026 @ 20:43:45.000	wazuh	114	wazuh	115	7609
/var/ossec/etc/shared/cis_rhel6_linux_rcl.txt	Mar 19, 2026 @ 20:43:45.000	wazuh	114	wazuh	115	7609
/var/ossec/etc/shared/cis_rhel7_linux_rcl.txt	Mar 19, 2026 @ 20:43:45.000	wazuh	114	wazuh	115	7609

3.2b - FIM Monitored Files List

Core router terminal output of `ls -la /etc/passwd /etc/shadow /etc/sudoers` showing /etc/shadow owned by root:shadow with permissions 640 and /etc/sudoers owned by root:root with permissions 440, enforcing restrictive data access controls.

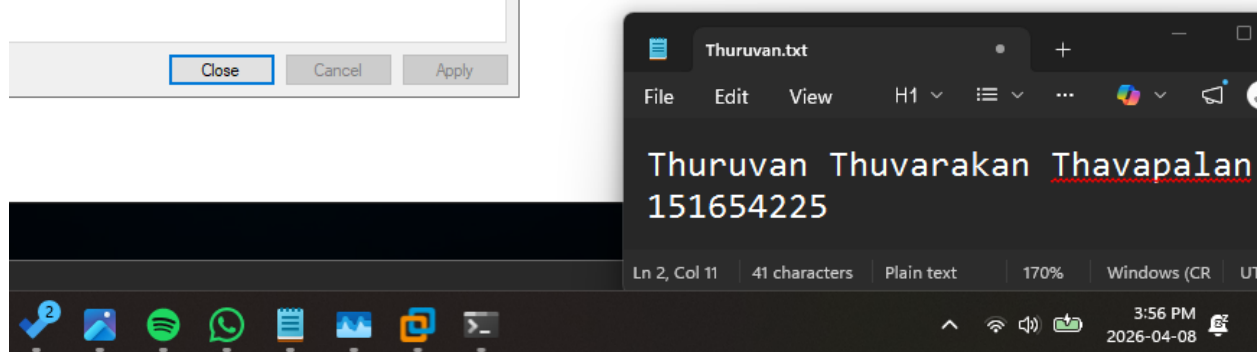
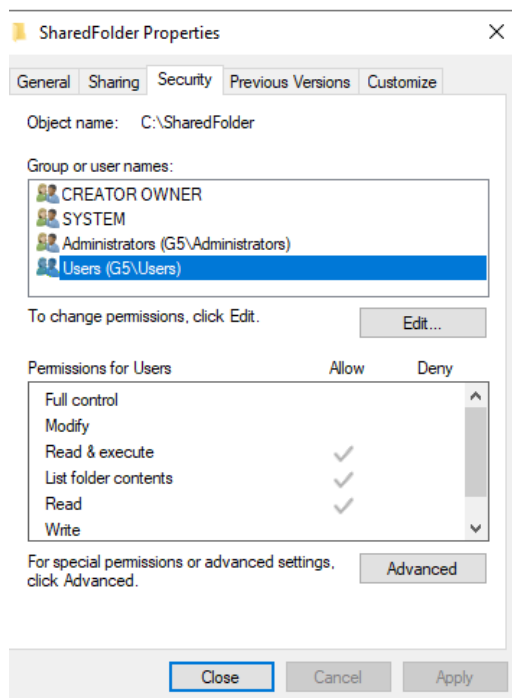


```

SPR600-L2CoreRouter-tthavapalan Wed Apr 08 - 10:51:36 $ ls -la /etc/passwd /etc/shadow /etc/sudoers
-rw-r--r-- 1 root root 2188 Feb  5 21:15 /etc/passwd
-rw-r----- 1 root shadow 1139 Feb  5 21:15 /etc/shadow
-r--r----- 1 root root 1800 Jan 29 2024 /etc/sudoers
SPR600-L2CoreRouter-tthavapalan Wed Apr 08 - 10:51:36 $
  
```

3.3a - Linux File Permissions

Windows SharedFolder Properties Security tab on GS-AD showing G5\Users group restricted to Read only, with Administrators holding full control, implementing role-based data access controls.



3.3b - AD Shared Folder ACL

Wazuh Manager /var/ossec/logs/alerts/ directory listing showing alerts.log (1.9MB) and alerts.json (3.0MB) with log rotation configured via logrotate.

```
SPR600-Wazuh-ttthavapalan Wed Apr 08 ~ 11:35:54 $ sudo -i
root@mgmt:~# grep -A5 'logall|max_output_size|logs' /var/ossec/etc/ossec.conf
<logall>no</logall>
<logall_json>no</logall_json>
<email_notification>no</email_notification>
<smtp_server>smtp.example.wazuh.com</smtp_server>
<email_from>wazuh@example.wazuh.com</email_from>
<email_to>recipient@example.wazuh.com</email_to>
<email_maxperhour>12</email_maxperhour>

--
<!-- Choose between "plain", "json", or "plain,json" for the format of internal logs -->
<logging>
<log_format>plain</log_format>
</logging>

<remote>

--
<ignore>/etc/httpd/logs</ignore>
<ignore>/etc/utmpx</ignore>
<ignore>/etc/wtmpx</ignore>
<ignore>/etc/cups/certs</ignore>
<ignore>/etc/dumpdates</ignore>
<ignore>/etc/svc/volatile</ignore>

--
<location>/var/ossec/logs/active-responses.log</location>
</localfile>

<localfile>
<log_format>syslog</log_format>
<location>/var/log/dpkg.log</location>
root@mgmt:~# cat /etc/logrotate.d/wazuh 2>/dev/null || ls /var/ossec/logs/
active-responses.log alerts api api.log archives cluster cluster.log firewall integrations.log ossec.log
root@mgmt:~# ls -lh /var/ossec/logs/alerts/
total 4.8M
drwxr-x--- 5 wazuh wazuh 4.0K Apr  8 10:23 2026
-rw-r----- 2 wazuh wazuh 3.0M Apr  8 15:57 alerts.json
-rw-r----- 2 wazuh wazuh 1.9M Apr  8 15:57 alerts.log
root@mgmt:~#
```

3.4 - Data Retention Logs

OpenVPN server configuration showing cipher AES-256-GCM, auth SHA256, tls-auth ta.key 0, and tls-verify script, implementing strong encryption for all data in transit over the remote access channel.

```
ttthavapalan@vpn:~$ cat /etc/openvpn/server/server.conf | grep -E 'cipher|tlsauth'
#auth-user-pass
tls-auth ta.key 0
cipher AES-256-GCM
auth SHA256
tls-verify /etc/openvpn/server/check_intermediate.sh
ttthavapalan@vpn:~$
ttthavapalan@vpn:~$
ttthavapalan@vpn:~$
```

3.10a - OpenVPN Encryption Config

Wazuh Manager ossec.conf showing agent communication configured on TCP port 1514, confirming all Wazuh agent-to-manager telemetry is transmitted over encrypted TCP channels.

```

root@mgmt:~# grep -E 'protocol|port' /var/ossec/etc/ossec.conf | head -10
<port>1514</port>
<protocol>tcp</protocol>
<check_ports>yes</check_ports>
<ports all="yes">yes</ports>
<alias>netstat listening ports</alias>
<port>1515</port>
<port>1516</port>

```

3.10b - Wazuh TLS Port Config

ls -la /var/ossec/etc/ output showing Wazuh security files with strict permissions: client.keys (rw-r----- root:wazuh), sslmanager.cert and sslmanager.key (rw-r----- root:root), protecting key material at rest.

```

root@mgmt:~# ls -la /var/ossec/etc/
total 76
drwxrwx--- 7 wazuh wazuh 4096 Apr  8 12:00 .
drwxr-x--- 20 root  wazuh 4096 Feb  5 15:30 ..
-rw-r----- 1 root  wazuh  572 Apr  8 12:00 client.keys
drwxrwx--- 2 root  wazuh 4096 Feb  5 15:30 decoders
-rw-r----- 1 root  wazuh 14961 Jan  8 14:06 internal_options.conf
drwxrwx--- 4 root  wazuh 4096 Feb  5 15:31 lists
-rw-r----- 1 root  wazuh  320 Jan  8 14:06 local_internal_options.conf
-rw-r----- 1 root  wazuh 3494 Apr 22 2025 localtime
-rw-rw---- 1 root  wazuh 9889 Apr  8 12:00 ossec.conf
drwxrwx--- 2 root  wazuh 4096 Feb  5 15:30 rootcheck
drwxrwx--- 2 root  wazuh 4096 Mar 19 20:19 rules
drwxrwx--- 4 root  wazuh 4096 Mar 20 00:41 shared
-rw-r----- 1 root  root  1164 Feb  5 15:30 sslmanager.cert
-rw-r----- 1 root  root  1704 Feb  5 15:30 sslmanager.key
root@mgmt:~# stat /var/ossec/etc/client.keys
  File: /var/ossec/etc/client.keys
  Size: 572          Blocks: 8          IO Block: 4096   regular file
Device: 252,0    Inode: 667582        Links: 1
Access: (0640/-rw-r-----)  Uid: (  0/   root)   Gid: ( 107/  wazuh)
Access: 2026-04-08 12:00:54.790281215 -0400
Modify: 2026-04-08 12:00:31.940946019 -0400
Change: 2026-04-08 12:00:31.940946019 -0400
 Birth: 2026-04-08 12:00:31.939946005 -0400
root@mgmt:~#

```

3.11 - Wazuh Config File Permissions

Core router ip -br a output showing all network interfaces UP with distinct subnet assignments across seven VLANs, demonstrating data segmentation by sensitivity.

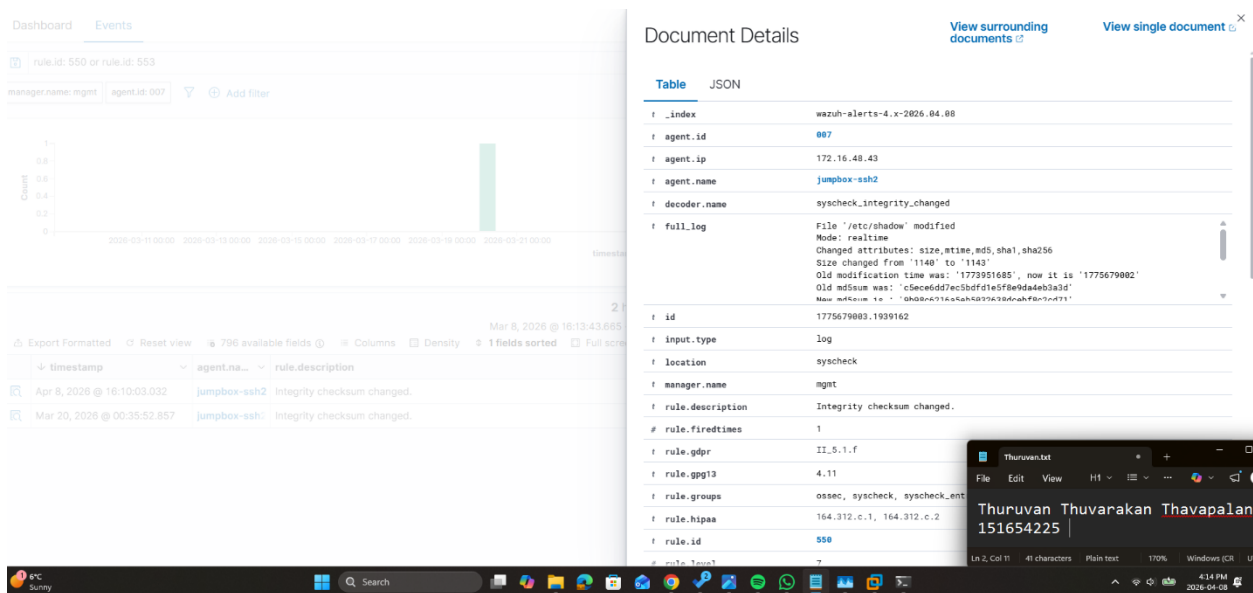
```

SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 10:51:36 $ ip -br a
lo                UNKNOWN      127.0.0.1/8 ::1/128
ens33             UP            172.16.33.1/26 fe80::20c:29ff:fe64:36eb/64
ens37            DOWN
ens38            UP            172.16.23.1/28 fe80::20c:29ff:fe64:36ff/64
ens39            UP            172.16.48.49/29 fe80::20c:29ff:fe64:3609/64
ens40            UP            fe80::20c:29ff:fe64:3613/64
ens41            UP            172.16.48.41/29 fe80::20c:29ff:fe64:361d/64
ens42            UP            172.16.23.17/28 fe80::20c:29ff:fe64:3627/64
ens43            UP            172.16.23.33/28 fe80::20c:29ff:fe64:3631/64
ens44            UP            192.168.64.136/24 metric 100 fe80::20c:29ff:fe64:363b/64
ens40.33@ens40   UP            172.16.48.25/29 fe80::20c:29ff:fe64:363d/64
ens40.31@ens40   UP            172.16.48.9/29 fe80::20c:29ff:fe64:363f/64
ens40.32@ens40   UP            172.16.48.17/29 fe80::20c:29ff:fe64:3641/64
ens40.30@ens40   UP            172.16.48.1/29 fe80::20c:29ff:fe64:3643/64
ens40.34@ens40   UP            172.16.48.33/29 fe80::20c:29ff:fe64:3645/64
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 10:51:36 $

```

3.12 - Router VLAN Segmentation

Wazuh FIM Events filtered by rule.id: 550 or rule.id: 553 for jumpbox-ssh2 showing integrity checksum change events, confirming FIM functions as a real-time data loss prevention mechanism.



3.13 - FIM DLP Alert Rules 550 553

Wazuh alert document detail for jumpbox-ssh2 showing /etc/shadow modified in realtime mode with changed size, mtime, md5, sha1, and sha256 attributes, with GDPR and HIPAA compliance tags confirming sensitive data access logging.

DOCUMENT DETAILS [documents](#)

Table JSON

t _index	wazuh-alerts-4.x-2026.04.08
t agent.id	007
t agent.ip	172.16.48.43
t agent.name	jumpbox-ssh2
t decoder.name	syscheck_integrity_changed
t full_log	File '/etc/shadow' modified Mode: realtime Changed attributes: size,mtime,md5,sha1,sha256 Size changed from '1140' to '1143' Old modification time was: '1773951685', now it is '1775679002' Old md5sum was: 'c5ece6dd7ec5bdfd1e5f8e9da4eb3a3d' New md5sum is: '0h0Rr6716a5ah5037638drahf8c7cd71'
t id	1775679003.1939162
t input.type	log
t location	syscheck
t manager.name	mgmt
t rule.description	Integrity checksum changed.
# rule.firedtimes	1
t rule.gdpr	II_5.1.f
t rule.gpg13	4.11
t rule.groups	ossec, syscheck, syscheck_ent
t rule.hipaa	164.312.c.1, 164.312.c.2
t rule.id	550
# rule.level	7

Thuruven.txt

File Edit View H1 H2 H3 H4 H5 H6 H7 H8 H9 H10 H11 H12 H13 H14 H15 H16 H17 H18 H19 H20 H21 H22 H23 H24 H25 H26 H27 H28 H29 H30 H31 H32 H33 H34 H35 H36 H37 H38 H39 H40 H41 H42 H43 H44 H45 H46 H47 H48 H49 H50 H51 H52 H53 H54 H55 H56 H57 H58 H59 H60 H61 H62 H63 H64 H65 H66 H67 H68 H69 H70 H71 H72 H73 H74 H75 H76 H77 H78 H79 H80 H81 H82 H83 H84 H85 H86 H87 H88 H89 H90 H91 H92 H93 H94 H95 H96 H97 H98 H99 H100 H101 H102 H103 H104 H105 H106 H107 H108 H109 H110 H111 H112 H113 H114 H115 H116 H117 H118 H119 H120 H121 H122 H123 H124 H125 H126 H127 H128 H129 H130 H131 H132 H133 H134 H135 H136 H137 H138 H139 H140 H141 H142 H143 H144 H145 H146 H147 H148 H149 H150 H151 H152 H153 H154 H155 H156 H157 H158 H159 H160 H161 H162 H163 H164 H165 H166 H167 H168 H169 H170 H171 H172 H173 H174 H175 H176 H177 H178 H179 H180 H181 H182 H183 H184 H185 H186 H187 H188 H189 H190 H191 H192 H193 H194 H195 H196 H197 H198 H199 H200 H201 H202 H203 H204 H205 H206 H207 H208 H209 H210 H211 H212 H213 H214 H215 H216 H217 H218 H219 H220 H221 H222 H223 H224 H225 H226 H227 H228 H229 H230 H231 H232 H233 H234 H235 H236 H237 H238 H239 H240 H241 H242 H243 H244 H245 H246 H247 H248 H249 H250 H251 H252 H253 H254 H255 H256 H257 H258 H259 H260 H261 H262 H263 H264 H265 H266 H267 H268 H269 H270 H271 H272 H273 H274 H275 H276 H277 H278 H279 H280 H281 H282 H283 H284 H285 H286 H287 H288 H289 H290 H291 H292 H293 H294 H295 H296 H297 H298 H299 H300 H301 H302 H303 H304 H305 H306 H307 H308 H309 H310 H311 H312 H313 H314 H315 H316 H317 H318 H319 H320 H321 H322 H323 H324 H325 H326 H327 H328 H329 H330 H331 H332 H333 H334 H335 H336 H337 H338 H339 H340 H341 H342 H343 H344 H345 H346 H347 H348 H349 H350 H351 H352 H353 H354 H355 H356 H357 H358 H359 H360 H361 H362 H363 H364 H365 H366 H367 H368 H369 H370 H371 H372 H373 H374 H375 H376 H377 H378 H379 H380 H381 H382 H383 H384 H385 H386 H387 H388 H389 H390 H391 H392 H393 H394 H395 H396 H397 H398 H399 H400 H401 H402 H403 H404 H405 H406 H407 H408 H409 H410 H411 H412 H413 H414 H415 H416 H417 H418 H419 H420 H421 H422 H423 H424 H425 H426 H427 H428 H429 H430 H431 H432 H433 H434 H435 H436 H437 H438 H439 H440 H441 H442 H443 H444 H445 H446 H447 H448 H449 H450 H451 H452 H453 H454 H455 H456 H457 H458 H459 H460 H461 H462 H463 H464 H465 H466 H467 H468 H469 H470 H471 H472 H473 H474 H475 H476 H477 H478 H479 H480 H481 H482 H483 H484 H485 H486 H487 H488 H489 H490 H491 H492 H493 H494 H495 H496 H497 H498 H499 H500 H501 H502 H503 H504 H505 H506 H507 H508 H509 H510 H511 H512 H513 H514 H515 H516 H517 H518 H519 H520 H521 H522 H523 H524 H525 H526 H527 H528 H529 H530 H531 H532 H533 H534 H535 H536 H537 H538 H539 H540 H541 H542 H543 H544 H545 H546 H547 H548 H549 H550 H551 H552 H553 H554 H555 H556 H557 H558 H559 H560 H561 H562 H563 H564 H565 H566 H567 H568 H569 H570 H571 H572 H573 H574 H575 H576 H577 H578 H579 H580 H581 H582 H583 H584 H585 H586 H587 H588 H589 H590 H591 H592 H593 H594 H595 H596 H597 H598 H599 H600 H601 H602 H603 H604 H605 H606 H607 H608 H609 H610 H611 H612 H613 H614 H615 H616 H617 H618 H619 H620 H621 H622 H623 H624 H625 H626 H627 H628 H629 H630 H631 H632 H633 H634 H635 H636 H637 H638 H639 H640 H641 H642 H643 H644 H645 H646 H647 H648 H649 H650 H651 H652 H653 H654 H655 H656 H657 H658 H659 H660 H661 H662 H663 H664 H665 H666 H667 H668 H669 H670 H671 H672 H673 H674 H675 H676 H677 H678 H679 H680 H681 H682 H683 H684 H685 H686 H687 H688 H689 H690 H691 H692 H693 H694 H695 H696 H697 H698 H699 H700 H701 H702 H703 H704 H705 H706 H707 H708 H709 H710 H711 H712 H713 H714 H715 H716 H717 H718 H719 H720 H721 H722 H723 H724 H725 H726 H727 H728 H729 H730 H731 H732 H733 H734 H735 H736 H737 H738 H739 H740 H741 H742 H743 H744 H745 H746 H747 H748 H749 H750 H751 H752 H753 H754 H755 H756 H757 H758 H759 H760 H761 H762 H763 H764 H765 H766 H767 H768 H769 H770 H771 H772 H773 H774 H775 H776 H777 H778 H779 H780 H781 H782 H783 H784 H785 H786 H787 H788 H789 H790 H791 H792 H793 H794 H795 H796 H797 H798 H799 H800 H801 H802 H803 H804 H805 H806 H807 H808 H809 H810 H811 H812 H813 H814 H815 H816 H817 H818 H819 H820 H821 H822 H823 H824 H825 H826 H827 H828 H829 H830 H831 H832 H833 H834 H835 H836 H837 H838 H839 H840 H841 H842 H843 H844 H845 H846 H847 H848 H849 H850 H851 H852 H853 H854 H855 H856 H857 H858 H859 H860 H861 H862 H863 H864 H865 H866 H867 H868 H869 H870 H871 H872 H873 H874 H875 H876 H877 H878 H879 H880 H881 H882 H883 H884 H885 H886 H887 H888 H889 H890 H891 H892 H893 H894 H895 H896 H897 H898 H899 H900 H901 H902 H903 H904 H905 H906 H907 H908 H909 H910 H911 H912 H913 H914 H915 H916 H917 H918 H919 H920 H921 H922 H923 H924 H925 H926 H927 H928 H929 H930 H931 H932 H933 H934 H935 H936 H937 H938 H939 H940 H941 H942 H943 H944 H945 H946 H947 H948 H949 H950 H951 H952 H953 H954 H955 H956 H957 H958 H959 H960 H961 H962 H963 H964 H965 H966 H967 H968 H969 H970 H971 H972 H973 H974 H975 H976 H977 H978 H979 H980 H981 H982 H983 H984 H985 H986 H987 H988 H989 H990 H991 H992 H993 H994 H995 H996 H997 H998 H999 H1000 H1001 H1002 H1003 H1004 H1005 H1006 H1007 H1008 H1009 H1010 H1011 H1012 H1013 H1014 H1015 H1016 H1017 H1018 H1019 H1020 H1021 H1022 H1023 H1024 H1025 H1026 H1027 H1028 H1029 H1030 H1031 H1032 H1033 H1034 H1035 H1036 H1037 H1038 H1039 H1040 H1041 H1042 H1043 H1044 H1045 H1046 H1047 H1048 H1049 H1050 H1051 H1052 H1053 H1054 H1055 H1056 H1057 H1058 H1059 H1060 H1061 H1062 H1063 H1064 H1065 H1066 H1067 H1068 H1069 H1070 H1071 H1072 H1073 H1074 H1075 H1076 H1077 H1078 H1079 H1080 H1081 H1082 H1083 H1084 H1085 H1086 H1087 H1088 H1089 H1090 H1091 H1092 H1093 H1094 H1095 H1096 H1097 H1098 H1099 H1100 H1101 H1102 H1103 H1104 H1105 H1106 H1107 H1108 H1109 H1110 H1111 H1112 H1113 H1114 H1115 H1116 H1117 H1118 H1119 H1120 H1121 H1122 H1123 H1124 H1125 H1126 H1127 H1128 H1129 H1130 H1131 H1132 H1133 H1134 H1135 H1136 H1137 H1138 H1139 H1140 H1141 H1142 H1143 H1144 H1145 H1146 H1147 H1148 H1149 H1150 H1151 H1152 H1153 H1154 H1155 H1156 H1157 H1158 H1159 H1160 H1161 H1162 H1163 H1164 H1165 H1166 H1167 H1168 H1169 H1170 H1171 H1172 H1173 H1174 H1175 H1176 H1177 H1178 H1179 H1180 H1181 H1182 H1183 H1184 H1185 H1186 H1187 H1188 H1189 H1190 H1191 H1192 H1193 H1194 H1195 H1196 H1197 H1198 H1199 H1200 H1201 H1202 H1203 H1204 H1205 H1206 H1207 H1208 H1209 H1210 H1211 H1212 H1213 H1214 H1215 H1216 H1217 H1218 H1219 H1220 H1221 H1222 H1223 H1224 H1225 H1226 H1227 H1228 H1229 H1230 H1231 H1232 H1233 H1234 H1235 H1236 H1237 H1238 H1239 H1240 H1241 H1242 H1243 H1244 H1245 H1246 H1247 H1248 H1249 H1250 H1251 H1252 H1253 H1254 H1255 H1256 H1257 H1258 H1259 H1260 H1261 H1262 H1263 H1264 H1265 H1266 H1267 H1268 H1269 H1270 H1271 H1272 H1273 H1274 H1275 H1276 H1277 H1278 H1279 H1280 H1281 H1282 H1283 H1284 H1285 H1286 H1287 H1288 H1289 H1290 H1291 H1292 H1293 H1294 H1295 H1296 H1297 H1298 H1299 H1300 H1301 H1302 H1303 H1304 H1305 H1306 H1307 H1308 H1309 H1310 H1311 H1312 H1313 H1314 H1315 H1316 H1317 H1318 H1319 H1320 H1321 H1322 H1323 H1324 H1325 H1326 H1327 H1328 H1329 H1330 H1331 H1332 H1333 H1334 H1335 H1336 H1337 H1338 H1339 H1340 H1341 H1342 H1343 H1344 H1345 H1346 H1347 H1348 H1349 H1350 H1351 H1352 H1353 H1354 H1355 H1356 H1357 H1358 H1359 H1360 H1361 H1362 H1363 H1364 H1365 H1366 H1367 H1368 H1369 H1370 H1371 H1372 H1373 H1374 H1375 H1376 H1377 H1378 H1379 H1380 H1381 H1382 H1383 H1384 H1385 H1386 H1387 H1388 H1389 H1390 H1391 H1392 H1393 H1394 H1395 H1396 H1397 H1398 H1399 H1400 H1401 H1402 H1403 H1404 H1405 H1406 H1407 H1408 H1409 H1410 H1411 H1412 H1413 H1414 H1415 H1416 H1417 H1418 H1419 H1420 H1421 H1422 H1423 H1424 H1425 H1426 H1427 H1428 H1429 H1430 H1431 H1432 H1433 H1434 H1435 H1436 H1437 H1438 H1439 H1440 H1441 H1442 H1443 H1444 H1445 H1446 H1447 H1448 H1449 H1450 H1451 H1452 H1453 H1454 H1455 H1456 H1457 H1458 H1459 H1460 H1461 H1462 H1463 H1464 H1465 H1466 H1467 H1468 H1469 H1470 H1471 H1472 H1473 H1474 H1475 H1476 H1477 H1478 H1479 H1480 H1481 H1482 H1483 H1484 H1485 H1486 H1487 H1488 H1489 H1490 H1491 H1492 H1493 H1494 H1495 H1496 H1497 H1498 H1499 H1500 H1501 H1502 H1503 H1504 H1505 H1506 H1507 H1508 H1509 H1510 H1511 H1512 H1513 H1514 H1515 H1516 H1517 H1518 H1519 H1520 H1521 H1522 H1523 H1524 H1525 H1526 H1527 H1528 H1529 H1530 H1531 H1532 H1533 H1534 H1535 H1536 H1537 H1538 H1539 H1540 H1541 H1542 H1543 H1544 H1545 H1546 H1547 H1548 H1549 H1550 H1551 H1552 H1553 H1554 H1555 H1556 H1557 H1558 H1559 H1560 H1561 H1562 H1563 H1564 H1565 H1566 H1567 H1568 H1569 H1570 H1571 H1572 H1573 H1574 H1575 H1576 H1577 H1578 H1579 H1580 H1581 H1582 H1583 H1584 H1585 H1586 H1587 H1588 H1589 H1590 H1591 H1592 H1593 H1594 H1595 H1596 H1597 H1598 H1599 H1600 H1601 H1602 H1603 H1604 H1605 H1606 H1607 H1608 H1609 H1610 H1611 H1612 H1613 H1614 H1615 H1616 H1617 H1618 H1619 H1620 H1621 H1622 H1623 H1624 H1625 H1626 H1627 H1628 H1629 H1630 H1631 H1632 H1633 H1634 H1635 H1636 H1637 H1638 H1639 H1640 H1641 H1642 H1643 H1644 H1645 H1646 H1647 H1648 H1649 H1650 H1651 H1652 H1653 H1654 H1655 H1656 H1657 H1658 H1659 H1660 H1661 H1662 H1663 H1664 H1665 H1666 H1667 H1668 H1669 H1670 H1671 H1672 H1673 H1674 H1675 H1676 H1677 H1678 H1679 H1680 H1681 H1682 H1683 H1684 H1685 H1686 H1687 H1688 H1689 H1690 H1691 H1692 H1693 H1694 H1695 H1696 H1697 H1698 H1699 H1700 H1701 H1702 H1703 H1704 H1705 H1706 H1707 H1708 H1709 H1710 H1711 H1712 H1713 H1714 H1715 H1716 H1717 H1718 H1719 H1720 H1721 H1722 H1723 H1724 H1725 H1726 H1727 H1728 H1729 H1730 H1731 H1732 H1733 H1734 H1735 H1736 H1737 H1738 H1739 H1740 H1741 H1742 H1743 H1744 H1745 H1746 H1747 H1748 H1749 H1750 H1751 H1752 H1753 H1754 H1755 H1756 H1757 H1758 H1759 H1760 H1761 H1762 H1763 H1764 H1765 H1766 H1767 H1768 H1769 H1770 H1771 H1772 H1773 H1774 H1775 H1776 H1777 H1778 H1779 H1780 H1781 H1782 H1783 H1784 H1785 H1786 H1787 H1788 H1789 H1790 H1791 H1792 H1793 H1794 H1795 H1796 H1797 H1798 H1799 H1800 H1801 H1802 H1803 H1804 H1805 H1806 H1807 H1808 H1809 H1810 H1811 H1812 H1813 H1814 H1815 H1816 H1817 H1818 H1819 H1820 H1821 H1822 H1823 H1824 H1825 H1826 H1827 H1828 H1829 H1830 H1831 H1832 H1833 H1834 H1835 H1836 H1837 H1838 H1839 H1840 H1841 H1842 H1843 H1844 H1845 H1846 H1847 H1848 H1849 H1850 H1851 H1852 H1853 H1854 H1855 H1856 H1857 H1858 H1859 H1860 H1861 H1862 H1863 H1864 H1865 H1866 H1867 H1868 H1869 H1870 H1871 H1872 H1873 H1874 H1875 H1876 H1877 H1878 H1879 H1880 H1881 H1882 H1883 H1884 H1885 H1886 H1887 H1888 H1889 H1890 H1891 H1892 H1893 H1894 H1895 H1896 H1897 H1898 H1899 H1900 H1901 H1902 H1903 H1904 H1905 H1906 H1907 H1908 H1909 H1910 H1911 H1912 H1913 H1914 H1915 H1916 H1917 H1918 H1919 H1920 H1921 H1922 H1923 H1924 H1925 H1926 H1927 H1928 H1929 H1930 H1931 H1932 H1933 H1934 H1935 H1936 H1937 H1938 H1939 H1940 H1941 H1942 H1943 H1944 H1945 H1946 H1947 H1948 H1949 H1950 H1951 H1952 H1953 H1954 H1955 H1956 H1957 H1958 H1959 H1960 H1961 H1962 H1963 H1964 H1965 H1966 H1967 H1968 H1969 H1970 H1971 H1972 H1973 H1974 H1975 H1976 H1977 H1978 H1979 H1980 H1981 H1982 H1983 H1984 H1985 H1986 H1987 H1988 H1989 H1990 H1991 H1992 H1993 H1994 H1995 H1996 H1997 H1998 H1999 H2000 H2001 H2002 H2003 H2004 H2005 H2006 H2007 H2008 H2009 H2010 H2011 H2012 H2013 H2014 H2015 H2016 H2017 H2018 H2019 H2020 H2021 H2022 H2023 H2024 H2025 H2026 H2027 H2028 H2029 H2030 H2031 H2032 H2033 H2034 H2035 H2036 H2037 H2038 H2039 H2040 H2041 H2042 H2043 H2044 H2045 H2046 H2047 H2048 H2049 H2050 H2051 H2052 H2053 H2054 H2055 H2056 H2057 H2058 H2059 H2060 H2061 H2062 H2063 H2064 H2065 H2066 H2067 H2068 H2069 H2070 H2071 H2072 H2073 H2074 H2075 H2076 H2077 H2078 H2079 H2080 H2081 H2082 H2083 H2084 H2085 H2086 H2087 H2088 H2089 H2090 H2091 H2092 H2093 H2094 H2095 H2096 H2097 H2098 H2099 H2100 H2101 H2102 H2103 H2104 H2105 H2106 H2107 H2108 H2109 H2110 H2111 H2112 H2113 H2114 H2115 H2116 H2117 H2118 H2119 H2120 H2121 H2122 H2123 H2124 H2125 H2126 H2127 H2128 H2129 H2130 H2131 H2132 H2133 H2134 H2135 H2136 H2137 H2138 H2139 H2140 H2141 H2142 H2143 H2144 H2145 H2146 H2147 H2148 H2149 H2150 H2151 H2152 H2153 H2154 H2155 H2156 H2157 H2158 H2159 H2160 H2161 H2162 H2163 H2164 H2165 H2166 H2167 H2168 H2169 H2170 H2171 H2172 H2173 H2174 H2175 H2176 H2177 H2178 H2179 H2180 H2181 H2182 H2183 H2184 H2185 H2186 H2187 H2188 H2189 H2190 H2191 H2192 H2193 H2194 H2195 H2196 H2197 H2198 H2199 H2200 H2201 H2202 H2203 H2204 H2205 H2206 H2207 H2208 H2209 H2210 H2211 H2212 H2213 H2214 H2215 H2216 H2217 H2218 H2219 H2220 H2221 H2222 H2223 H2224 H2225 H2226 H2227 H2228 H2229 H2230 H2231 H2232 H2233 H2234 H2235 H2236 H2237 H2238 H2239 H2240 H2241 H2242 H2243 H2244 H2245 H2246 H2247 H2248 H2249 H2250 H2251 H2252 H2253 H2254 H2255 H2256 H2257 H2258 H2259 H2260 H2261 H2262 H2263 H2264 H2265 H2266 H2267 H2268 H2269 H2270 H2271 H2272 H2273 H2274 H2275 H2276 H2277 H2278 H2279 H2280 H2281 H2282 H2283 H2284 H2285 H2286 H2287 H2288 H2289 H2290 H2291 H2292 H2293 H2294 H2295 H2296 H2297 H2298 H2299 H2300 H2301 H2302 H2303 H2304 H2305 H2306 H2307 H2308 H2309 H2310 H2311 H2312 H2313 H2314 H2315 H2316 H2317 H2318 H2319 H2320 H2321 H2322 H2323 H2324 H2325 H2326 H2327 H2328 H2329 H2330 H2331 H2332 H2333 H2334 H2335 H2336 H2337 H2338 H2339 H2340 H2341 H2342 H2343 H2344 H2345 H2346 H2347 H2348 H2349 H2350 H2351 H2352 H2353 H2354 H2355 H2356 H2357 H2358 H2359 H2360 H2361 H2362 H2363 H2364 H2365 H2366 H2367 H2368 H2369 H2370 H2371 H2372 H2373 H2374 H2375 H2376 H2377 H2378 H2379 H2380 H2381 H2382 H2383 H2384 H2385 H2386 H2387 H2388 H2389 H2390 H2391 H2392 H2393 H2394 H2395 H2396 H2397 H2398 H2399 H2400 H2401 H2402 H2403 H2404 H2405 H2406 H2407 H2408 H2409 H2410 H2411 H2412 H2413 H2414 H2415 H2416 H2417 H2418 H2419 H2420 H2421 H2422 H2423 H2424 H2425 H2426 H2427 H2428 H2429 H2430 H2431 H2432 H2433 H2434 H2435 H2436 H2437 H2438 H2439 H2440 H2441 H2442 H2443 H2444 H2445 H2446 H2447 H2448 H2449 H2450 H2451 H2452 H2453 H2454 H2455 H2456 H2457 H2458 H2459 H2460 H2461 H2462 H2463 H2464 H2465 H2466 H2467 H2468 H2469 H2470 H2471 H2472 H2473 H2474 H2475 H2476 H2477 H2478 H2479 H2480 H2481 H2482 H2483 H2484 H2485 H2486 H2487 H2488 H2489 H2490 H2491 H2492 H2493 H2494 H2495 H2496 H2497 H2498 H2499 H2500 H2501 H2502 H2503 H2504 H2505 H2506 H2507 H2508 H2509 H2510 H2511 H2512 H2513 H2514 H2515 H2516 H2517 H2518 H2519 H2520 H2521 H2522 H2523 H2524 H2525 H2526 H2527 H2528 H2529 H2530 H2531 H25

Extended FIM alert detail showing complete before/after cryptographic hash comparison for the modified credential file, confirming precise integrity verification.

t rule.tsc	PI1.4, PI1.5, CC6.1, CC6.8, CC7.2, CC7.3
t syscheck.changed_attributes	size, mtime, md5, sha1, sha256
t syscheck.event	modified
t syscheck.gid_after	42
t syscheck.gname_after	shadow
t syscheck.inode_after	395106
t syscheck.md5_after	9b98c6216a5eb5032638dcebf8c2cd71
t syscheck.md5_before	c5ece6dd7ec5bdfd1e5f8e9da4eb3a3d
t syscheck.mode	realtime
syscheck.mtime_after	Apr 8, 2026 @ 12:10:02.000
syscheck.mtime_before	Mar 19, 2026 @ 12:21:25.000
t syscheck.path	/etc/shadow
t syscheck.perm_after	rw-r-----
t syscheck.sha1_after	abe818373ca8f39108ee68f8d3a92dc32d7c4ee5
t syscheck.sha1_before	9a46a99092dd7e742d06ceff228611bfc61aca75
t syscheck.sha256_after	8eb57c0c0ca97e0367bff670f394d95921e97fd4c242fa5b9c302e8b22c4fa3f
t syscheck.sha256_before	0d521c245a85953a054c4e903f1f0f002ae97e8aed32c8634d3a45e6828b6d22
# syscheck.size_after	1,143
# syscheck.size_before	1,140
t syscheck.uid_after	0
t syscheck.username_after	root
timestamp	Apr 8, 2026 @ 16:10:03.032

Thuruvaran Thavarakan Thavapalan
151654225

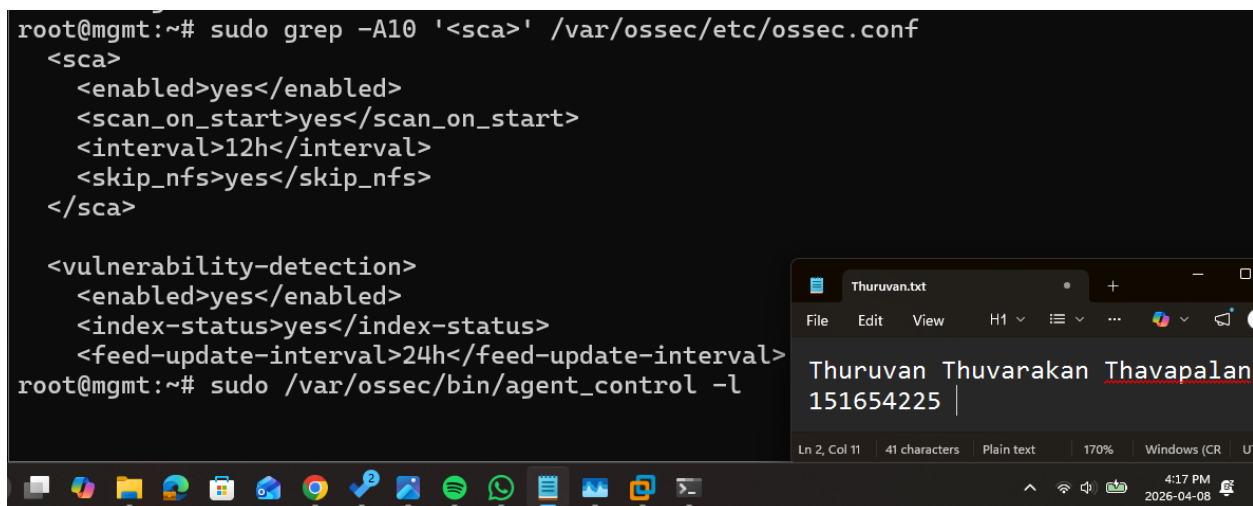
3.14b - FIM Hash Comparison Detail

Control 4 — Secure Configuration of Enterprise Assets and Software

Secure configuration is enforced through Wazuh's Security Configuration Assessment module, which continuously evaluates all agents against CIS benchmarks. The CIS Microsoft Windows Server 2019 Benchmark v2.0.0 is applied to G5-AD (score: 24%, 84 passed, 263 failed) and the CIS Ubuntu Linux 24.04 LTS Benchmark v1.0.0 is applied to Linux agents (CoreRouter: 47%, 116 passed, 128 failed). Host-level firewalls are deployed as systemd services on both the Wazuh Manager and jumpbox-ssh with default DROP policies. SSH hardening and default account controls are applied across all Linux hosts.

Wazuh Manager ossec.conf SCA configuration

showing `<enabled>yes</enabled>`, `<scan_on_start>yes</scan_on_start>`, and `<interval>12h</interval>`, confirming automated CIS benchmark scanning is enabled across all agents.

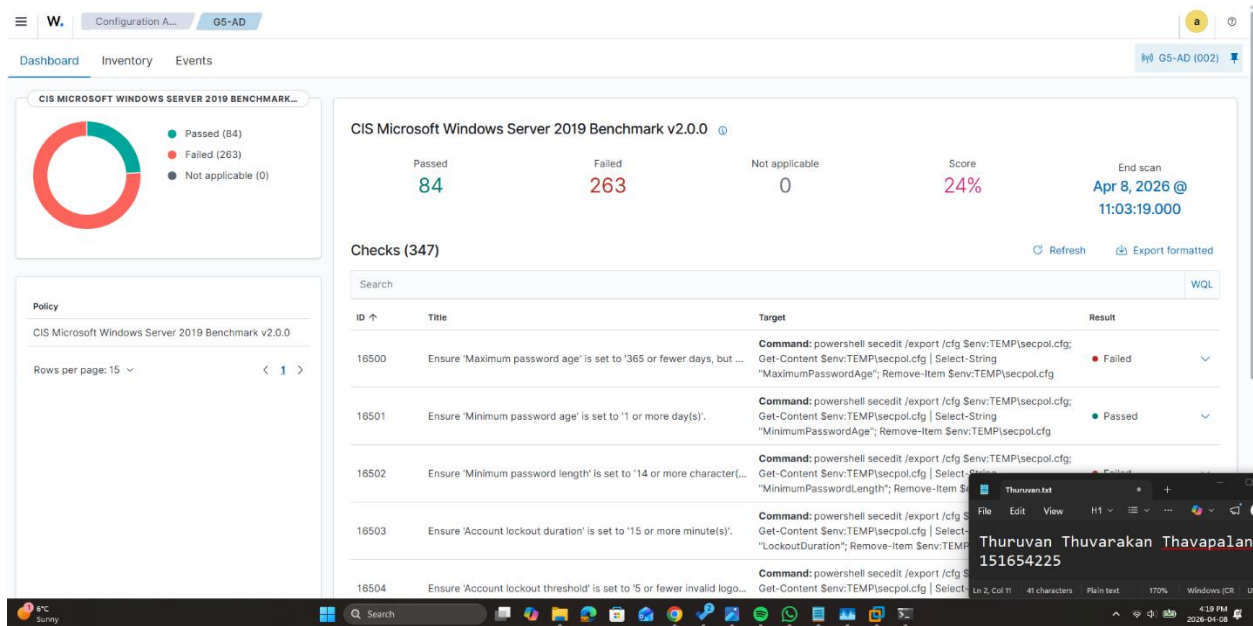


```
root@mgmt:~# sudo grep -A10 '<sca>' /var/ossec/etc/ossec.conf
<sca>
  <enabled>yes</enabled>
  <scan_on_start>yes</scan_on_start>
  <interval>12h</interval>
  <skip_nfs>yes</skip_nfs>
</sca>

<vulnerability-detection>
  <enabled>yes</enabled>
  <index-status>yes</index-status>
  <feed-update-interval>24h</feed-update-interval>
root@mgmt:~# sudo /var/ossec/bin/agent_control -l
```

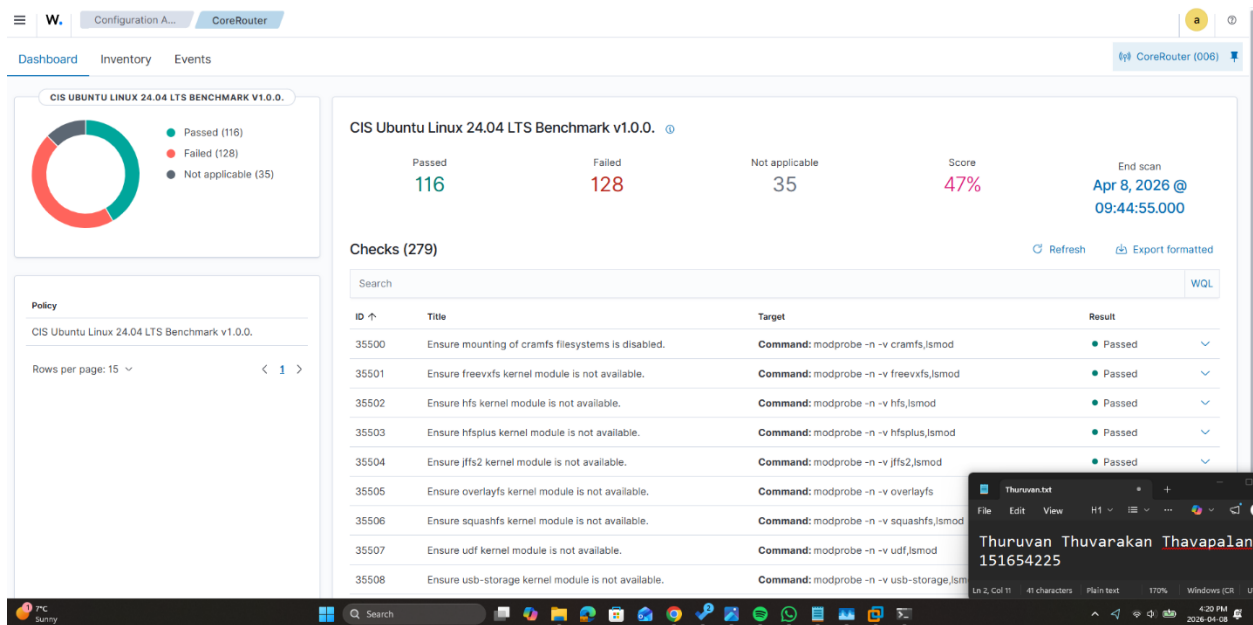
4.1a - SCA Config Enabled

Wazuh Configuration Assessment dashboard for G5-AD showing CIS Windows Server 2019 Benchmark v2.0.0 with 84 passed, 263 failed, score 24%, with individual check results and remediation guidance visible.



4.1b - CIS SCA Dashboard AD

Wazuh Configuration Assessment dashboard for CoreRouter showing CIS Ubuntu Linux 24.04 LTS Benchmark v1.0.0 with 116 passed, 128 failed, score 47%.



4.1c - CIS SCA Dashboard CoreRouter

Core router iptables FORWARD and OUTPUT chain rules showing VLAN-specific subnet access controls, DNS forwarding rules restricted to 172.16.33.0/26, and HTTP/HTTPS access controls for container subnets.

```

0 0 ACCEPT 6 -- * * 172.16.48.0/29 172.16.48.16/29 tcp dpt:80 ctstate NEW,ESTA
BLISHED
0 0 ACCEPT 6 -- * * 172.16.48.0/29 172.16.48.32/29 tcp dpt:80 ctstate NEW,ESTA
BLISHED
0 0 ACCEPT 6 -- * * 0.0.0.0/0 172.16.48.24/29 match-set INTERNAL_SUBNETS
src tcp dpt:445 ctstate NEW,ESTABLISHED
0 0 ACCEPT 17 -- * * ens37 172.16.33.0/26 0.0.0.0/0 udp dpt:53 ctstate NEW,ESTA
BLISHED
0 0 ACCEPT 17 -- * * 172.16.2.0/27 172.16.33.0/26 udp dpt:53 STRING match "I
0267350865787465726e616c001" ALGO name bm
0 0 ACCEPT 6 -- * * 172.16.2.0/27 172.16.33.0/26 tcp dpt:53 STRING match "I
0267350865787465726e616c001" ALGO name bm
0 0 ACCEPT 6 -- * * ens37 0.0.0.0/0 0.0.0.0/0 match-set INTERNET_ALLOWED_
SUBNETS src multiport dports 80,443 ctstate NEW,ESTABLISHED
0 0 ACCEPT 17 -- * * ens37 0.0.0.0/0 0.0.0.0/0 match-set INTERNET_ALLOWED_
SUBNETS src udp dpt:53 ctstate NEW,ESTABLISHED
0 0 ACCEPT 17 -- * * 172.16.2.0/27 172.16.48.48/29 udp dpt:1194 ctstate NEW,ES
TABLISHED
25 1780 LOG 0 -- * * 0.0.0.0/0 0.0.0.0/0 LOG flags 0 level 4 prefix
"FORWARD-DROP: "

Chain OUTPUT (policy DROP 587 packets, 816K bytes)
pkts bytes target prot opt in out source destination
2 690 ACCEPT 0 -- * * 0.0.0.0/0 0.0.0.0/0 ctstate RELATED,ESTABLISHED
6 450 ACCEPT 0 -- * * lo 0.0.0.0/0 0.0.0.0/0
0 0 ACCEPT 6 -- * * 0.0.0.0/0 172.16.48.0/29 tcp dpt:443
0 0 ACCEPT 6 -- * * 0.0.0.0/0 172.16.48.0/29 tcp dpt:80
0 0 ACCEPT 6 -- * * 172.16.48.0/29 0.0.0.0/0 tcp spt:443 flags:0x12/0x12
0 0 ACCEPT 6 -- * * 172.16.48.0/29 0.0.0.0/0 tcp spt:80 flags:0x12/0x12
0 0 ACCEPT 89 -- * * ens37 0.0.0.0/0 0.0.0.0/0
0 0 ACCEPT 17 -- * * 0.0.0.0/0 0.0.0.0/0
h-set INTERNAL_SUBNETS dst limit: avg 5/min burst 10
0 0 ACCEPT 6 -- * * 0.0.0.0/0 172.16.23.16/28
596 816K LOG 0 -- * * 0.0.0.0/0 0.0.0.0/0
"OUTPUT-DROP: "
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 09:45:18 $

```

4.2a - Core Router IPTables Rules

jumpbox-ssh sshd_config output showing PermitRootLogin prohibit-password, confirming SSH hardening prevents direct root login via password.

```

ttthavapalan@jumpbox-ssh2:~$ cat /etc/ssh/sshd_config | grep -E 'PermitRootLogin|PasswordAuthentication|MaxAuthTries|Protocol'
#PermitRootLogin prohibit-password
#MaxAuthTries 6
PasswordAuthentication yes
# PasswordAuthentication, Depending on your PAM configuration,
# the setting of "PermitRootLogin prohibit-password".
# PAM authentication, then enable this but set PasswordAuthentication
ttthavapalan@jumpbox-ssh2:~$

```

4.2b - SSH Server Hardening

Wazuh Manager iptables -L -n -v --line-numbers output showing INPUT chain with DROP default policy, rules permitting only required ports (22, 1514, 1515, 1516, 55000, 443), with LOG rule for dropped packets prefixed [CIS-4.4-DROP].

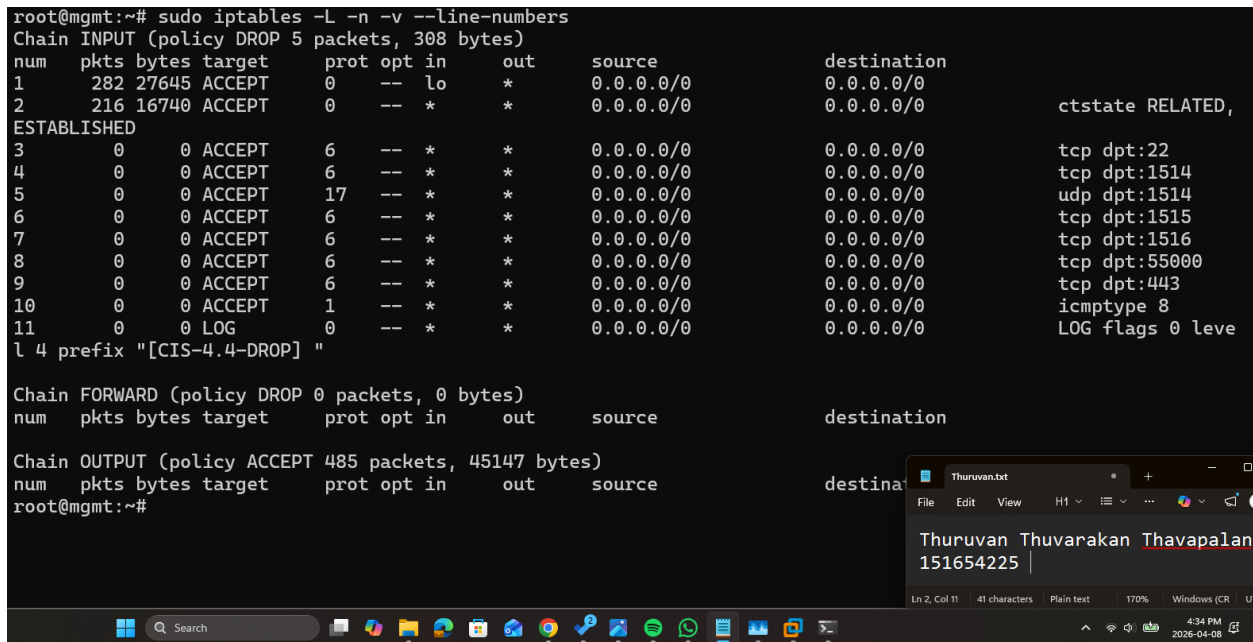
```

root@mgmt:~# sudo iptables -L -n -v --line-numbers
Chain INPUT (policy DROP 5 packets, 308 bytes)
num  pkts bytes target    prot opt in     out     source    destination
1      282 27645 ACCEPT    0     --  lo     *        0.0.0.0/0
2     216 16740 ACCEPT    0     --  *      *        0.0.0.0/0
ESTABLISHED
3        0      0 ACCEPT    6     --  *      *        0.0.0.0/0
4        0      0 ACCEPT    6     --  *      *        0.0.0.0/0
5        0      0 ACCEPT   17     --  *      *        0.0.0.0/0
6        0      0 ACCEPT    6     --  *      *        0.0.0.0/0
7        0      0 ACCEPT    6     --  *      *        0.0.0.0/0
8        0      0 ACCEPT    6     --  *      *        0.0.0.0/0
9        0      0 ACCEPT    6     --  *      *        0.0.0.0/0
10       0      0 ACCEPT    1     --  *      *        0.0.0.0/0
11       0      0 LOG       0     --  *      *        0.0.0.0/0
l 4 prefix "[CIS-4.4-DROP] "

Chain FORWARD (policy DROP 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source    destination

Chain OUTPUT (policy ACCEPT 485 packets, 45147 bytes)
num  pkts bytes target    prot opt in     out     source    destination
root@mgmt:~#

```



The screenshot shows a terminal window with the output of the command `sudo iptables -L -n -v --line-numbers`. The output displays the INPUT chain with a default policy of DROP and 11 rules. Rules 1-10 are ACCEPT rules for various ports (22, 1514, 1515, 1516, 55000, 443). Rule 11 is a LOG rule for dropped packets. A text editor window titled 'Thuruvan.txt' is open in the foreground, showing the text 'Thuruvan Thuvarakan Thavapalan 151654225'.

4.4a - Wazuh Manager Firewall Rules

jumpbox-ssh jumpbox-firewall.service systemd status showing active, alongside iptables output confirming INPUT DROP default policy with rules permitting only SSH, Wazuh agent port, and ICMP.

```

• jumpbox-firewall.service - CIS Control 4.4 - jumpbox-ssh Host Firewall
   Loaded: loaded (/etc/systemd/system/jumpbox-firewall.service; enabled; preset: enabled)
   Active: active (exited) since Wed 2026-04-08 16:36:58 EDT; 18s ago
   Process: 30657 ExecStart=/usr/local/bin/jumpbox_firewall.sh (code=exited, status=0/SUCCESS)
   Main PID: 30657 (code=exited, status=0/SUCCESS)
   CPU: 62ms

Apr 08 16:36:58 jumpbox-ssh2 jumpbox-firewall[30672]: 3      0      0 ACCEPT      6      -- *      *      0.0.0.0/0
Apr 08 16:36:58 jumpbox-ssh2 jumpbox-firewall[30672]: 4      0      0 ACCEPT      6      -- *      *      0.0.0.0/0
Apr 08 16:36:58 jumpbox-ssh2 jumpbox-firewall[30672]: 5      0      0 ACCEPT     17      -- *      *      0.0.0.0/0
Apr 08 16:36:58 jumpbox-ssh2 jumpbox-firewall[30672]: 6      0      0 ACCEPT      1      -- *      *      0.0.0.0/0
Apr 08 16:36:58 jumpbox-ssh2 jumpbox-firewall[30672]: 7      0      0 LOG          0      -- *      *      0.0.0.0/0
Apr 08 16:36:58 jumpbox-ssh2 jumpbox-firewall[30672]: Chain FORWARD (policy DROP 0 packets, 0 bytes)
Apr 08 16:36:58 jumpbox-ssh2 jumpbox-firewall[30672]: num  pkts bytes target  prot opt in  out  source
Apr 08 16:36:58 jumpbox-ssh2 jumpbox-firewall[30672]: Chain OUTPUT (policy ACCEPT 1 packets, 226 bytes)
Apr 08 16:36:58 jumpbox-ssh2 jumpbox-firewall[30672]: num  pkts bytes target  prot opt in  out  source
Apr 08 16:36:58 jumpbox-ssh2 systemd[1]: Finished jumpbox-firewall.service - CIS Control 4.4 - jumpbox-ssh Host Fir
ttthavapalan@jumpbox-ssh2:~$ sudo iptables -L -n -v --line-numbers
Chain INPUT (policy DROP 2 packets, 124 bytes)
num  pkts bytes target  prot opt in  out  source      destination
1      36 3426 ACCEPT      0  --  lo  *      0.0.0.0/0    0.0.0.0/0
2     110 20674 ACCEPT      0  --  *  *      0.0.0.0/0    0.0.0.0/0
ctstate RELATED,ESTABLISHED
3      0      0 ACCEPT      6  --  *  *      0.0.0.0/0    0.0.0.0/0
4      0      0 ACCEPT      6  --  *  *      0.0.0.0/0    0.0.0.0/0
5      0      0 ACCEPT     17  --  *  *      0.0.0.0/0    0.0.0.0/0
6      0      0 ACCEPT      1  --  *  *      0.0.0.0/0    0.0.0.0/0
7      1      72 LOG          0  --  *  *      0.0.0.0/0    0.0.0.0/0
LOG flags 0 level 4 pr
efix "[CIS-4.4-DROP]"

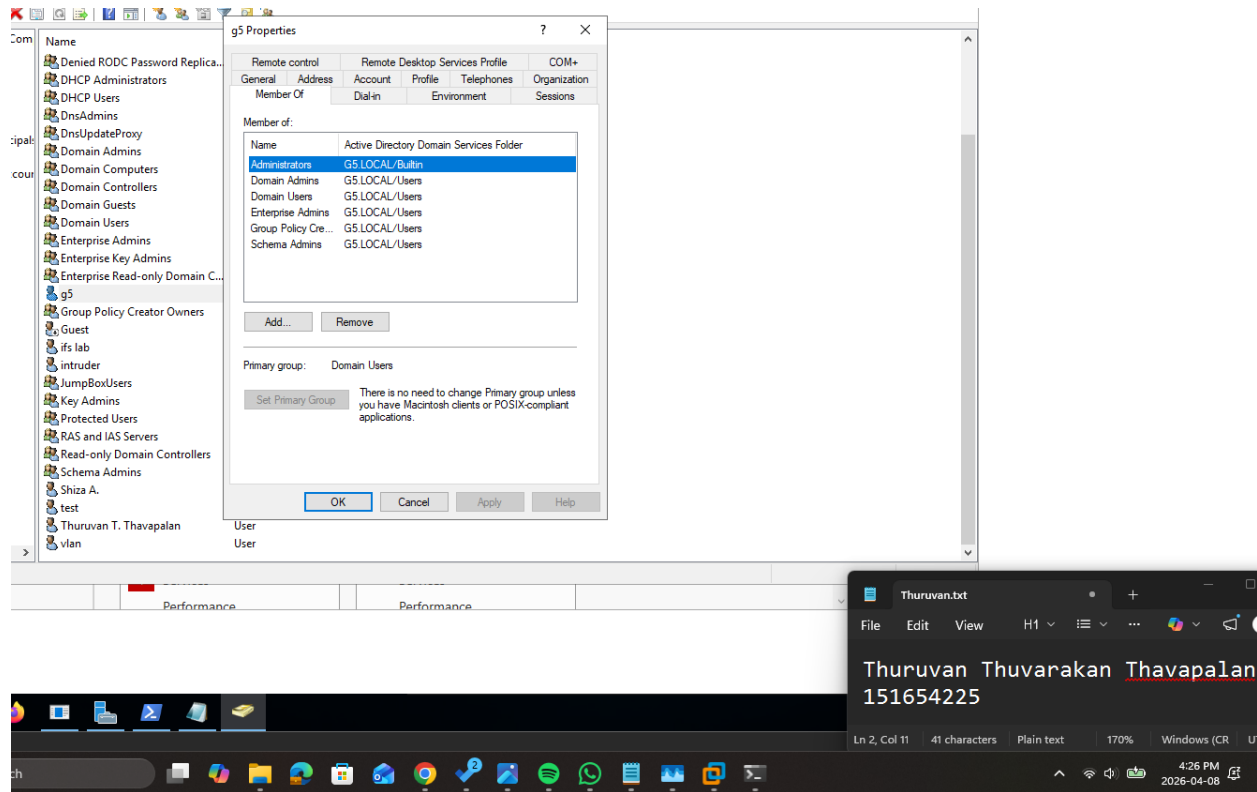
Chain FORWARD (policy DROP 0 packets, 0 bytes)
num  pkts bytes target  prot opt in  out  source      destination

Chain OUTPUT (policy ACCEPT 165 packets, 19501 bytes)
num  pkts bytes target  prot opt in  out  source      destination
ttthavapalan@jumpbox-ssh2:~$

```

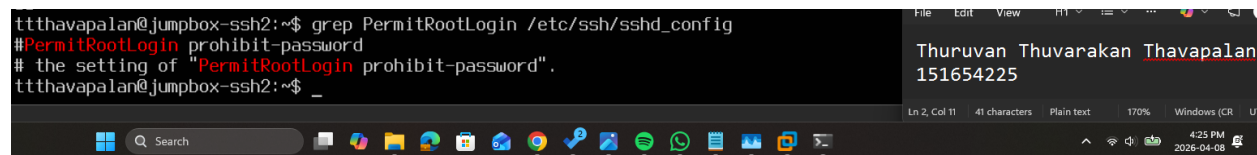
4.4b - Jumpbox Firewall Service and Rules

Active Directory Users and Computers showing disabled Guest and intruder accounts, confirming default account management per CIS Control 4.7.



4.7a - AD Admin Account Configuration

Jumpbox-ssh grep PermitRootLogin /etc/ssh/sshd_config showing PermitRootLogin prohibit-password, preventing root login via password authentication.



4.7b - SSH No Root Login

jumpbox-ssh system account listing showing all service accounts (UIDs 0 through 111) with /usr/sbin/nologin or /bin/false shells, confirming no unexpected interactive system accounts exist.

```

root 0 /bin/bash
daemon 1 /usr/sbin/nologin
bin 2 /usr/sbin/nologin
sys 3 /usr/sbin/nologin
sync 4 /bin/sync
games 5 /usr/sbin/nologin
man 6 /usr/sbin/nologin
lp 7 /usr/sbin/nologin
mail 8 /usr/sbin/nologin
news 9 /usr/sbin/nologin
uucp 10 /usr/sbin/nologin
proxy 13 /usr/sbin/nologin
www-data 33 /usr/sbin/nologin
backup 34 /usr/sbin/nologin
list 38 /usr/sbin/nologin
irc 39 /usr/sbin/nologin
_apt 42 /usr/sbin/nologin
systemd-network 998 /usr/sbin/nologin
systemd-timesync 997 /usr/sbin/nologin
dhcpcd 100 /bin/false
messagebus 101 /usr/sbin/nologin
systemd-resolve 992 /usr/sbin/nologin
pollinate 102 /bin/false
polkitd 991 /usr/sbin/nologin
syslog 103 /usr/sbin/nologin
uidd 104 /usr/sbin/nologin
tcpdump 105 /usr/sbin/nologin
tss 106 /bin/false
landscape 107 /usr/sbin/nologin
fwupd-refresh 989 /usr/sbin/nologin
usbmux 108 /usr/sbin/nologin
bind 109 /usr/sbin/nologin
sshd 110 /usr/sbin/nologin
dnsmasq 999 /usr/sbin/nologin
cockpit-ws 111 /usr/sbin/nologin
:

```

4.7c - System Accounts nologin

jumpbox-ssh ss -tlnp output showing only port 22 (sshd) and port 9090 (systemd internal), confirming no unnecessary network services are exposed.

```

tthavapalan@jumpbox-ssh2:~$ ss -tlnp
State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
LISTEN 0 4096 0.0.0.0:22 0.0.0.0:* users:(("sshd",pid=1292,fd=3))
LISTEN 0 4096 127.0.0.54:53 0.0.0.0:* users:(("systemd-resolve",pid=1,fd=3))
LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:* users:(("systemd-resolve",pid=1,fd=4))
LISTEN 0 4096 *:*9090 *:* users:(("systemd",pid=1,fd=23))
LISTEN 0 4096 [::]:22 [::]:* users:(("sshd",pid=1292,fd=4))
tthavapalan@jumpbox-ssh2:~$

```

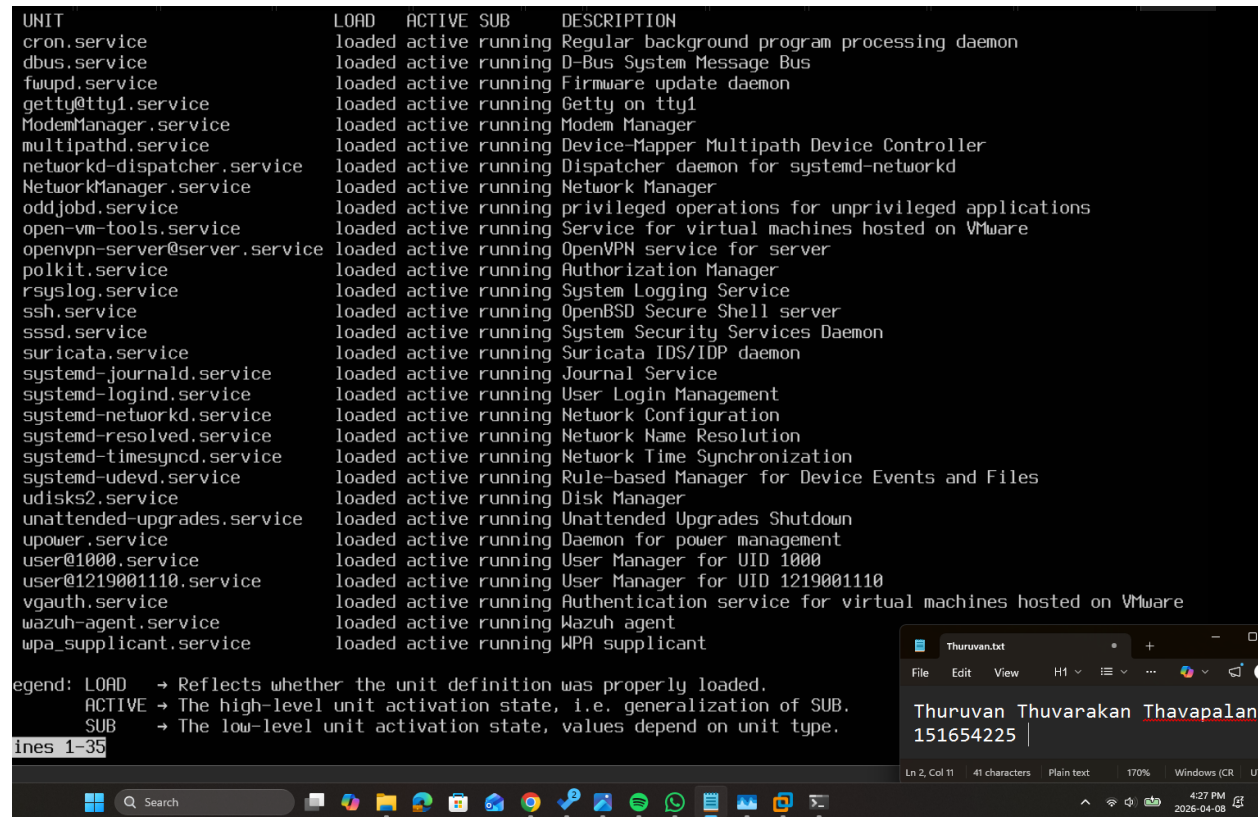
4.8a - Only Expected Ports Listening

jumpbox-ssh systemctl list-units --type=service --state=running showing only essential services: sssd, wazuh-agent, ssh, suricata, rsyslog, with no unnecessary services present.

```
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
cron.service                       loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
fwupd.service                      loaded active running Firmware update daemon
getty@tty1.service                 loaded active running Getty on tty1
ModemManager.service              loaded active running Modem Manager
multipathd.service                loaded active running Device-Mapper Multipath Device Controller
networkd-dispatcher.service        loaded active running Dispatcher daemon for systemd-networkd
NetworkManager.service            loaded active running Network Manager
oddjobd.service                   loaded active running privileged operations for unprivileged applications
open-vm-tools.service              loaded active running Service for virtual machines hosted on VMware
openvpn-server@server.service      loaded active running OpenVPN service for server
polkit.service                    loaded active running Authorization Manager
rsyslog.service                   loaded active running System Logging Service
ssh.service                       loaded active running OpenBSD Secure Shell server
sssd.service                      loaded active running System Security Services Daemon
suricata.service                  loaded active running Suricata IDS/IDP daemon
systemd-journald.service           loaded active running Journal Service
systemd-logind.service             loaded active running User Login Management
systemd-networkd.service           loaded active running Network Configuration
systemd-resolved.service           loaded active running Network Name Resolution
systemd-timesyncd.service          loaded active running Network Time Synchronization
systemd-udev.service               loaded active running Rule-based Manager for Device Events and Files
udisks2.service                   loaded active running Disk Manager
unattended-upgrades.service        loaded active running Unattended Upgrades Shutdown
upower.service                    loaded active running Daemon for power management
user@1000.service                  loaded active running User Manager for UID 1000
user@1219001110.service            loaded active running User Manager for UID 1219001110
vgauth.service                    loaded active running Authentication service for virtual machines hosted on VMware
wazuh-agent.service                loaded active running Wazuh agent
wpa_supplicant.service             loaded active running WPA supplicant
```

Legend: LOAD → Reflects whether the unit definition was properly loaded.
 ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
 SUB → The low-level unit activation state, values depend on unit type.

lines 1-35



4.8b - Minimal Services Running

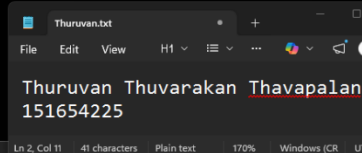
jumpbox-ssh resolved.conf showing DNS=172.16.33.2 (internal AD DNS) as primary with FallbackDNS=1.1.1.1 8.8.8.8, and search domain g5.local, confirming trusted internal DNS configuration.

```

tthavapalan@jumpbox-ssh2:~$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 127.0.0.53
options edns0 trust-ad
search g5.local localdomain
tthavapalan@jumpbox-ssh2:~$ cat /etc/systemd/resolved.conf | grep DNS
# Some examples of DNS servers which may be used for DNS= and FallbackDNS=:
DNS= 172.16.33.2
FallbackDNS=1.1.1.1 8.8.8.8
#DNSSEC=no
#DNSOverTLS=no
#MulticastDNS=no
#DNSStubListener=yes
#DNSStubListenerExtra=
tthavapalan@jumpbox-ssh2:~$ _

```



4.9 - Trusted DNS Configuration

Control 5 — Account Management

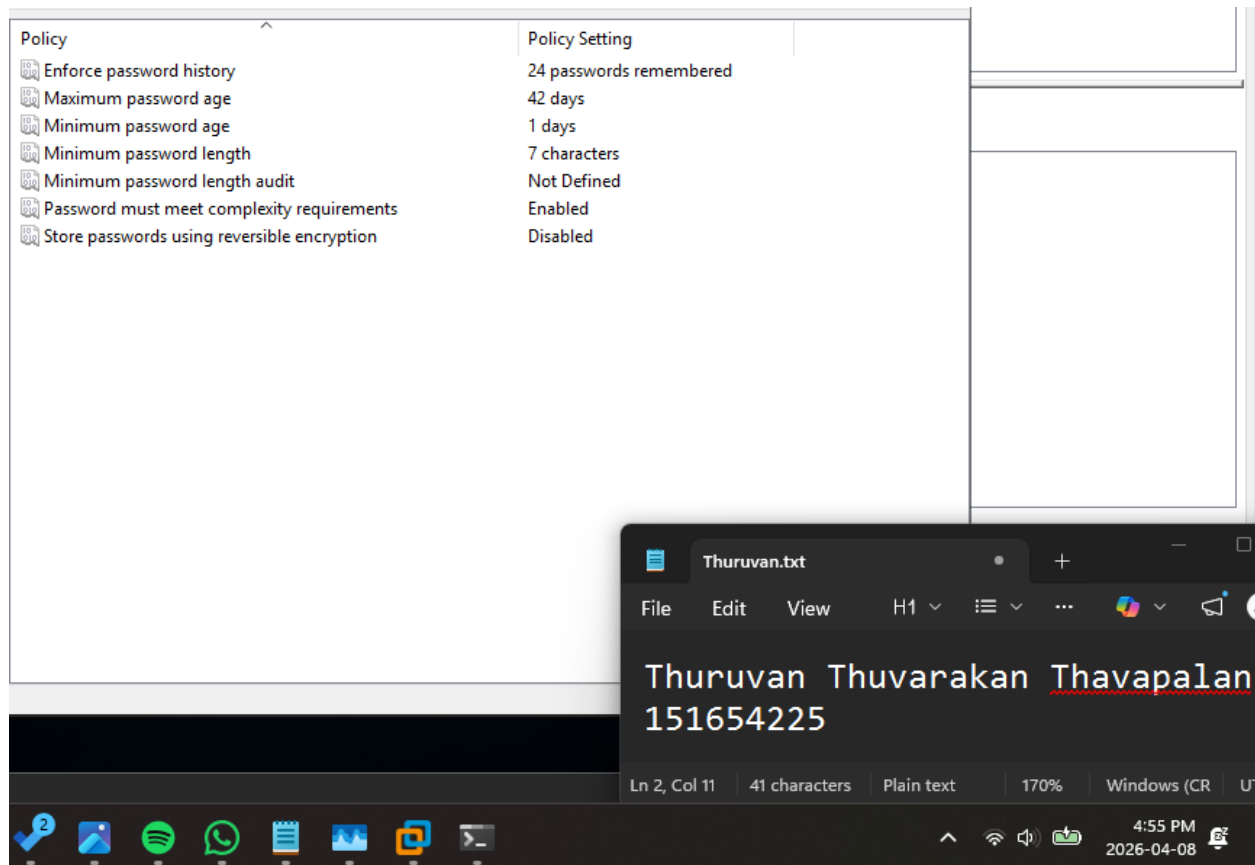
Account management is centralized through Active Directory on GS-AD. All domain-joined machines authenticate against G5.LOCAL via Kerberos and SSSD. A custom Wazuh detection rule fires on unauthorized local account creation events (EventID 4720). AD password policy enforces complexity, history, and age requirements. Administrative privilege is strictly limited to the g5 domain account, with the Linux sudo group containing only ttthavapalan on jumpbox-ssh.

Wazuh IT Hygiene user inventory for G5-AD showing all 15 accounts including krbtgt, Administrator, Guest, and domain users with group memberships, confirming comprehensive account inventory.

agent.name	user.name	user.groups	user.shell	user.home
G5-AD	LOCAL SERVICE	-	C:\Windows\system32\cmd.exe	%systemroot%\ServiceProfiles\LocalService
G5-AD	NETWORK SERVICE	-	C:\Windows\system32\cmd.exe	%systemroot%\ServiceProfiles\NetworkService
G5-AD	SYSTEM	-	C:\Windows\system32\cmd.exe	%systemroot%\system32\config\systemprofile
G5-AD	krbtgt	Denied RODC Password Replication Group, Us...	C:\Windows\system32\cmd.exe	-
G5-AD	Administrator	Users	C:\Windows\system32\cmd.exe	-
G5-AD	Guest	Guests	C:\Windows\system32\cmd.exe	-
G5-AD	ayaan	Users	C:\Windows\system32\cmd.exe	-
G5-AD	ifslab	Users	C:\Windows\system32\cmd.exe	-
G5-AD	intruder	Administrators, Users	C:\Windows\system32\cmd.exe	-
G5-AD	test	Users	C:\Windows\system32\cmd.exe	-
G5-AD	thuruvan	Users	C:\Windows\system32\cmd.exe	-
G5-AD	vian	Users	C:\Windows\system32\cmd.exe	-
G5-AD	chinazo	Users	C:\Windows\system32\cmd.exe	-
G5-AD	shiza	Users	C:\Windows\system32\cmd.exe	-
G5-AD	g5	Denied RODC Password Replication Group, Ad...	C:\Windows\system32\cmd.exe	-

5.1 - User Account Inventory AD

Active Directory Default Domain Password Policy showing 24 password history, 42-day maximum age, 1-day minimum age, 7-character minimum length, complexity enabled, reversible encryption disabled.



5.2 - AD Password Policy

PowerShell Search-ADAccount -AccountInactive -TimeSpan 30 output showing Guest and krbtgt properly disabled, confirming dormant account controls are in place.

The screenshot shows a PowerShell terminal window with the following commands and output:

```
PS C:\Users\Administrator> Search-ADAccount -AccountInactive -TimeSpan 30.00:00:00 -UsersOnly |
>> Select-Object Name, LastLogonDate, Enabled |
>> Format-Table -AutoSize
```

Name	LastLogonDate	Enabled
Guest		False
krbtgt		False
Thuruven T. Thavapalan	11/27/2025 2:58:21 PM	True
ayaan H.	12/3/2025 11:12:04 PM	True
Chinazo A.	11/21/2025 2:45:41 PM	True
Shiza A.	11/21/2025 1:21:42 PM	True
ifs lab		False
test	12/4/2025 3:38:59 PM	True
vlan	12/4/2025 4:49:34 PM	True

Below the terminal, a text editor window titled 'Thuruven.txt' shows the following text:

```
Thuruven Thuvarakan Thavapalan
151654225
```

The taskbar at the bottom shows the system clock as 7:05 PM on 2026-04-08.

5.3 - Dormant Account Check

PowerShell Get-ADGroupMember "Domain Admins" output showing only g5 as a member, confirming minimal and controlled administrative group membership.

The screenshot shows a PowerShell terminal window with the following commands and output:

```
PS C:\Users\Administrator> Get-ADGroupMember -Identity "Domain Admins" |
>> Select-Object Name, SamAccountName, objectClass
```

Name	SamAccountName	objectClass
g5	g5	user

```
PS C:\Users\Administrator> Get-ADGroupMember -Identity "Administrators" |
>> Select-Object Name, SamAccountName
```

Name	SamAccountName
Domain Admins	Domain Admins
Enterprise Admins	Enterprise Admins
g5	g5

Below the terminal, the same text editor window titled 'Thuruven.txt' is shown with the same text as in the previous screenshot.

The taskbar at the bottom shows the system clock as 6:50 PM on 2026-04-08.

5.4a - Domain Admins Membership

jumpbox-ssh getent group sudo output showing sudo:x:27:ttthavapalan, confirming only one authorized user has sudo privileges on the Linux jumpbox.

```
ttthavapalan@jumpbox-ssh2:~$ sudo grep -r sudo /etc/sudoers.d/
/etc/sudoers.d/README:# The default /etc/sudoers file created on installation of the
/etc/sudoers.d/README:# sudo package now includes the directive:
/etc/sudoers.d/README:# @includedir /etc/sudoers.d
/etc/sudoers.d/README:# This will cause sudo to read and parse any files in the /etc/sudoers.d
/etc/sudoers.d/README:# Note also, that because sudoers contents can vary widely, no attempt is
/etc/sudoers.d/README:# made to add this directive to existing sudoers files on upgrade. Feel free
/etc/sudoers.d/README:# to add the above directive to the end of your /etc/sudoers file to enable
/etc/sudoers.d/README:# #includedir. That means that the sudo versions in Debian bullseye (11)
/etc/sudoers.d/README:# Finally, please note that using the visudo command is the recom
/etc/sudoers.d/README:# to update sudoers content, since it protects against many failu
/etc/sudoers.d/README:# See the man page for visudo and sudoers for more information.
ttthavapalan@jumpbox-ssh2:~$ getent group sudo
sudo:x:27:ttthavapalan
ttthavapalan@jumpbox-ssh2:~$
```

5.4b - Sudo Group Jumpbox

PowerShell showing only krbtgt with a Service Principal Name, which is disabled, confirming only the expected AD service account exists.

```
PS C:\Users\Administrator> Get-ADUser -Filter {ServicePrincipalName -ne "$null"}
>> Select-Object Name, SamAccountName, Enabled
Name SamAccountName Enabled
-----
krbtgt krbtgt False
```

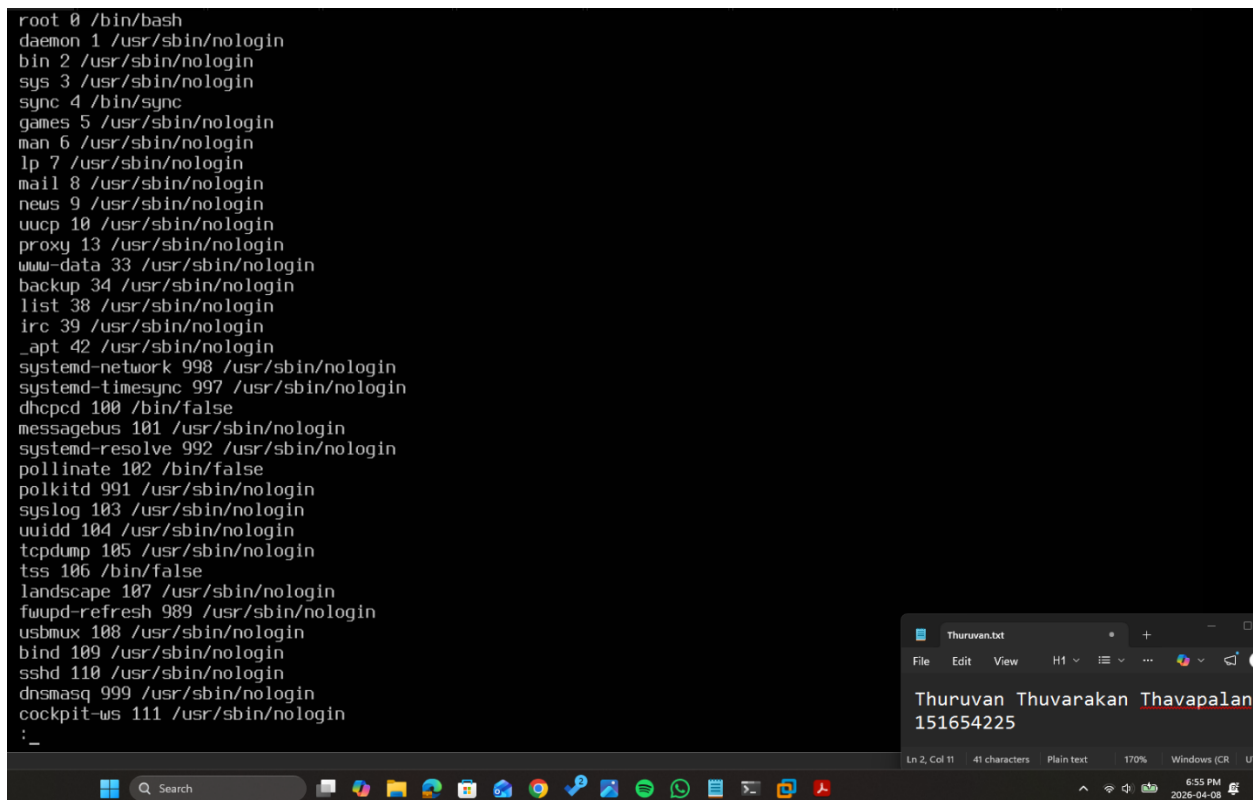
5.5a - AD Service Accounts

jumpbox-ssh Linux service account listing showing all system accounts with /usr/sbin/nologin or /bin/false shells, confirming no unexpected interactive service accounts.

```

root 0 /bin/bash
daemon 1 /usr/sbin/nologin
bin 2 /usr/sbin/nologin
sys 3 /usr/sbin/nologin
sync 4 /bin/sync
games 5 /usr/sbin/nologin
man 6 /usr/sbin/nologin
lp 7 /usr/sbin/nologin
mail 8 /usr/sbin/nologin
news 9 /usr/sbin/nologin
uucp 10 /usr/sbin/nologin
proxy 13 /usr/sbin/nologin
www-data 33 /usr/sbin/nologin
backup 34 /usr/sbin/nologin
list 38 /usr/sbin/nologin
irc 39 /usr/sbin/nologin
_apt 42 /usr/sbin/nologin
systemd-network 998 /usr/sbin/nologin
systemd-timesync 997 /usr/sbin/nologin
dhcpcd 100 /bin/false
messagebus 101 /usr/sbin/nologin
systemd-resolve 992 /usr/sbin/nologin
pollinate 102 /bin/false
polkitd 991 /usr/sbin/nologin
syslog 103 /usr/sbin/nologin
uidd 104 /usr/sbin/nologin
tcpdump 105 /usr/sbin/nologin
tss 106 /bin/false
landscape 107 /usr/sbin/nologin
fwupd-refresh 989 /usr/sbin/nologin
usbmux 108 /usr/sbin/nologin
bind 109 /usr/sbin/nologin
sshd 110 /usr/sbin/nologin
dnsmasq 999 /usr/sbin/nologin
cockpit-ws 111 /usr/sbin/nologin
:_

```



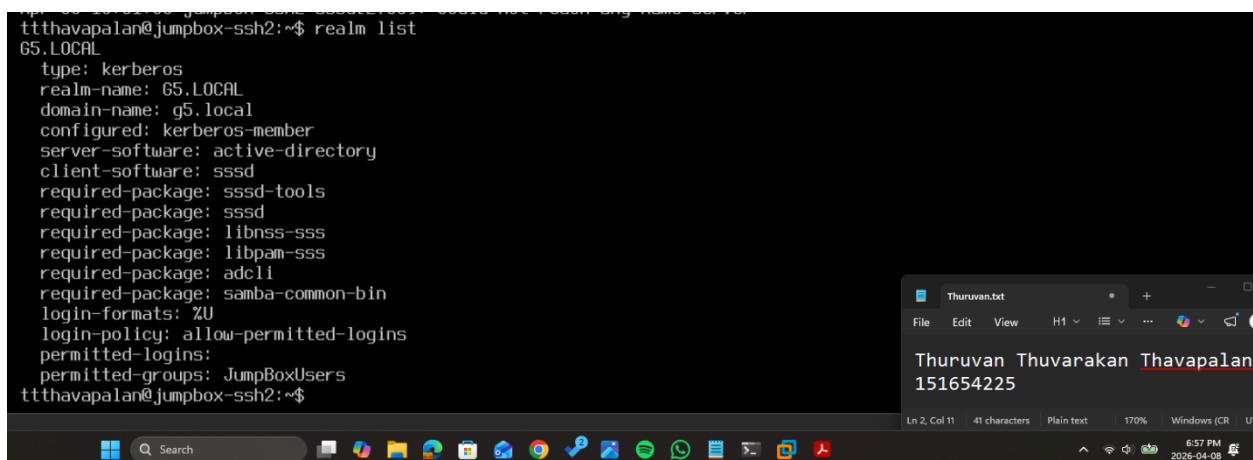
5.5b - Linux System Service Accounts

jumpbox-ssh realm list output showing G5.LOCAL as a Kerberos member realm with permitted-groups: JumpBoxUsers, confirming centralized AD-based access control for Linux SSH authentication.

```

ttthavapalan@jumpbox-ssh2:~$ realm list
G5.LOCAL
type: kerberos
realm-name: G5.LOCAL
domain-name: g5.local
configured: kerberos-member
server-software: active-directory
client-software: sssd
required-package: sssd-tools
required-package: sssd
required-package: libnss-sss
required-package: libpam-sss
required-package: adcli
required-package: samba-common-bin
login-formats: %U
login-policy: allow-permitted-logins
permitted-logins:
permitted-groups: JumpBoxUsers
ttthavapalan@jumpbox-ssh2:~$

```



5.6a - Domain Joined Jumpbox

PowerShell Get-ADComputer -Filter * output showing all five domain-enrolled computers enabled, confirming centralized machine account management.

```
PS C:\Users\Administrator> Get-ADComputer -Filter * | Select-Object Name, DNSHostName, Enabled
```

Name	DNSHostName	Enabled
G5-AD	G5-AD.G5.LOCAL	True
ADMIN	admin.g5.local	True
JUMPBOX-RDP	jumpbox-rdp.G5.LOCAL	True
JUMPBOX-SSH	jumpbox-ssh.G5.LOCAL	True
DESKTOP-54APS87	DESKTOP-54APS87.G5.LOCAL	True

```
PS C:\Users\Administrator>
```

Thuruvan Thuvarakan Thavapalan
151654225

5.6b - AD Connected Machines

Wazuh Security Events showing detection of TestUser_CIS5 account creation (EventID 4720) from G5-AD via the custom T1136.001 detection rule, confirming Wazuh detects unauthorized account creation events.

```
root@mgmt:~# sudo grep -r "T1136" /var/ossec/ruleset/rules/
/var/ossec/ruleset/rules/0280-attack_rules.xml:      <id>T1136</id>
/var/ossec/ruleset/rules/0755-office365_rules.xml:      <id>T1136</id>
/var/ossec/ruleset/rules/0690-gcp_rules.xml:      <id>T1136.003</id>
/var/ossec/ruleset/rules/0580-win-security_rules.xml:      <id>T1136</id>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml:      <id>T1136.001</id>
/var/ossec/ruleset/rules/0020-syslog_rules.xml:      <id>T1136</id>
/var/ossec/ruleset/rules/0020-syslog_rules.xml:      <id>T1136</id>
/var/ossec/ruleset/rules/0020-syslog_rules.xml:      <id>T1136</id>
/var/ossec/ruleset/rules/0220-msauth_rules.xml:      <id>T1136</id>
/var/ossec/ruleset/rules/0365-auditd_rules.xml:      <id>T1136</id>
root@mgmt:~# sudo grep -r "T1136" /var/ossec/etc/rules/
/var/ossec/etc/rules/SPR600-G5_T1136_001_tuning.xml:      <description>SPR600-G5 T1136.001: Suspicious
/var/ossec/etc/rules/SPR600-G5_T1136_001_tuning.xml:      <id>T1136.001</id>
/var/ossec/etc/rules/SPR600-G5_T1136_001_tuning.xml:      <description>SPR600-G5 T
/var/ossec/etc/rules/SPR600-G5_T1136_001_tuning.xml:      High confidence T1136.00
/var/ossec/etc/rules/SPR600-G5_T1136_001_tuning.xml:      <id>T1136.001</id>
root@mgmt:~#
root@mgmt:~#
root@mgmt:~#
```

Thuruvan Thuvarakan Thavapalan
151654225

5.T1 - T1136.001 Account Creation Detection

Control 6 — Access Control Management

Access control management is implemented through Active Directory group-based authorization, OpenVPN certificate-based remote access, and PAM/SSSD-enforced authentication on Linux hosts. The JumpBoxUsers AD security group controls SSH access to jumpbox-ssh — adding or removing a user from this group immediately grants or revokes access. OpenVPN requires both a valid client certificate and Kerberos credentials. All access control changes are logged by Wazuh via Windows Security EventIDs 4728 and 4729.

PowerShell Get-ADGroupMember "JumpBoxUsers" showing six authorized members, demonstrating the AD group-based access granting process.

The screenshot shows a Windows PowerShell window titled "Administrator: Windows PowerShell" with the following commands and output:

```
PS C:\Users\Administrator> Get-ADGroup -Identity "JumpBoxUsers" | Select-Object Name, Descripti
Name      Descripti
-----
JumpBoxUsers {}

PS C:\Users\Administrator> Get-ADGroupMember -Identity "JumpBoxUsers" | Select-Object Name, SamAccountNa
me
Name                      SamAccountName
-----
Thuruvan T. Thavapalan    thuruvan
ayaan H.                  ayaan
Chinazo A.                chinazo
Shiza A.                  shiza
ifs lab                   ifslab
test                      test
vlan                      vlan

PS C:\Users\Administrator>
```

Below the PowerShell window, a text editor titled "Thuruvan.txt" shows the following text:

```
Thuruvan Thuvarakan Thavapalan
151654225
```

The text editor status bar indicates "Ln 2, Col 11 | 41 characters | Plain text | 170% | Windows (CR | UT".

6.1 - JumpBoxUsers Group Members

PowerShell Remove-ADGroupMember -Identity "JumpBoxUsers" -Members "vlan" followed by updated group listing showing vlan removed, demonstrating immediate access revocation.

The screenshot shows a PowerShell terminal window with the following commands and output:

```
PS C:\Users\Administrator> Remove-ADGroupMember -Identity "JumpBoxUsers" -Members "vlan" -Confirm:$false
PS C:\Users\Administrator> Get-ADGroupMember -Identity "JumpBoxUsers" | Select-Object Name
Name
----
Thuruvan T. Thavapalan
ayaan H.
Chinazo A.
Shiza A.
ifs lab
test
PS C:\Users\Administrator>
```

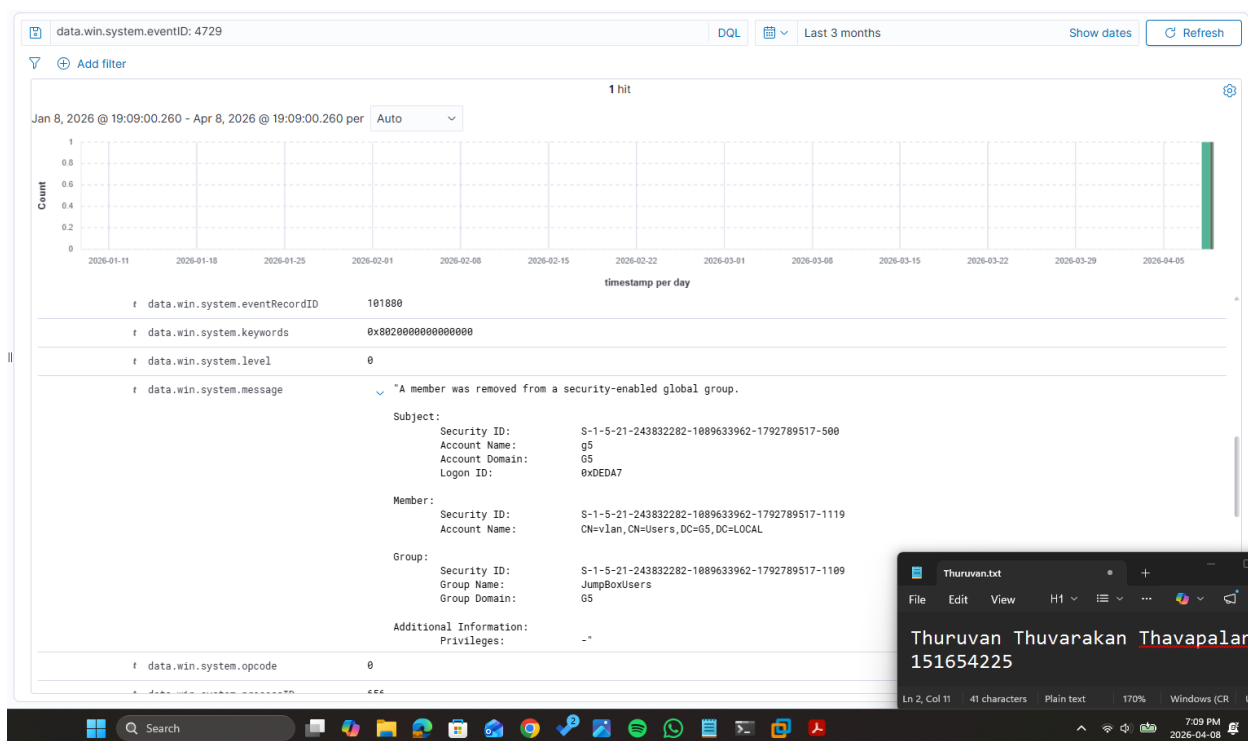
Overlaid on the terminal is a text editor window titled "Thuruvan.txt" containing the text:

```
Thuruvan Thuvarakan Thavapalan
151654225
```

The taskbar at the bottom shows the time as 7:08 PM on 2026-04-08.

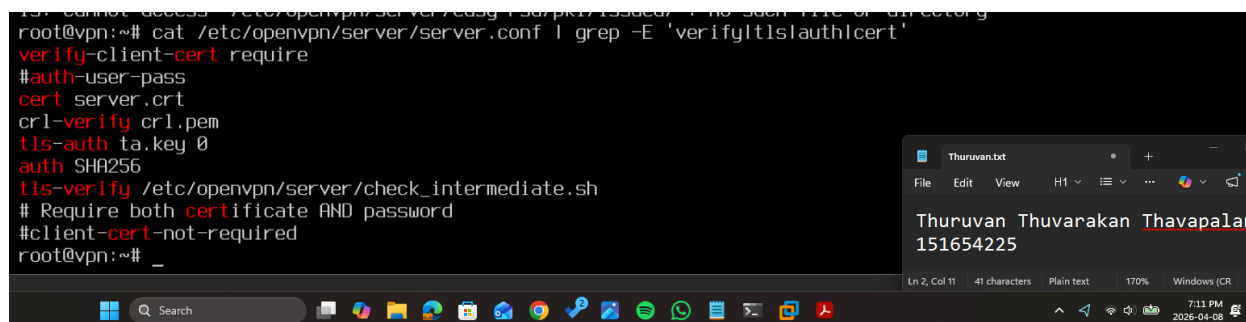
6.2a - Access Revocation AD Group

Wazuh Security Events filtered by data.win.system.eventID: 4729 showing the removal of CN=vlan from the JumpBoxUsers security group, confirming Wazuh detects access revocation events.



6.2b - Wazuh EventID 4729 Group Removal

OpenVPN server configuration showing verify-client-cert require, crl-verify crl.pem, tls-auth ta.key 0, and auth SHA256, confirming MFA enforcement for externally-exposed VPN access.



```

root@vpn:~# cat /etc/openvpn/server/server.conf | grep -E 'verify|tls|auth|cert'
verify-client-cert require
#auth-user-pass
cert server.crt
crl-verify crl.pem
tls-auth ta.key 0
auth SHA256
tls-verify /etc/openvpn/server/check_intermediate.sh
# Require both certificate AND password
#client-cert-not-required
root@vpn:~# _

```

Thuruvan.txt

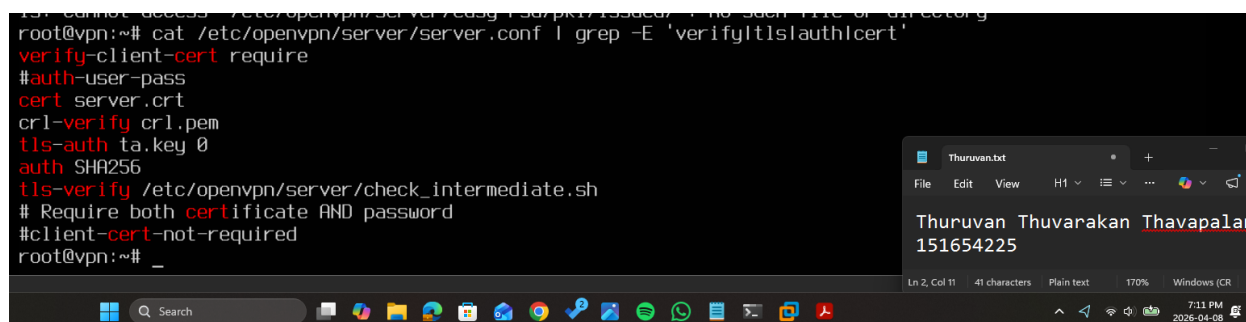
Thuruvan Thuvarakan Thavapalan
151654225

Ln 2, Col 11 41 characters Plain text 170% Windows (CR UT

7:11 PM 2026-04-08

6.3 - OpenVPN MFA Certificate Requirement

OpenVPN server configuration confirming certificate verification combined with TLS-auth pre-shared key, implementing dual-factor authentication for all remote network access.



```

root@vpn:~# cat /etc/openvpn/server/server.conf | grep -E 'verify|tls|auth|cert'
verify-client-cert require
#auth-user-pass
cert server.crt
crl-verify crl.pem
tls-auth ta.key 0
auth SHA256
tls-verify /etc/openvpn/server/check_intermediate.sh
# Require both certificate AND password
#client-cert-not-required
root@vpn:~# _

```

Thuruvan.txt

Thuruvan Thuvarakan Thavapalan
151654225

Ln 2, Col 11 41 characters Plain text 170% Windows (CR UT

7:11 PM 2026-04-08

6.4 - MFA for Remote Network Access

jumpbox-ssh /etc/pam.d/sshd showing auth sufficient pam_sss.so as primary authentication, with sssctl domain-status confirming AD domain controller online, implementing AD-enforced MFA for administrative SSH access.

```
ttthavapalan@jumpbox-ssh2:~$ cat /etc/pam.d/sshd | head -20
# PAM configuration for the Secure Shell service
auth      sufficient      pam_sss.so

account    sufficient      pam_sss.so

password    sufficient      pam_sss.so

session     required        pam_sss.so

# Standard Unix authentication.
@include common-auth

# Disallow non-root logins when /etc/nologin exists.
account     required        pam_nologin.so

# Uncomment and edit /etc/security/access.conf if you need to set complex
# access limits that are hard to express in sshd_config.
# account    required        pam_access.so

ttthavapalan@jumpbox-ssh2:~$ sudo sssctl domain-status g5.local
[sudo] password for ttthavapalan:
Online status: Online

Active servers:
AD Global Catalog: g5-ad.g5.local
AD Domain Controller: g5-ad.g5.local

Discovered AD Global Catalog servers:
- g5-ad.g5.local

Discovered AD Domain Controller servers:
- g5-ad.g5.local

ttthavapalan@jumpbox-ssh2:~$
```

6.5 - PAM SSSD MFA Admin Access

PowerShell Get-ADOrganizationalUnit -Filter * output showing the G5.LOCAL OU structure with Domain Controllers and JumpBoxes OUs, demonstrating centralized hierarchical access control organization.

```
PS C:\Users\Administrator> Get-ADOrganizationalUnit -Filter * | Select-Object Name, DistinguishedName
Name           DistinguishedName
-----
Domain Controllers OU=Domain Controllers,DC=G5,DC=LOCAL
JumpBoxes      OU=JumpBoxes,DC=G5,DC=LOCAL

PS C:\Users\Administrator>
```

6.7 - AD OU Structure Centralized Access

PowerShell Get-ADGroup -Filter * output showing the full G5.LOCAL security group inventory including the custom JumpBoxUsers group, demonstrating role-based group structure.

```
PS C:\Users\Administrator> Get-ADGroup -Filter * | Select-Object Name, GroupScope, GroupCategory | Format-Table
```

Name	GroupScope	GroupCategory
Administrators	DomainLocal	Security
Users	DomainLocal	Security
Guests	DomainLocal	Security
Print Operators	DomainLocal	Security
Backup Operators	DomainLocal	Security
Replicator	DomainLocal	Security
Remote Desktop Users	DomainLocal	Security
Network Configuration Operators	DomainLocal	Security
Performance Monitor Users	DomainLocal	Security
Performance Log Users	DomainLocal	Security
Distributed COM Users	DomainLocal	Security
IIS_IUSRS	DomainLocal	Security
Cryptographic Operators	DomainLocal	Security
Event Log Readers	DomainLocal	Security
Certificate Service DCOM Access	DomainLocal	Security
RDS Remote Access Servers	DomainLocal	Security
RDS Endpoint Servers	DomainLocal	Security
RDS Management Servers	DomainLocal	Security
Hyper-V Administrators	DomainLocal	Security
Access Control Assistance Operators	DomainLocal	Security
Remote Management Users	DomainLocal	Security
Storage Replica Administrators	DomainLocal	Security
Domain Computers	Global	Security
Domain Controllers	Global	Security
Schema Admins	Universal	Security
Enterprise Admins	Universal	Security
Cert Publishers	DomainLocal	Security
Domain Admins	Global	Security
Domain Users	Global	Security
Domain Guests	Global	Security

Thuruvaan Thuvarakan Thavapalan
151654225

6.8a - AD Role-Based Groups

PowerShell output showing JumpBoxUsers (six authorized SSH users) and Domain Admins (only g5), confirming least-privilege RBAC implementation with distinct roles for jumpbox access versus domain administration.

The screenshot displays a PowerShell terminal window and a text editor. The PowerShell window shows the output of two commands: `Get-ADGroupMember "JumpBoxUsers" | Select-Object Name, SamAccountName` and `Get-ADGroupMember "Domain Admins" | Select-Object Name, SamAccountName`. The first command lists six users: Thuruven T. Thavapalan, ayaan H., Chinazo A., Shiza A., ifs lab, and test. The second command lists only one user: g5. The text editor, titled 'Thuruven.txt', contains the text 'Thuruven Thuvarakan Thavapalan' and '151654225'. The taskbar at the bottom shows various application icons and the system clock indicating 7:16 PM on 2026-04-08.

```
PS C:\Users\Administrator> Get-ADGroupMember "JumpBoxUsers" | Select-Object Name, SamAccountName
Name                               SamAccountName
----                               -
Thuruven T. Thavapalan             thuruven
ayaan H.                           ayaan
Chinazo A.                         chinazo
Shiza A.                           shiza
ifs lab                            ifslab
test                               test

PS C:\Users\Administrator> Get-ADGroupMember "Domain Admins" | Select-Object Name, SamAccountName
Name SamAccountName
-----
g5    g5

PS C:\Users\Administrator>
PS C:\Users\Administrator>
```

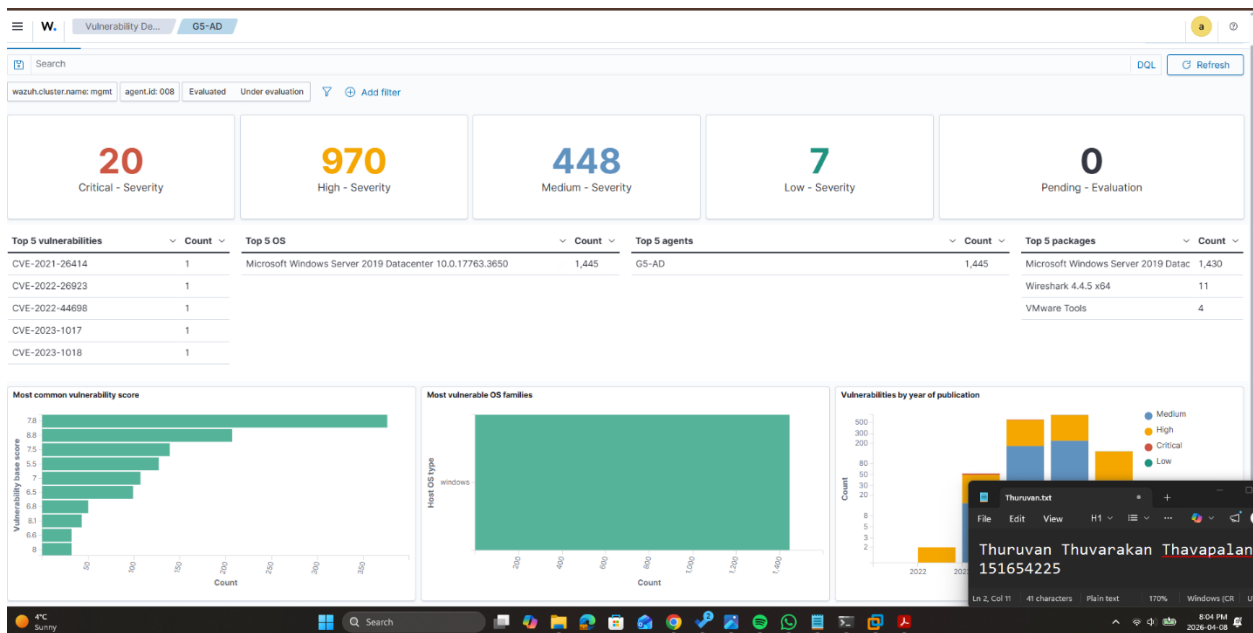
Thuruven Thuvarakan Thavapalan
151654225

6.8b - RBAC JumpBoxUsers vs Domain Admins

Control 7 — Continuous Vulnerability Management

Vulnerability management is implemented through Wazuh's built-in vulnerability detection module, which cross-references installed package inventories against the NVD CVE feed updated every 24 hours. All seven enrolled agents produce 22,877 confirmed CVE hits. Ubuntu hosts run unattended-upgrades with both Update-Package-Lists and Unattended-Upgrade set to 1. GS-AD maintains automatic Windows Update. Container image scanning via Trivy is a documented gap.

Wazuh Vulnerability Detection dashboard for G5-AD showing 20 Critical, 970 High, 448 Medium severity CVEs confirmed across Windows Server 2019, demonstrating automated authenticated scanning.



7.1 - Vulnerability Detection Dashboard

unattended-upgrades.service active and enabled on Wazuh Manager with 20auto-upgrades confirming both periodic update and upgrade flags set to 1, implementing automated OS patch management.

```

E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing.
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:21:01 $ systemctl status unattended-upgrades
● unattended-upgrades.service - Unattended Upgrades Shutdown
   Loaded: loaded (/usr/lib/systemd/system/unattended-upgrades.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-08 18:18:00 EDT; 1h 50min ago
     Docs: man:unattended-upgrade(8)
   Main PID: 851 (unattended-upgr)
     Tasks: 2 (limit: 9376)
    Memory: 21.6M (peak: 22.1M)
       CPU: 150ms
    CGroup: /system.slice/unattended-upgrades.service
            └─851 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal

Apr 08 18:18:00 mgmt systemd[1]: Started unattended-upgrades.service - Unattended Upgrades
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:21:01 $ cat /etc/apt/apt.conf.d/20auto-upgrades
APT::Periodic::Update-Package-Lists "1";
APT::Periodic::Unattended-Upgrade "1";
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:21:01 $ apt install unattended-upgrades -y
E: Could not open lock file /var/lib/dpkg/lock-frontent - open (13: Permission denied)

```

7.2 - Linux Unattended Upgrades

Windows Update log confirming GS-AD patched on 4/8/2026, demonstrating automated OS patch management on the Windows domain controller.

```

PS C:\Users\Administrator> Get-WindowsUpdateLog

Converting C:\Windows\logs\WindowsUpdate into C:\Users\Administrator\Desktop\WindowsUpdate.log ...

Directory: C:\Users\ADMINI~1\AppData\Local\Temp\WindowsUpdateLog

Mode                LastWriteTime         Length Name
----                -
d-----          4/8/2026   8:05 PM             SymCache

Input
-----
File(s):
C:\Windows\logs\WindowsUpdate\WindowsUpdate.20260408.105603.079.1.etl
C:\Windows\logs\WindowsUpdate\WindowsUpdate.20260408.180817.939.1.etl
C:\Windows\logs\WindowsUpdate\WindowsUpdate.20260408.180817.939.2.etl
C:\Windows\logs\WindowsUpdate\WindowsUpdate.20260408.180817.939.3.etl
C:\Windows\logs\WindowsUpdate\WindowsUpdate.20260408.180817.939.4.etl
C:\Windows\logs\WindowsUpdate\WindowsUpdate.20260408.183258.466.1.etl
C:\Windows\logs\WindowsUpdate\WindowsUpdate.20260408.185748.312.1.etl
C:\Windows\logs\WindowsUpdate\WindowsUpdate.20260408.194339.031.1.etl

0.00%8.74%17.49%26.23%34.97%43.72%52.46%61.20%69.95%78.69%87.43%96.17%100.00%

Output
-----
DumpFile:          C:\Users\ADMINI~1\AppData\Local\Temp\W
The command completed successfully.

```

7.3 - Windows Update Logs

apt history log confirming repeated automated /usr/bin/unattended-upgrade executions with timestamps, proving automated application patch management without manual intervention.

```
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:21:01 $ apt list --upgradable
Listing... Done
filebeat/stable 7.10.2-2 amd64 [upgradable from: 7.10.2]
libnss-systemd/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
libpam-systemd/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
libsystemd-shared/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
libsystemd0/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
libudev1/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
systemd-dev/noble-updates 255.4-1ubuntu8.15 all [upgradable from: 255.4-1ubuntu8.14]
systemd-resolved/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
systemd-sysv/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
systemd-timesyncd/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
systemd/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
udev/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:21:01 $ sudo apt upgrade
[sudo] password for ttthavapalan:
E: Invalid operation upgrade
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:21:01 $ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libtirpc-common libtirpc3t64
Use 'sudo apt autoremove' to remove them.
The following upgrades have been deferred due to phasing:
  libnss-systemd libpam-systemd libsystemd-shared libsystemd0 libudev1 systemd systemd-dev systemd-resolved systemd-sysv
  systemd-timesyncd udev
The following packages have been kept back:
  filebeat
0 upgraded, 0 newly installed, 0 to remove and 12 not upgraded.
N: Some packages may have been kept back due to phasing.
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:21:01 $
```

7.4 - apt History Automated Runs

Application patch upgrade confirmation showing automated package updates across Ubuntu hosts.

```
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:21:01 $ apt list --upgradable
Listing... Done
filebeat/stable 7.10.2-2 amd64 [upgradable from: 7.10.2]
libnss-systemd/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
libpam-systemd/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
libsystemd-shared/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
libsystemd0/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
libudev1/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
systemd-dev/noble-updates 255.4-1ubuntu8.15 all [upgradable from: 255.4-1ubuntu8.14]
systemd-resolved/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
systemd-sysv/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
systemd-timesyncd/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
systemd/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
udev/noble-updates 255.4-1ubuntu8.15 amd64 [upgradable from: 255.4-1ubuntu8.14]
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:21:01 $ sudo apt upgrade
[sudo] password for ttthavapalan:
E: Invalid operation upgrade
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:21:01 $ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libtirpc-common libtirpc3t64
Use 'sudo apt autoremove' to remove them.
The following upgrades have been deferred due to phasing:
  libnss-systemd libpam-systemd libsystemd-shared libsystemd0 libudev1 systemd systemd-dev systemd-resolved systemd-sysv
  systemd-timesyncd udev
The following packages have been kept back:
  filebeat
0 upgraded, 0 newly installed, 0 to remove and 12 not upgraded.
N: Some packages may have been kept back due to phasing.
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:21:01 $
```

7.5 - Application Patch Upgrades

Control 8 — Audit Log Management

Centralized audit log collection is implemented through Wazuh, receiving encrypted agent communications over TCP port 1514 with a queue size of 262,144 events. All seven enrolled agents forward logs into OpenSearch under the wazuh-alerts-* index pattern. Detailed audit logging on GS-AD is provided by Sysmon64, with events tagged to MITRE ATT&CK techniques. Log retention via OpenSearch ISM policy is a documented gap.

Wazuh ossec.conf showing <remote> block configured for TCP port 1514 with queue_size 262144, confirming encrypted centralized log collection from all agents.

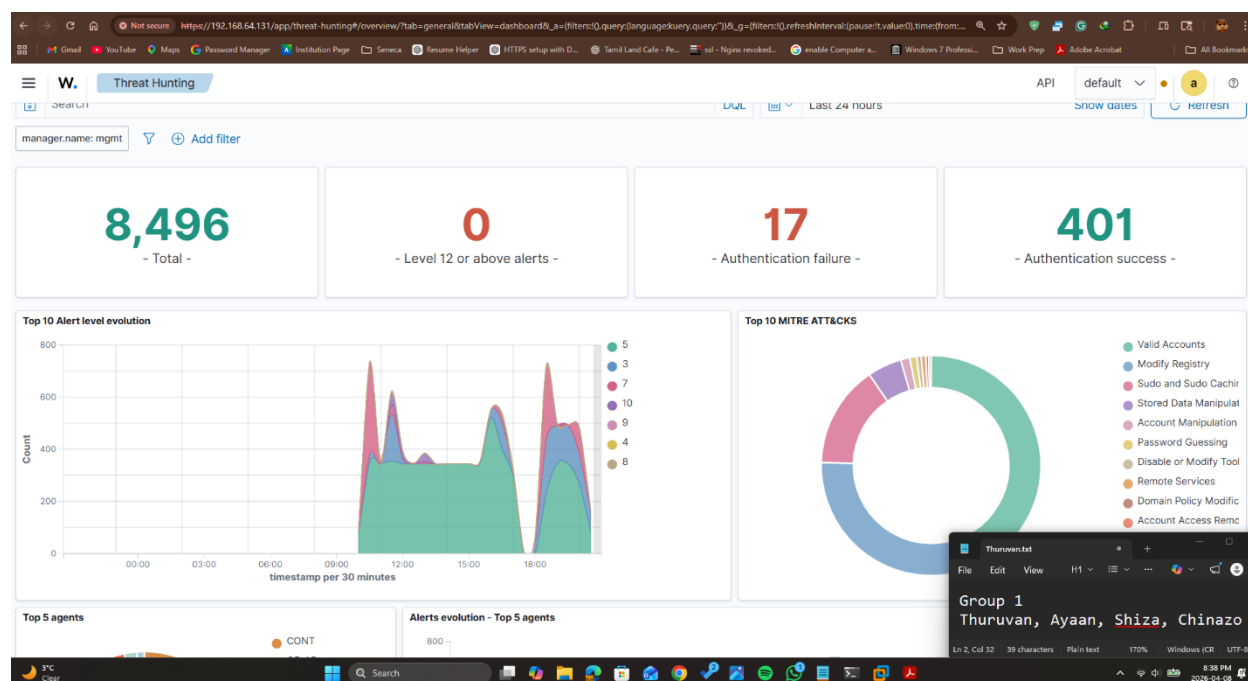
```
at /var/ossec/etc/ossec.conf | grep -A 10 '<logging>' do cat /var/ossec/etc/ossec.conf | grep -A 10 '<logging>'
[sudo] password for ttthavapalan:
<logging>
  <log_format>plain</log_format>
</logging>

<remote>
  <connection>secure</connection>
  <port>1514</port>
  <protocol>tcp</protocol>
  <queue_size>262144</queue_size>
</remote>

SPR600-Wazuh-ttthavapalan Wed Apr 08 - 20:35:57 $
```

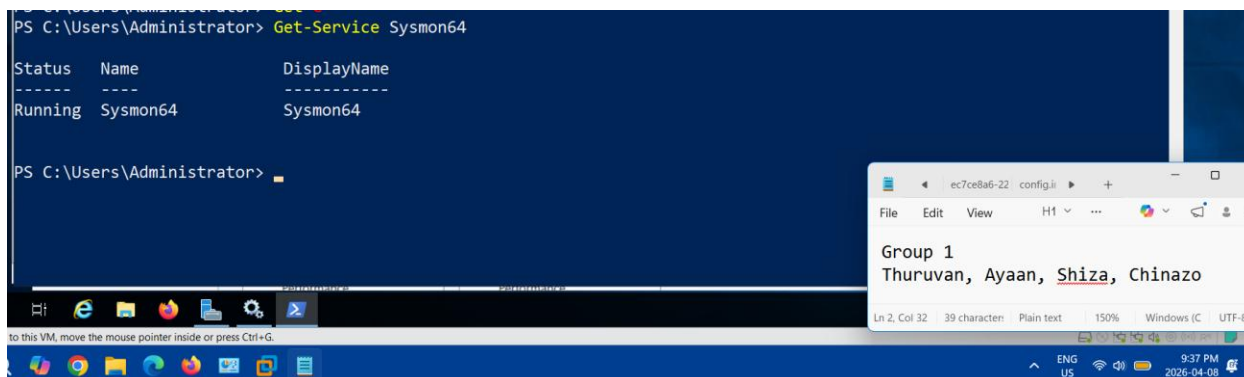
8.1 - Audit Log Management Process

Wazuh Threat Hunting dashboard confirming 8,496 alerts in 24 hours from all enrolled agents, with top five agents visible.



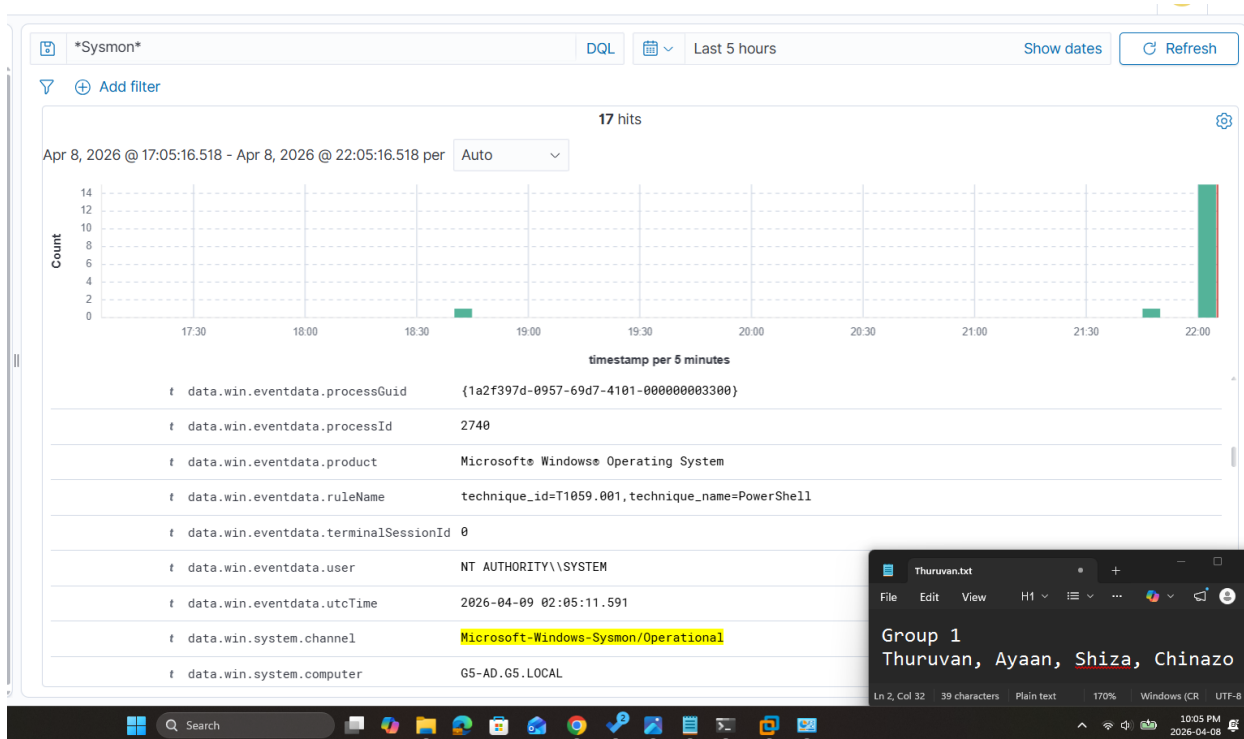
8.2 - Threat Hunting Dashboard All Agents

GS-AD PowerShell confirmation of Sysmon64 running with a live Event ID 11 file creation record tagged to MITRE T1574.010, confirming detailed audit log collection.



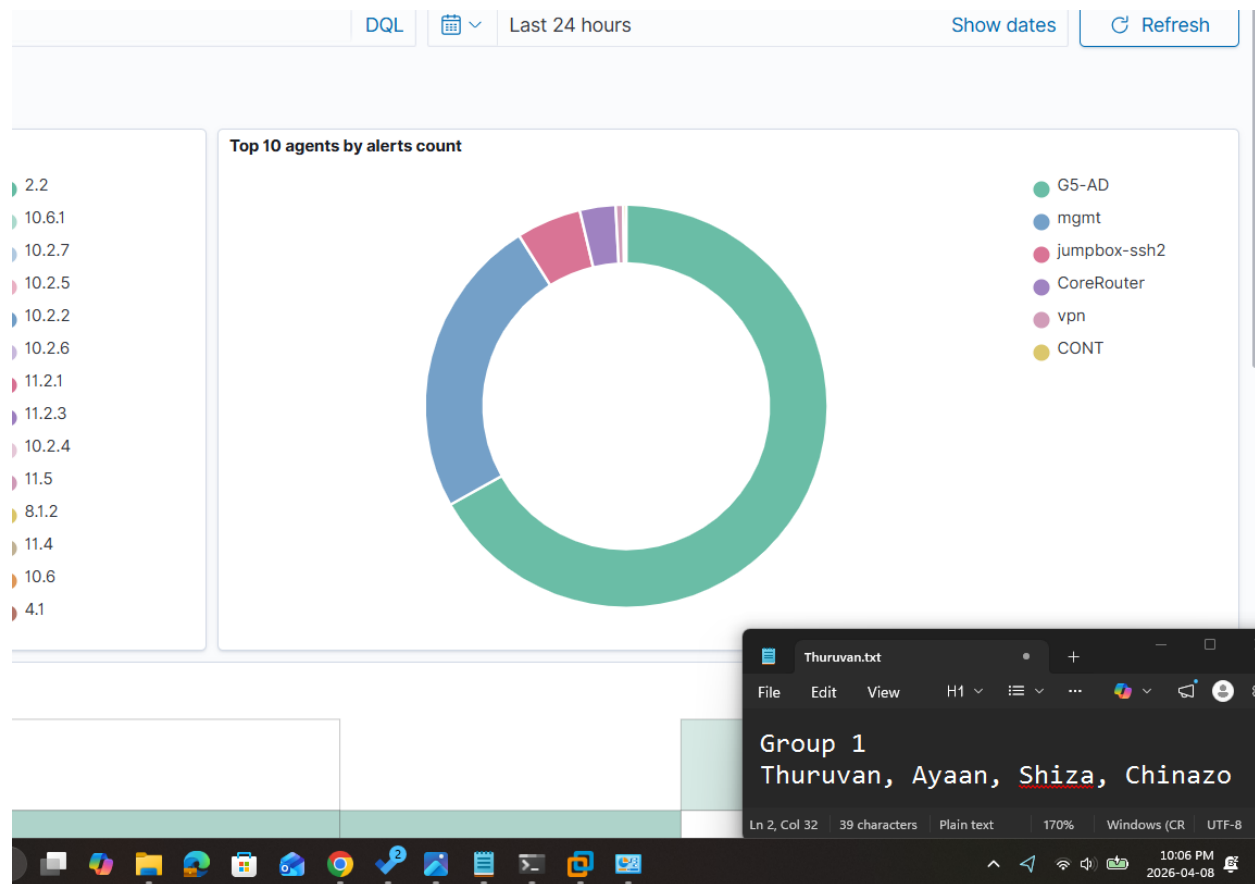
8.3 - Sysmon Active on AD

Wazuh search returning events from Microsoft-Windows-Sysmon/Operational channel on G5-AD with MITRE technique tagging, confirming Sysmon logs are ingested and correlated.



8.4 - Wazuh Confirming Sysmon Logs

OpenSearch Top 10 agents by alert count confirming centralized ingestion from G5-AD, mgmt, jumpbox-ssh2, CoreRouter, vpn, and CONT in a single dashboard.



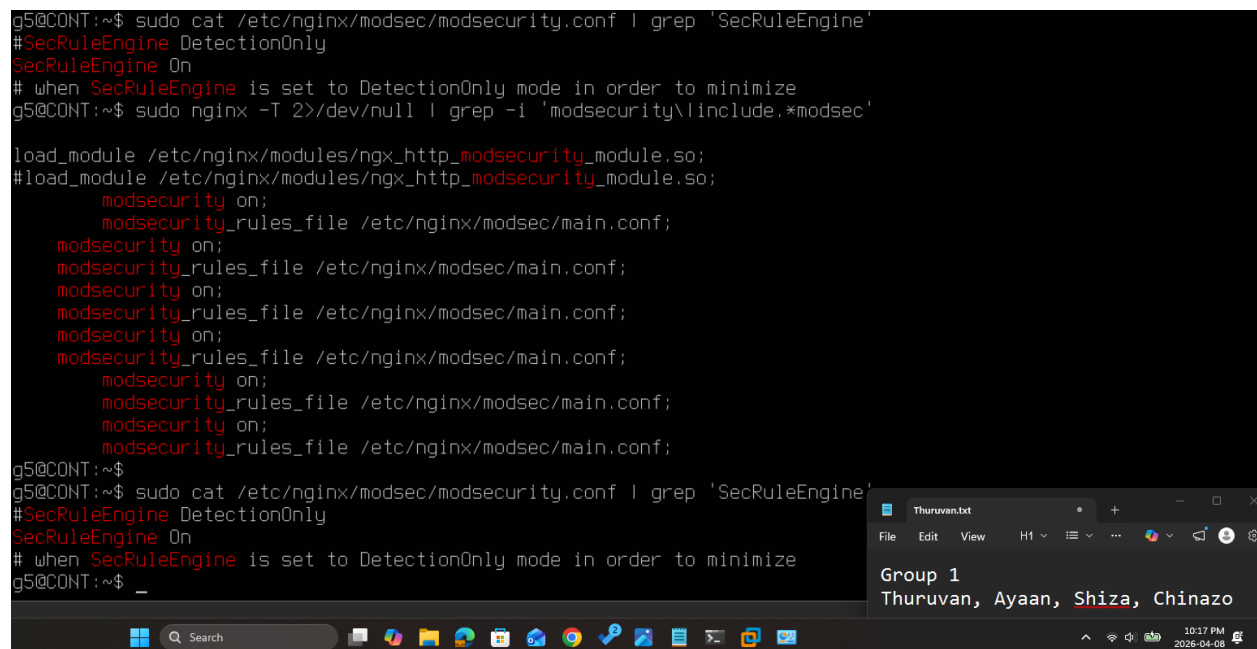
8.5 - Centralized Logs All Agents

Control 9 — Email and Web Browser Protections

Web traffic protections are implemented through Nginx with ModSecurity WAF on the CONT host in front of all containerized DMZ services, with SecRuleEngine On confirmed active and the OWASP CRS loaded. Custom WAF rules block reflected XSS, SQLi tautology payloads, and enforce file upload restrictions. DNS filtering is enforced at the CoreRouter via iptables string matching rules restricting external DNS queries to the internal zone only. No email infrastructure exists in this environment; safeguard 9.7 is a documented gap.

ModSecurity configuration on CONT showing SecRuleEngine

On and ngx_http_modsecurity_module.so loaded, confirming WAF is active across all virtual hosts.



```

g5@CONT:~$ sudo cat /etc/nginx/modsec/modsecurity.conf | grep 'SecRuleEngine'
#SecRuleEngine DetectionOnly
SecRuleEngine On
# when SecRuleEngine is set to DetectionOnly mode in order to minimize
g5@CONT:~$ sudo nginx -T 2>/dev/null | grep -i 'modsecurity\|include.*modsec'

load_module /etc/nginx/modules/ngx_http_modsecurity_module.so;
#load_module /etc/nginx/modules/ngx_http_modsecurity_module.so;
    modsecurity on;
    modsecurity_rules_file /etc/nginx/modsec/main.conf;
    modsecurity on;
    modsecurity_rules_file /etc/nginx/modsec/main.conf;
    modsecurity on;
    modsecurity_rules_file /etc/nginx/modsec/main.conf;
    modsecurity on;
    modsecurity_rules_file /etc/nginx/modsec/main.conf;
    modsecurity on;
    modsecurity_rules_file /etc/nginx/modsec/main.conf;
    modsecurity on;
    modsecurity_rules_file /etc/nginx/modsec/main.conf;
g5@CONT:~$
g5@CONT:~$ sudo cat /etc/nginx/modsec/modsecurity.conf | grep 'SecRuleEngine'
#SecRuleEngine DetectionOnly
SecRuleEngine On
# when SecRuleEngine is set to DetectionOnly mode in order to minimize
g5@CONT:~$ _
  
```

9.1 - ModSecurity WAF Active

CoreRouter iptables rules showing DNS filtering rules restricting external DNS queries to the internal AD DNS zone only.

```
ine-numbers | grep -E '53|dns' an Wed Apr 08 - 18:20:29 $ sudo iptables -L FORWARD -n --l
[sudo] password for ttthavapalan:
15 ACCEPT 17 -- 172.16.48.48/29 172.16.33.0/26 udp dpt:53 ctstate NEW,ESTABLISHED
16 ACCEPT 17 -- 172.16.48.40/29 172.16.33.0/26 udp dpt:53 ctstate NEW,ESTABLISHED
25 ACCEPT 17 -- 0.0.0.0/0 172.16.33.0/26 match-set INTERNAL_SUBNETS src udp dpt:53 ctstate
NEW,ESTABLISHED
26 ACCEPT 17 -- 172.16.33.0/26 0.0.0.0/0 udp dpt:53 ctstate NEW,ESTABLISHED
31 ACCEPT 17 -- 172.16.33.0/26 0.0.0.0/0 udp dpt:53 ctstate NEW,ESTABLISHED
32 ACCEPT 17 -- 172.16.2.0/27 172.16.33.0/26 udp dpt:53 STRING match "10267350865787465726e61
6c001" ALGO name bm
33 ACCEPT 6 -- 172.16.2.0/27 172.16.33.0/26 tcp dpt:53 STRING match "10267350865787465726e61
6c001" ALGO name bm
35 ACCEPT 17 -- 0.0.0.0/0 0.0.0.0/0 match-set INTERNET_ALLOWED_SUBNETS src udp dpt:5
ctstate NEW,ESTABLISHED
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 18:20:29 $
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 18:20:29 $
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 18:20:29 $
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 18:20:29 $
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 18:20:29 $
```

9.2 - DNS Filtering Rules

ModSecurity blocking a reflected XSS payload on the /search endpoint, returning HTTP 403, confirming WAF blocking mode is operational.

```
deny,status:403,msg:'XSS-REFLECTED: HTML tags with event handlers in search'
SecRule REQUEST_URI "@rx ^/search(?:$|\\?)" "id:10006,phase:2,deny,log,status:403,msg:'XSS: event handler i
n searchString on /search',chain,t:none" SecRule ARGS_GET:searchString "@rx (?i)\\bon(?:mouse\\w+load|error
|click)\\s*=\\s*[\\'\\"]([\\^\\"]*(?:alert|confirm|prompt)\\s*\\(" t:urlDecodeUni,t:htmlEntityDecode"

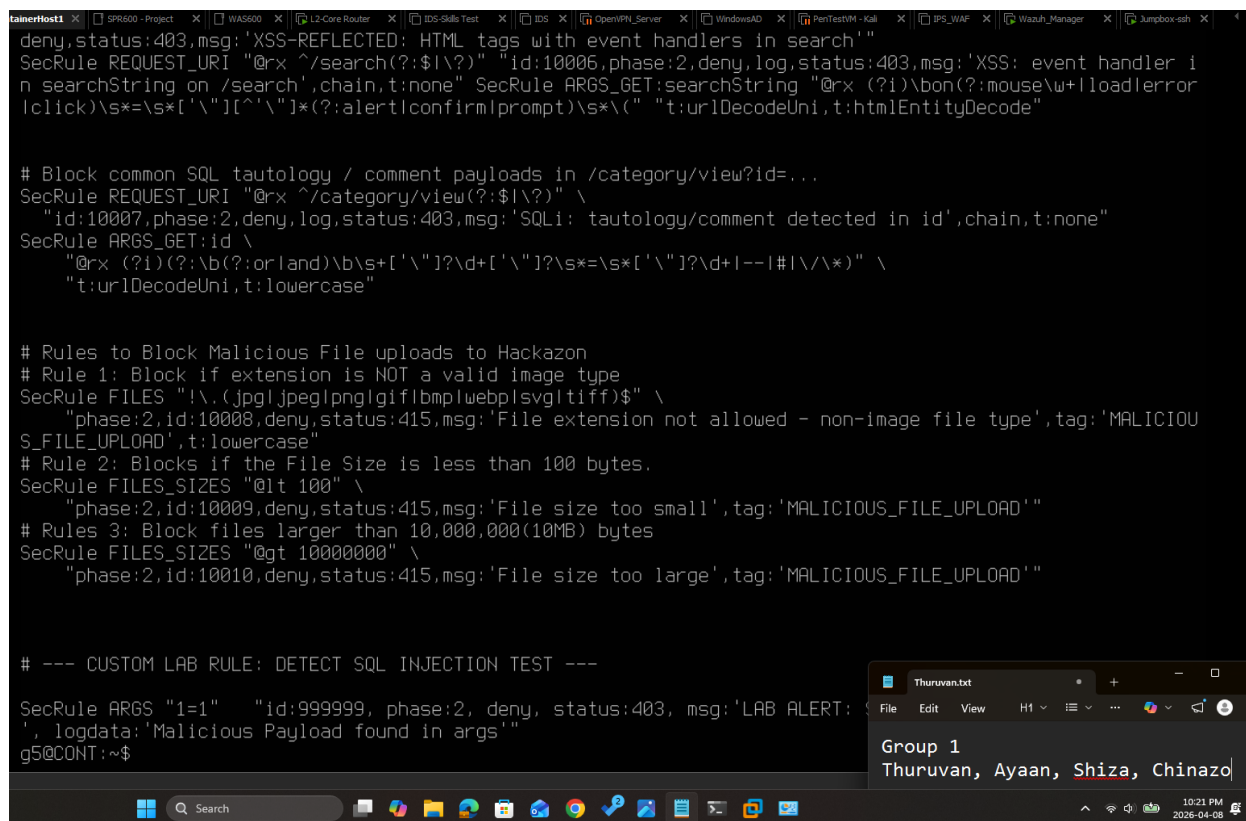
# Block common SQL tautology / comment payloads in /category/view?id=...
SecRule REQUEST_URI "@rx ^/category/view(?:$|\\?)" \
  "id:10007,phase:2,deny,log,status:403,msg:'SQLi: tautology/comment detected in id',chain,t:none"
SecRule ARGS_GET:id \
  "@rx (?i)(?:\\b(?:or|and)\\b\\s+[\\'\\"]?\\d+[\\'\\"]?\\s*=\\s*[\\'\\"]?\\d+|--|#!\\s*\\s*)" \
  "t:urlDecodeUni,t:lowercase"

# Rules to Block Malicious File uploads to Hackazon
# Rule 1: Block if extension is NOT a valid image type
SecRule FILES "@\\. (jpg|jpeg|png|gif|bmp|webp|svg|tiff)$" \
  "phase:2,id:10008,deny,status:415,msg:'File extension not allowed - non-image file type',tag:'MALICIOU
S_FILE_UPLOAD',t:lowercase"
# Rule 2: Blocks if the File Size is less than 100 bytes.
SecRule FILES_SIZES "@lt 100" \
  "phase:2,id:10009,deny,status:415,msg:'File size too small',tag:'MALICIOUS_FILE_UPLOAD'"
# Rule 3: Block files larger than 10,000,000(10MB) bytes
SecRule FILES_SIZES "@gt 10000000" \
  "phase:2,id:10010,deny,status:415,msg:'File size too large',tag:'MALICIOUS_FILE_UPLOAD'"

# --- CUSTOM LAB RULE: DETECT SQL INJECTION TEST ---
SecRule ARGS "1=1" "id:999999, phase:2, deny, status:403, msg:'LAB ALERT: $
', logdata:'Malicious Payload found in args'"
g5@CONT:~$
```

9.3 - ModSecurity XSS Block

ModSecurity blocking a SQLi tautology payload, confirming injection attack prevention at the application layer.



```

deny,status:403,msg:'XSS-REFLECTED: HTML tags with event handlers in search'
SecRule REQUEST_URI "@rx ^/search(?:$|\\?)" "id:10006,phase:2,deny,log,status:403,msg:'XSS: event handler i
n searchString on /search',chain,t:none" SecRule ARGS_GET:searchString "@rx (?i)\\bon(?:mouse\\w+load|error
|click)\\s*=\\s*[\\'\\"](?:\\'\\")*(?:alert|confirm|prompt)\\s*\\(" "t:urlDecodeUni,t:htmlEntityDecode"

# Block common SQL tautology / comment payloads in /category/view?id=...
SecRule REQUEST_URI "@rx ^/category/view(?:$|\\?)" \
    "id:10007,phase:2,deny,log,status:403,msg:'SQLi: tautology/comment detected in id',chain,t:none"
SecRule ARGS_GET:id \
    "@rx (?i)(?:\\b(?:or|and)\\b\\s+[\\'\\"]?\\d+[\\'\\"]?\\s*=\\s*[\\'\\"]?\\d+|--|_|#|\\|\\|\\*)" \
    "t:urlDecodeUni,t:lowercase"

# Rules to Block Malicious File uploads to Hackazon
# Rule 1: Block if extension is NOT a valid image type
SecRule FILES "!\\. (jpg|jpeg|png|gif|bmp|webp|svg|tiff)$" \
    "phase:2,id:10008,deny,status:415,msg:'File extension not allowed - non-image file type',tag:'MALICIOU
S_FILE_UPLOAD',t:lowercase"
# Rule 2: Blocks if the File Size is less than 100 bytes.
SecRule FILES_SIZES "@lt 100" \
    "phase:2,id:10009,deny,status:415,msg:'File size too small',tag:'MALICIOUS_FILE_UPLOAD'"
# Rules 3: Block files larger than 10,000,000(10MB) bytes
SecRule FILES_SIZES "@gt 10000000" \
    "phase:2,id:10010,deny,status:415,msg:'File size too large',tag:'MALICIOUS_FILE_UPLOAD'"

# --- CUSTOM LAB RULE: DETECT SQL INJECTION TEST ---
SecRule ARGS "1=1" "id:999999, phase:2, deny, status:403, msg:'LAB ALERT:
', logdata:'Malicious Payload found in args'"
g5@CONT:~$
  
```

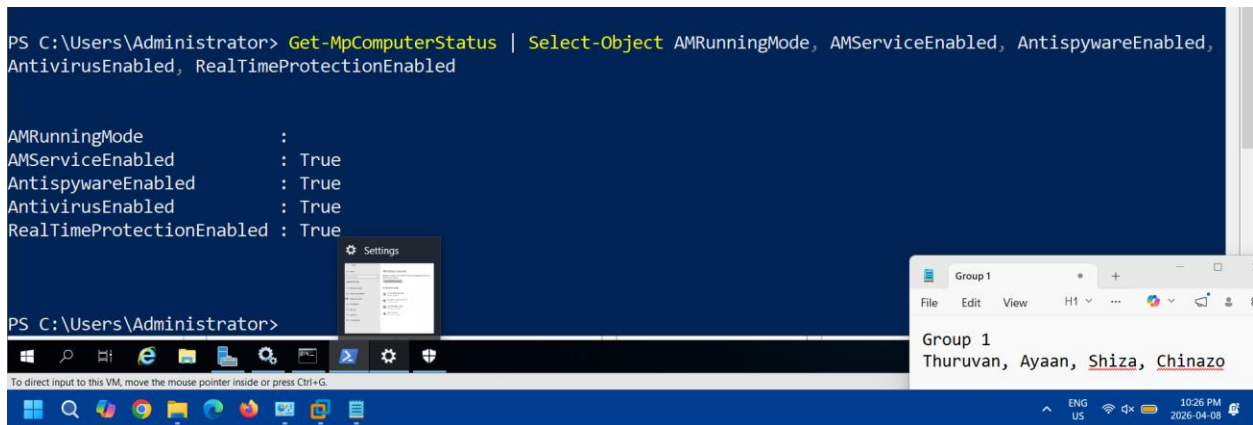
The screenshot shows a Windows desktop environment. The taskbar at the bottom includes icons for the Start menu, Search, File Explorer, and various applications. A terminal window is open, displaying ModSecurity rule configurations. The rules include XSS protection, SQLi tautology detection, and file upload restrictions. A file explorer window titled 'Thuruvan.txt' is also visible, showing a list of files with names like 'Thuruvan, Ayaan, Shiza, Chinazo'.

9.4 - ModSecurity SQLi Block

Control 10 — Malware Defenses

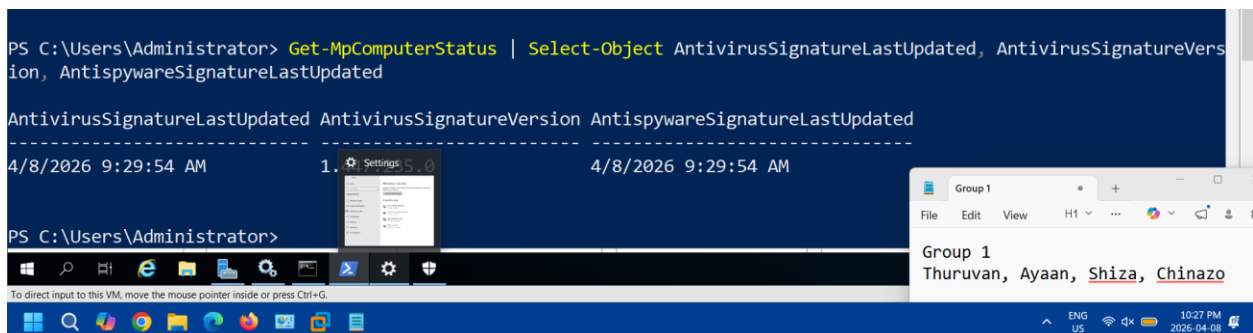
Malware defenses are implemented through Windows Defender on GS-AD with real-time protection and automatic signature updates. Wazuh FIM with realtime monitoring on critical paths and AppArmor with 30 profiles in enforce mode on jumpbox-ssh2 provide behavior-based detection and prevention at the host level.

Windows Defender status on GS-AD showing AMServiceEnabled, AntivirusEnabled, AntispywareEnabled, and RealTimeProtectionEnabled all True, confirming anti-malware is deployed and active.



10.1 - Windows Defender Active

Get-MpComputerStatus confirming antivirus and antispyware signatures last updated 4/8/2026, demonstrating automatic anti-malware signature updates.



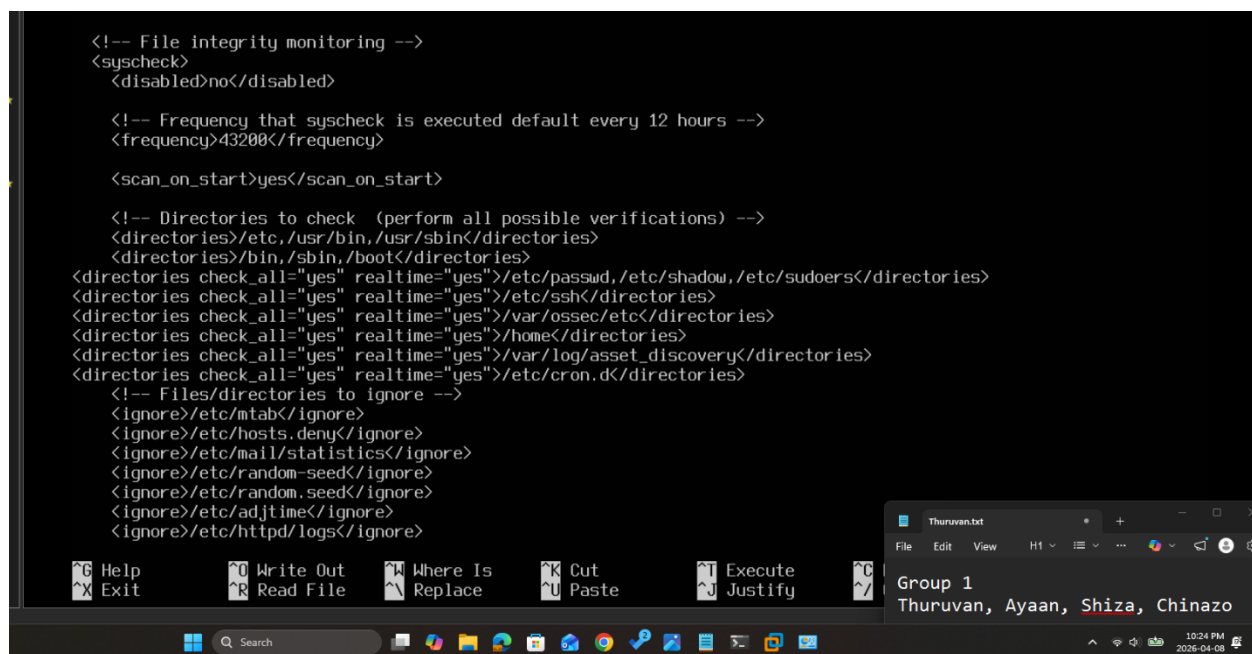
10.2 - Antimalware Signature Updates

Registry query on GS-AD confirming NoDriveTypeAutoRun = 255, disabling autorun for all drive types including removable media.



10.3 - Autorun Disabled

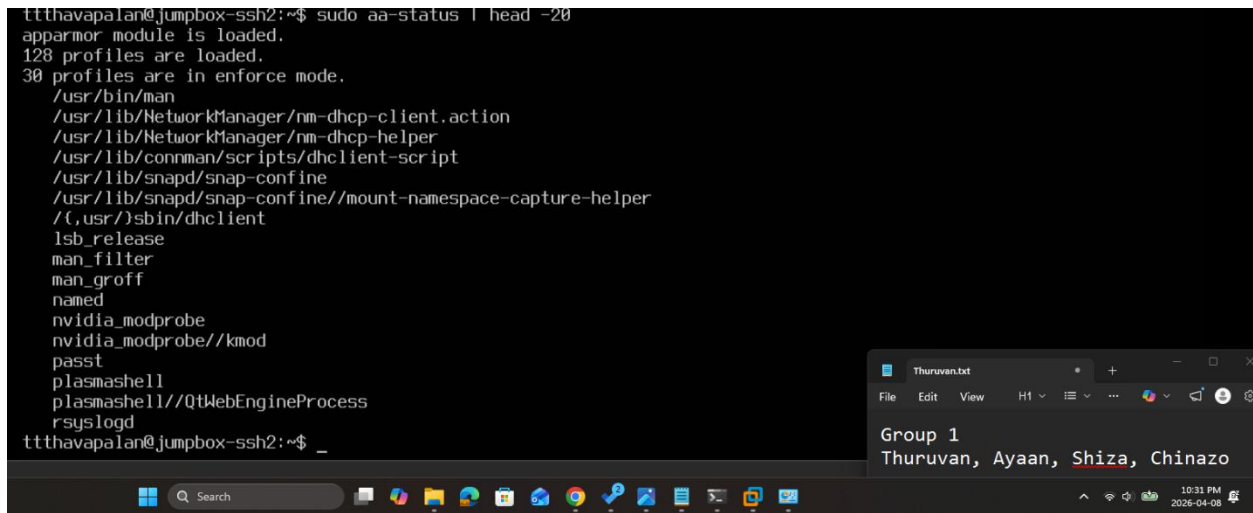
Wazuh FIM syscheck configuration on jumpbox-ssh2 showing realtime monitoring enabled on /etc/passwd, /etc/shadow, /etc/sudoers, /etc/ssh, /home, and /etc/cron.d, implementing behavior-based file integrity monitoring.



10.4 - FIM Realtime Monitoring Config

AppArmor aa-status output on jumpbox-ssh2 confirming 30 profiles in enforce mode, restricting process behavior at the kernel level.

```
ttthavapalan@jumpbox-ssh2:~$ sudo aa-status | head -20
apparmor module is loaded.
128 profiles are loaded.
30 profiles are in enforce mode.
/usr/bin/man
/usr/lib/NetworkManager/nm-dhcp-client.action
/usr/lib/NetworkManager/nm-dhcp-helper
/usr/lib/connman/scripts/dhclient-script
/usr/lib/snapd/snap-confine
/usr/lib/snapd/snap-confine/mount-namespace-capture-helper
/(/usr/sbin/dhclient
lsb_release
man_filter
man_groff
named
nvidia_modprobe
nvidia_modprobe//kmod
passt
plasmashell
plasmashell//QtWebEngineProcess
rsyslogd
ttthavapalan@jumpbox-ssh2:~$ _
```

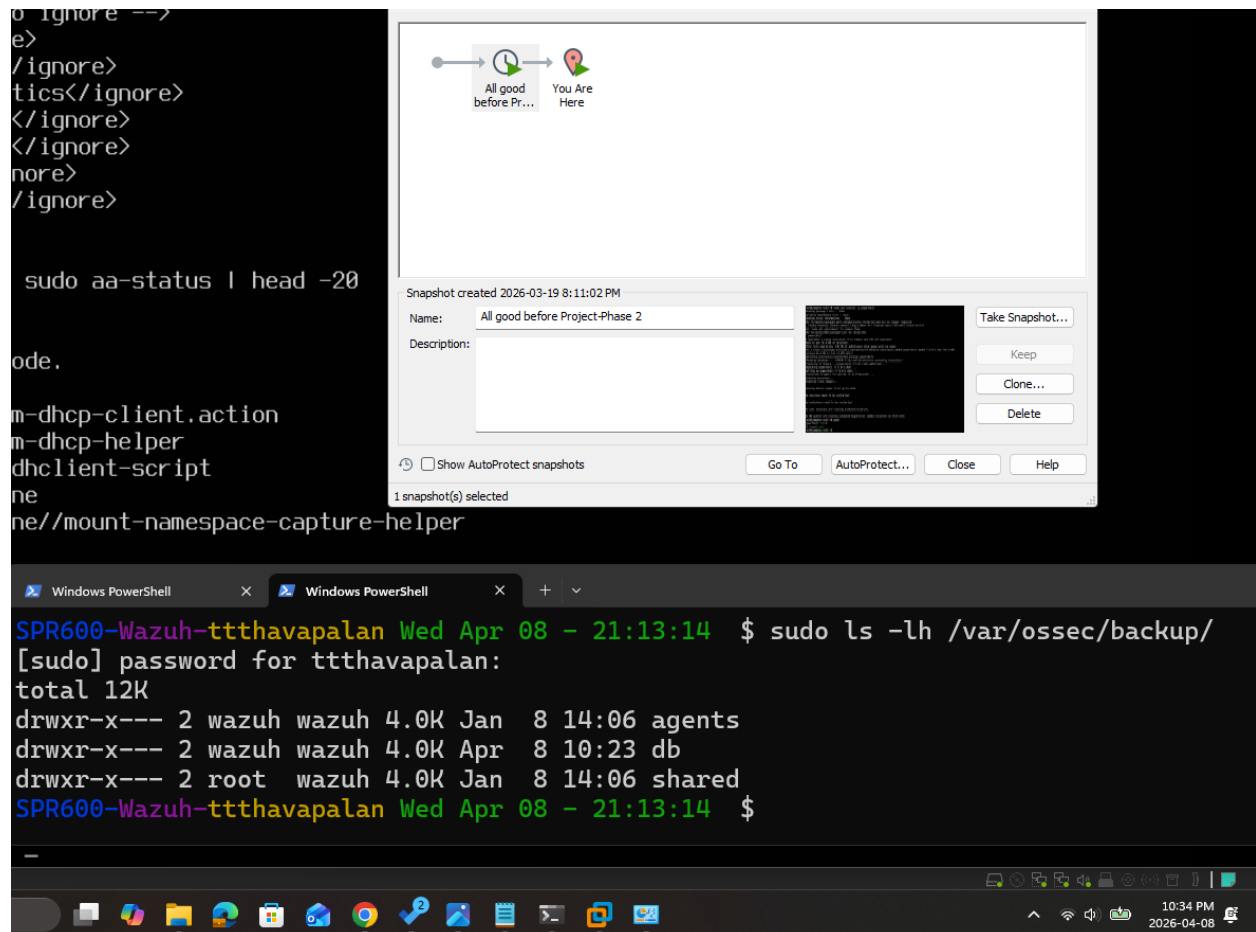


10.5 - AppArmor 30 Profiles Enforced

Control 11 — Data Recovery

Data recovery capability is implemented through VMware snapshot management across all lab VMs stored on the host filesystem at E:\Semester 5, isolated from guest OS filesystems. Wazuh maintains configuration backups at /var/ossec/backup/ with subdirectories for agents, db, and shared, owned by root:wazuh with permissions 0750. The recovery process was validated during Phase 2 when a T1070.004 Atomic Red Team test destroyed jumpbox-ssh binaries, requiring a full VM restore from snapshot.

VMware Snapshot Manager showing snapshots for all lab VMs alongside Wazuh /var/ossec/backup/ directory confirming agent and db backup subdirectories exist.



11.1 - VMware Snapshots and Wazuh Backup

/var/ossec/backup/ directory permissions confirmed as 0750 root:wazuh, inaccessible to unprivileged users, protecting recovery data.

```

stat /var/ossec/backup/lan Wed Apr 08 - 21:13:14 $ sudo stat /var/ossec/backup/
File: /var/ossec/backup/
Size: 4096      Blocks: 8      IO Block: 4096   directory
Device: 252,0  Inode: 277801   Links: 5
Access: (0750/drwxr-x---)  Uid: (  0/   root)   Gid: (107/  wazuh)
Access: 2026-04-08 22:30:11.065412110 -0400
Modify: 2026-02-05 15:30:33.955499318 -0500
Change: 2026-04-08 20:21:38.455062905 -0400
Birth: 2026-02-05 15:30:33.954499260 -0500
SPR600-Wazuh-tttthavapalan Wed Apr 08 - 21:13:14 $

```

11.2 - Backup Directory Permissions

PowerShell enumeration of .vmxn snapshot files on E:\Semester 5 confirming isolated snapshot storage on the hypervisor host for all lab VMs including ContainerHost1, jumpbox-ssh, OpenVPN, and IPS-WAF.

```

PS C:\WINDOWS\system32> Get-ChildItem "E:\Semester 5" -Recurse -Filter "*.vmxn" | Select-Object Name, LastWriteTime
-----
Name                                     LastWriteTime
-----
Kali-Scanner-Snapshot1.vmsn             2025-10-09 1:06:54 PM
Kali-Scanner-Snapshot3.vmsn             2025-11-04 11:16:07 PM
ContainerHost1-Snapshot1.vmsn           2025-11-01 12:08:58 AM
ContainerHost1-Snapshot2.vmsn           2025-11-02 1:01:50 AM
ContainerHost1-Snapshot3.vmsn           2025-11-14 12:49:42 PM
ContainerHost1-Snapshot4.vmsn           2025-11-23 8:44:11 PM
Jumpbox-ssh-Snapshot1.vmsn              2026-03-19 8:17:42 PM
UbuntuMachine-Snapshot1.vmsn            2025-10-01 8:21:13 PM
OpenVPN_Server-Snapshot1.vmsn           2025-11-16 7:47:23 PM
WindowsJumpbox-RDP-Snapshot2.vmsn       2025-11-27 2:58:01 PM
IPS_WAF-Snapshot1.vmsn                  2026-01-15 6:15:19 PM

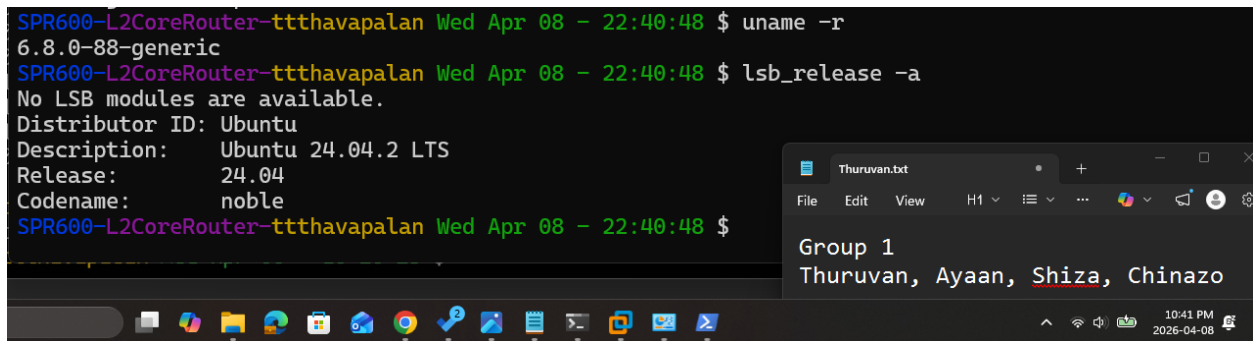
```

11.3 - Isolated Snapshot Storage

Control 12 — Network Infrastructure Management

Network infrastructure management is implemented through a CoreRouter running Ubuntu 24.04.2 LTS enforcing VLAN segmentation across 13 distinct subnets via iptables with a default-DROP policy on all chains. Firewall rules are reboot-persistent via netfilter-persistent.service. Remote administrative access is provided exclusively through OpenVPN with AES-256-GCM and certificate plus Kerberos MFA. The Wazuh agent on CoreRouter maintains audit visibility over network infrastructure changes.

CoreRouter `uname -r` and `lsb_release -a` confirming Ubuntu 24.04.2 LTS kernel 6.8.0-88-generic, demonstrating up-to-date network infrastructure.



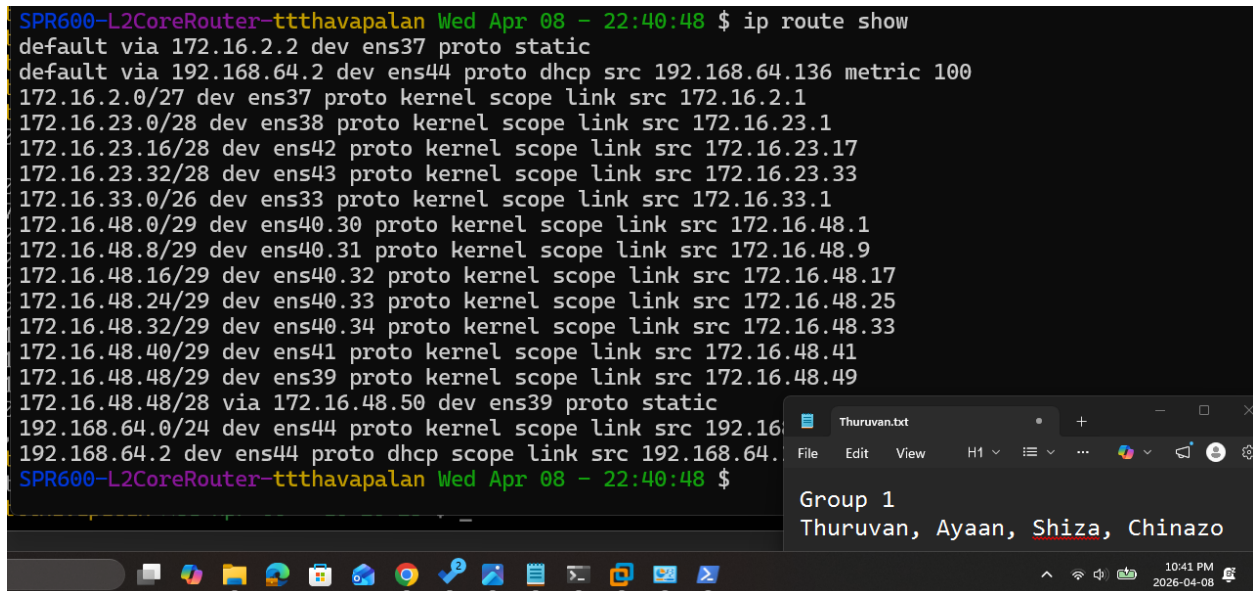
```

SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 22:40:48 $ uname -r
6.8.0-88-generic
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 22:40:48 $ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:   Ubuntu 24.04.2 LTS
Release:      24.04
Codename:     noble
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 22:40:48 $
  
```

The terminal output shows the kernel version and Ubuntu release details. A file manager window titled 'Thuruvan.txt' is open, displaying the text: 'Group 1', 'Thuruvan, Ayaan, Shiza, Chinazo'.

12.1 - CoreRouter OS Version

`ip route show` confirming 13 active subnet routes across 7 VLANs on CoreRouter, with zone-based segmentation enforced via iptables with default-DROP.



```

SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 22:40:48 $ ip route show
default via 172.16.2.2 dev ens37 proto static
default via 192.168.64.2 dev ens44 proto dhcp src 192.168.64.136 metric 100
172.16.2.0/27 dev ens37 proto kernel scope link src 172.16.2.1
172.16.23.0/28 dev ens38 proto kernel scope link src 172.16.23.1
172.16.23.16/28 dev ens42 proto kernel scope link src 172.16.23.17
172.16.23.32/28 dev ens43 proto kernel scope link src 172.16.23.33
172.16.33.0/26 dev ens33 proto kernel scope link src 172.16.33.1
172.16.48.0/29 dev ens40.30 proto kernel scope link src 172.16.48.1
172.16.48.8/29 dev ens40.31 proto kernel scope link src 172.16.48.9
172.16.48.16/29 dev ens40.32 proto kernel scope link src 172.16.48.17
172.16.48.24/29 dev ens40.33 proto kernel scope link src 172.16.48.25
172.16.48.32/29 dev ens40.34 proto kernel scope link src 172.16.48.33
172.16.48.40/29 dev ens41 proto kernel scope link src 172.16.48.41
172.16.48.48/29 dev ens39 proto kernel scope link src 172.16.48.49
172.16.48.48/28 via 172.16.48.50 dev ens39 proto static
192.168.64.0/24 dev ens44 proto kernel scope link src 192.168.64.1
192.168.64.2 dev ens44 proto dhcp scope link src 192.168.64.136 metric 100
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 22:40:48 $
  
```

The terminal output lists 13 active subnet routes across 7 VLANs. A file manager window titled 'Thuruvan.txt' is open, displaying the text: 'Group 1', 'Thuruvan, Ayaan, Shiza, Chinazo'.

12.2 - Secure Network Architecture Routes

netfilter-persistent.service status showing active and enabled with iptables.service symlinked, confirming firewall rules survive reboots on CoreRouter.

```

sudo tail -f /var/log/kern.log | grep -E FORWARD-DROPIINPUT-DROPIOUTPUT-DROP
2>/dev/null || systemctl status netfilter-persistent29 $ systemctl status iptables-save
• netfilter-persistent.service - netfilter persistent configuration
  Loaded: loaded (/usr/lib/systemd/system/netfilter-persistent.service; enabled; preset: enabled)
  Drop-In: /usr/lib/systemd/system/netfilter-persistent.service.d
           └─iptables.conf
  Active: active (exited) since Wed 2026-04-08 18:18:45 EDT; 4h 23min ago
  Docs: man:netfilter-persistent(8)
  Main PID: 761 (code=exited, status=0/SUCCESS)
  CPU: 23ms

Apr 08 18:18:45 CoreRouter systemd[1]: Starting netfilter-persistent.service - netfilter persistent configuration...
Apr 08 18:18:45 CoreRouter netfilter-persistent[770]: run-parts: executing /usr/share/netfilter-persistent/plugins.>
Apr 08 18:18:45 CoreRouter netfilter-persistent[770]: run-parts: executing /usr/share/netfilter-persistent/plugins.>
Apr 08 18:18:45 CoreRouter systemd[1]: Finished netfilter-persistent.service - netfilter persistent configuration.
rep iptablesRouter-ttthavapalan Wed Apr 08 - 18:20:29 $ ls -lh /etc/systemd/system/ | g
-rw-r--r-- 1 root root 490 Oct 30 14:49 iptables-restore-late.service
lrwxrwxrwx 1 root root 52 Oct 28 18:15 iptables.service -> /usr/lib/systemd/system/netfilter-persistent.service
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 18:20:29 $
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 18:20:29 $
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 18:20:29 $
SPR600-L2CoreRouter-ttthavapalan Wed Apr 08 - 18:20:29 $

```

12.3 - Firewall Rules Reboot Persistent

OpenVPN service enabled and active on vpn host with AES-256-GCM and Wazuh agent listening on port 1514, confirming secure network management and communication protocols.

```

Active: active (running) since Wed 2026-04-08 18:20:31 EDT; 4h 24min ago
Process: 1382 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
Tasks: 32 (limit: 2132)
Memory: 459.7M (peak: 466.5M)
CPU: 22.492s
CGroup: /system.slice/wazuh-agent.service
├─1431 /var/ossec/bin/wazuh-execd
├─1441 /var/ossec/bin/wazuh-agentd
├─1454 /var/ossec/bin/wazuh-syscheckd
├─1464 /var/ossec/bin/wazuh-logcollector
└─1479 /var/ossec/bin/wazuh-modulesd

Apr 08 18:20:16 vpn systemd[1]: Starting wazuh-agent.service - Wazuh agent...
Apr 08 18:20:21 vpn env[1382]: Starting Wazuh v4.14.2...
Apr 08 18:20:22 vpn env[1382]: Started wazuh-execd...
Apr 08 18:20:28 vpn env[1382]: Started wazuh-agentd...
Apr 08 18:20:28 vpn env[1382]: Started wazuh-syscheckd...
Apr 08 18:20:28 vpn env[1382]: Started wazuh-logcollector...
Apr 08 18:20:29 vpn env[1382]: Started wazuh-modulesd...
Apr 08 18:20:31 vpn env[1382]: Completed.
Apr 08 18:20:31 vpn systemd[1]: Started wazuh-agent.service - Wazuh agent.
ttthavapalan@vpn:~$ systemctl status openvpn
• openvpn.service - OpenVPN service
  Loaded: loaded (/usr/lib/systemd/system/openvpn.service; enabled; preset: enabled)
  Active: active (exited) since Wed 2026-04-08 18:18:18 EDT; 4h 26min ago
  Docs: man:openvpn(8)
  Process: 1299 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
  Main PID: 1299 (code=exited, status=0/SUCCESS)
  CPU: 5ms

Apr 08 18:18:18 vpn systemd[1]: Starting openvpn.service - OpenVPN service..
Apr 08 18:18:18 vpn systemd[1]: Finished openvpn.service - OpenVPN service.
ttthavapalan@vpn:~$ _

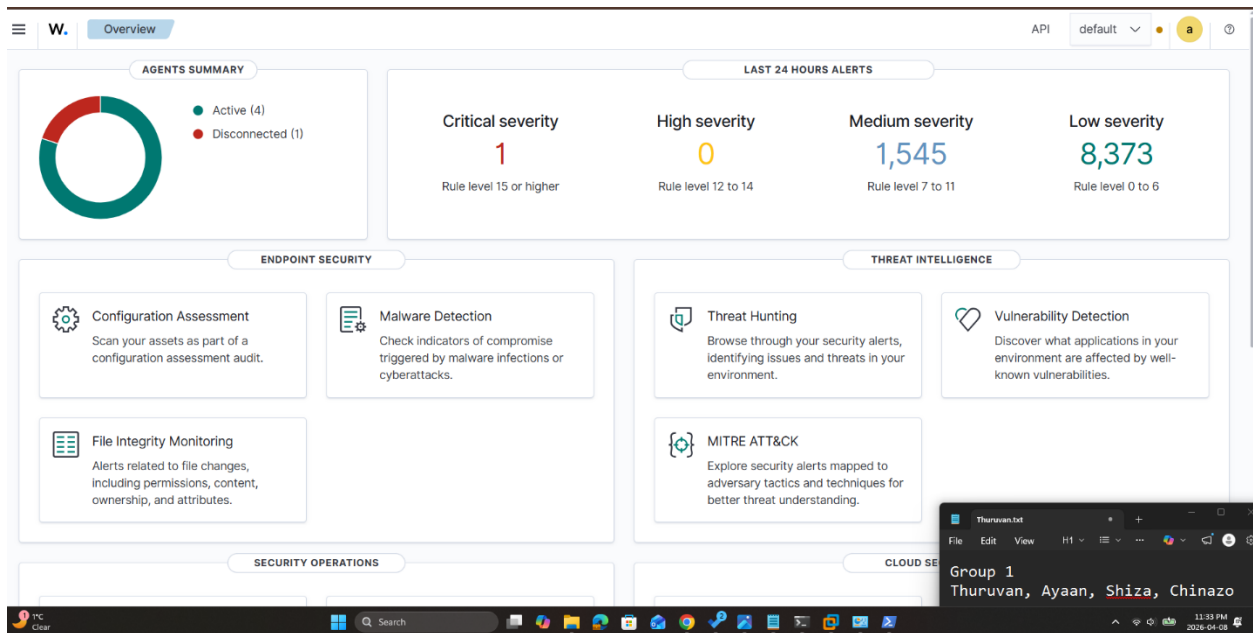
```

12.4 - VPN and Wazuh Agent Ports

Control 13 — Network Monitoring and Defense

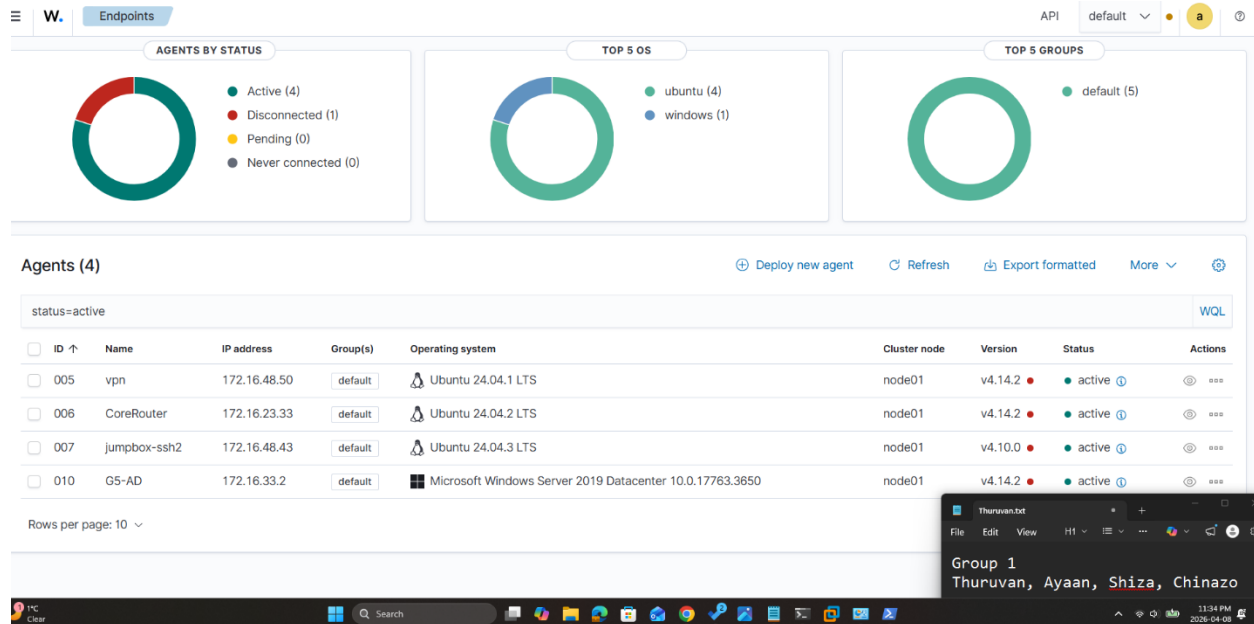
Network monitoring is implemented through a layered stack: Wazuh with OpenSearch provides centralized SIEM alerting across all enrolled agents; Suricata 7.0.3 and Zeek run on CoreRouter monitoring the VLAN trunk and three additional interfaces; AppArmor enforces 30 mandatory access control profiles on jumpbox-ssh2; and ModSecurity WAF operates in blocking mode with the OWASP CRS loaded. Inter-VLAN traffic filtering is enforced by iptables with a default-DROP policy on both INPUT and FORWARD chains.

Wazuh Overview dashboard showing 24-hour alert totals across all severity bands and enrolled agents, confirming centralized security event alerting.



13.1a - Centralized Security Event Alerting

Wazuh Endpoints panel confirming agents vpn-005, CoreRouter-006, jumpbox-ssh2-007, and G5-AD-010 in active status.



13.1b - Active Agents on Wazuh

GS-AD PowerShell confirmation of Sysmon64 running with a live Event ID 11 file creation record tagged to MITRE T1574.010, confirming host-based intrusion detection is operational.

```
PS C:\Users\Administrator> Get-Service Sysmon64

Status Name      DisplayName
-----
Running Sysmon64 Sysmon64

PS C:\Users\Administrator> Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" -MaxEvents 5 | Select-Object
TimeCreated, Id, Message | Format-List

TimeCreated : 4/8/2026 11:37:08 PM
Id           : 11
Message      : File created:
               RuleName: technique_id=T1574.010,technique_name=Services File Permissions Weakness
               UtcTime: 2026-04-09 03:37:08.428
               ProcessGuid: {1a2f397d-0890-69d7-1c00-000000003300}
               ProcessId: 1296
               Image: C:\Windows\System32\svchost.exe
```

13.2a - Sysmon Active on AD

Sysmon rule files and syscheck configuration loaded on the Wazuh Manager confirming HIDS deployment across the environment.

[illegible]

13.2b - Wazuh FIM and Sysmon Rules

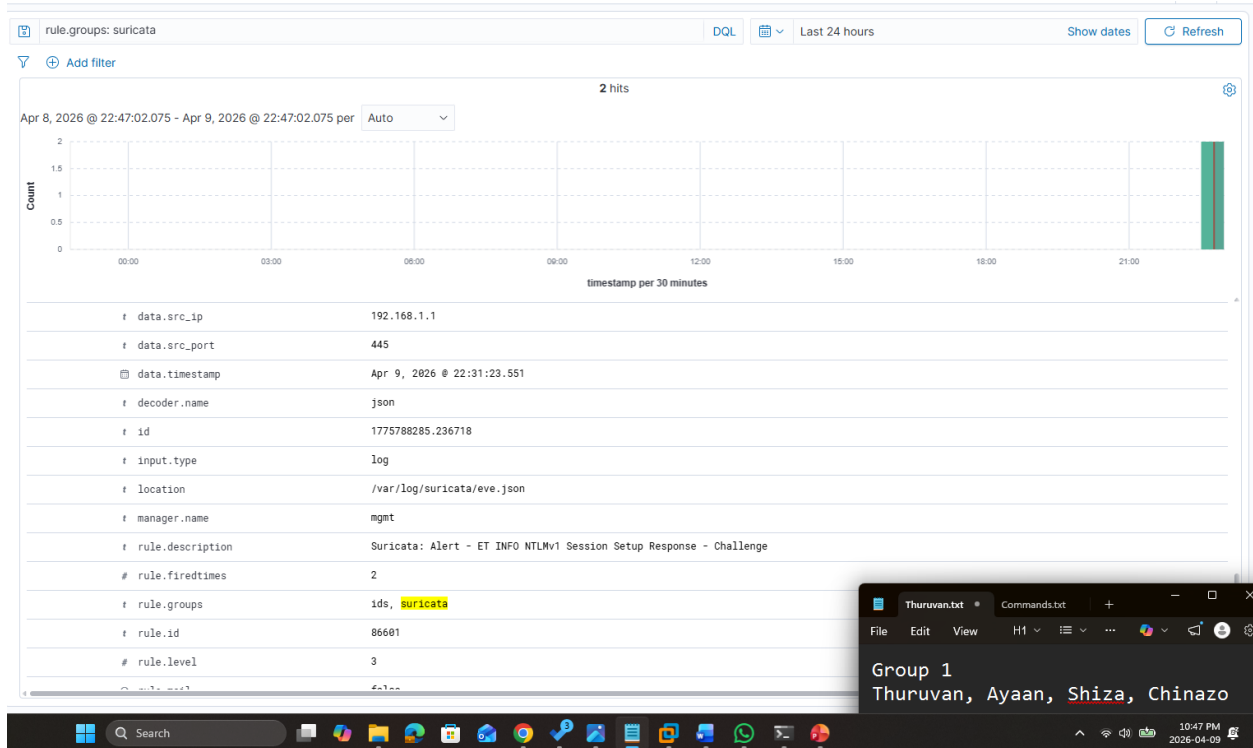
Suricata 7.0.3 active on CoreRouter with the IDS/IDP daemon running across interfaces ens40, ens33, ens38, and ens39, confirming network intrusion detection coverage.

```
SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 21:24:52 $ sudo systemctl status suricata --no-pager
[sudo] password for ttthavapalan:
● suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-04-09 21:06:49 EDT; 1h 37min ago
     Docs: man:suricata(8)
           man:suricatasc(8)
           https://suricata.io/documentation/
  Main PID: 14765 (Suricata-Main)
    Tasks: 14 (Limit: 9379)
  Memory: 548.7M (peak: 549.2M)
    CPU: 1min 13.364s
   CGroup: /system.slice/suricata.service
           └─14765 /usr/bin/suricata -D -af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid

Apr 09 21:06:49 CoreRouter systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Apr 09 21:06:49 CoreRouter suricata[14764]: i: suricata: This is Suricata version 7.0.3 RELEASE running in SYS
Apr 09 21:06:49 CoreRouter systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
udo tail -20 /var/log/suricata/eve.json | python3 -m json.tool | head -40&ar/log/suricata/eve.json | python3
Extra data: line 2 column 1 (char 331)
SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 21:24:52 $
```

13.3a - Suricata Running on CoreRouter

OpenSearch event confirming a live Suricata alert ingested from /var/log/suricata/eve.json via CoreRouter agent 006, demonstrating end-to-end NIDS alert pipeline.



13.3b - Suricata Alert in OpenSearch

CoreRouter iptables INPUT and FORWARD chains with default-DROP policy, SYNPROXY rules, bogon drops, and LOG entries confirming inter-VLAN traffic filtering.

```
Chain INPUT (policy DROP 3 packets, 156 bytes)
  pkts bytes target     prot opt in     out     source               destination
  10 3070 ACCEPT     0    --  *      *       0.0.0.0/0            0.0.0.0/0           ctstate RELATED,ESTABLISHED
  30 2282 ACCEPT     0    --  lo      *       0.0.0.0/0            0.0.0.0/0
  0 0 ACCEPT     6    --  *      *       172.16.48.0/29        0.0.0.0/0           tcp dpt:443
  0 0 ACCEPT     6    --  *      *       172.16.23.16/28       0.0.0.0/0           tcp dpt:22
  0 0 ACCEPT     89   -- ens37  *       0.0.0.0/0            0.0.0.0/0           multiport dports 57,68 matc
  0 0 ACCEPT     17   --  *      *       0.0.0.0/0            0.0.0.0/0           LOG flags 0 level 4 prefix
h-set INTERNAL_SUBNETS src limit: avg 5/min burst 10
  3 156 LOG      0    --  *      *       0.0.0.0/0            0.0.0.0/0           LOG flags 0 level 4 prefix
"INPUT-DROP: "
  0 0 DROP      0    -- ens37  *       10.0.0.0/8           0.0.0.0/0
  0 0 DROP      0    -- ens37  *       192.168.0.0/16        0.0.0.0/0
  0 0 DROP      0    -- ens37  *       127.0.0.0/8           0.0.0.0/0
  0 0 DROP      0    -- ens37  *       169.254.0.0/16        0.0.0.0/0
  0 0 DROP      0    -- ens37  *       224.0.0.0/4           0.0.0.0/0

Chain FORWARD (policy DROP 22 packets, 1200 bytes)
  pkts bytes target     prot opt in     out     source               destination
  0 0 SYNPROXY   6    --  *      *       0.0.0.0/0            172.16.48.0/29      tcp dpt:80 ctstate INVALID,
UNTRACKED SYNPROXY sack-perm timestamp wscale 7 mss 1460
  0 0 SYNPROXY   6    --  *      *       0.0.0.0/0            172.16.48.0/29      tcp dpt:443 ctstate INVALID
,UNTRACKED SYNPROXY sack-perm timestamp wscale 7 mss 1460
  0 0 ACCEPT     6    --  *      *       0.0.0.0/0            172.16.48.0/29      tcp dpt:80 ctstate UNTRACKE
D
  0 0 ACCEPT     6    --  *      *       0.0.0.0/0            172.16.48.0/29      tcp dpt:443 ctstate UNTRACK
ED
  39 21472 DROP      0    --  *      *       0.0.0.0/0            0.0.0.0/0           ctstate INVALID
  25 5655 ACCEPT     0    --  *      *       0.0.0.0/0            0.0.0.0/0           ctstate RELATED,ESTABLISHED
  0 0 ACCEPT     6    --  *      *       0.0.0.0/0            172.16.48.0/29      tcp dpt:80 ctstate NEW
  0 0 ACCEPT     6    --  *      *       0.0.0.0/0            172.16.48.0/29      tcp dpt:443 ctstate NEW
  0 0 DROP      0    -- ens37  *       0.0.0.0/0            255.255.255.255
  0 0 DROP      0    -- ens37  *       0.0.0.0/0            224.0.0.0/4
  0 0 ACCEPT     6    --  *      *       172.16.23.16/28      0.0.0.0/0
dst tcp dpt:22 ctstate NEW,ESTABLISHED
:_
```

13.4 - Default Drop and Forward Rules

OpenVPN service status confirming active state with AES-256-GCM cipher and RADIUS MFA plugin, managing access control for remote assets.

```

ttthavapalan@vpn:~$ sudo systemctl status openvpn-server@server --no-pager
• openvpn-server@server.service - OpenVPN service for server
   Loaded: loaded (/usr/lib/systemd/system/openvpn-server@.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-04-09 22:33:25 EDT; 19min ago
     Docs: man:openvpn(8)
           https://openvpn.net/community-resources/reference-manual-for-openvpn-2-6/
           https://community.openvpn.net/openvpn/wiki/HOWTO
   Main PID: 1378 (openvpn)
   Status: "Initialization Sequence Completed"
   Tasks: 3 (limit: 2132)
  Memory: 3.9M (peak: 4.1M)
    CPU: 140ms
   CGroup: /system.slice/system-openvpn\x2dservice.slice/openvpn-server@server.service
           └─1378 /usr/sbin/openvpn --status /run/openvpn-server/status-server.log --status-version 2 -...
             └─1394 /usr/sbin/openvpn --status /run/openvpn-server/status-server.log --status-version 2 -...
               └─1395 /usr/sbin/openvpn --status /run/openvpn-server/status-server.log --status-version 2 -...

Apr 09 22:33:25 vpn systemd[1]: Starting openvpn-server@server.service - OpenVPN service for server...
Apr 09 22:33:25 vpn systemd[1]: Started openvpn-server@server.service - OpenVPN service for server.
ttthavapalan@vpn:~$ sudo grep -E "cipher|auth|tls|plugin|mfalplugin|krb" /etc/openvpn/server/server.conf
plugin /usr/lib/openvpn/radiusplugin.so /etc/openvpn/radiusplugin.cnf
#auth-user-pass
tls-auth ta.key 0
cipher AES-256-GCM
auth SHA256
tls-verify /etc/openvpn/server/check_intermediate.sh
ttthavapalan@vpn:~$ _

```

13.5 - OpenVPN Status and Cipher

zeekctl status with all four workers running on ens40, ens33, ens38, and ens39, alongside /opt/zeek/logs/current/ directory confirming active network traffic flow log generation.

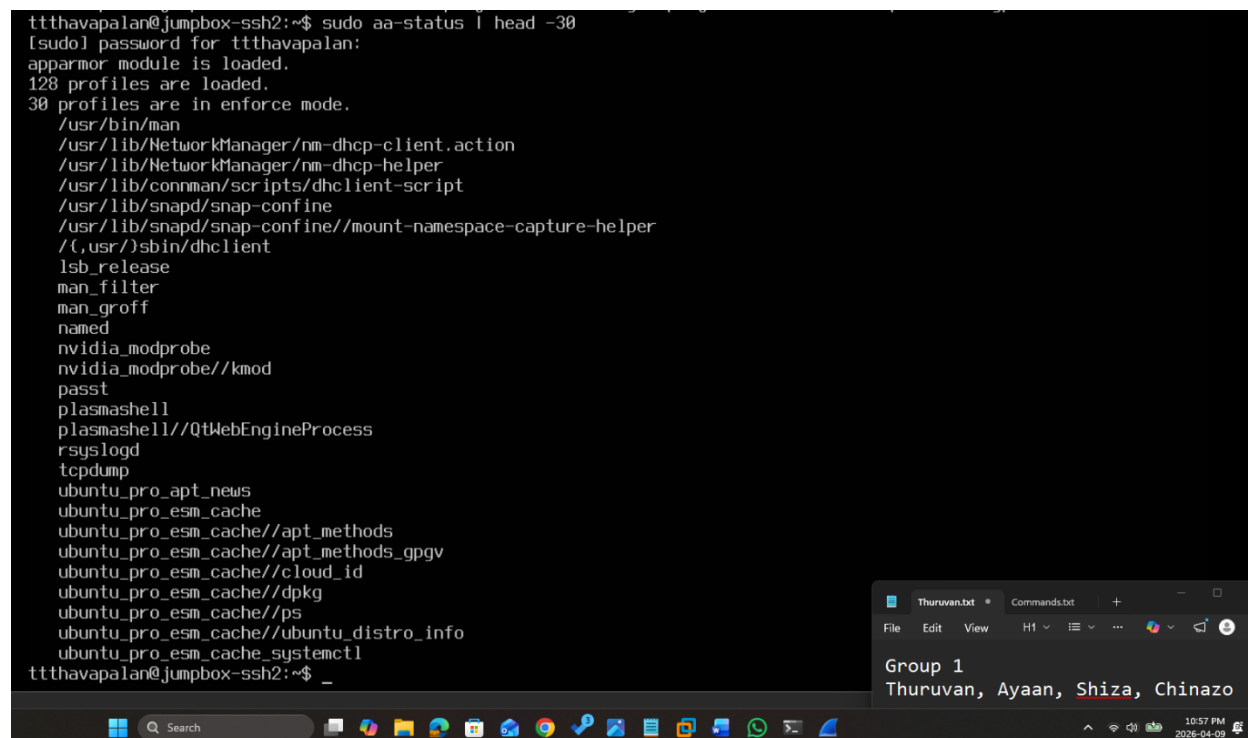
```

SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 23:08:14 $ sudo zeekctl status
Warning: zeekctl option UseWebSocket is set, but websockets non-functional: Failed to import websockets module - is it installed? (ModuleNotFound
Error("No module named 'websockets.sync'"))
Name      Type      Host      Status  Pid   Started
logger    logger    localhost running  5082  09 Apr 23:09:28
manager   manager   localhost running  5192  09 Apr 23:09:29
proxy-1    proxy     localhost running  5254  09 Apr 23:09:31
worker-ens40 worker localhost running  5353  09 Apr 23:09:32
worker-ens33 worker localhost running  5348  09 Apr 23:09:32
worker-ens38 worker localhost running  5352  09 Apr 23:09:32
worker-ens39 worker localhost running  5351  09 Apr 23:09:32
SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 23:08:14 $ ls -lh /opt/zeek/logs/current/
ls: cannot access '/opt/zeek/logs/current/': Permission denied
SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 23:08:14 $ sudo ls -lh /opt/zeek/logs/current/
total 460K
-rw-r--r-- 1 root zeek 405 Apr 9 23:10 capture_loss.log
-rw-r--r-- 1 root zeek 4.5K Apr 9 23:09 cluster.log
-rw-r--r-- 1 root zeek 1.7K Apr 9 23:10 conn.log
-rw-r--r-- 1 root zeek 2.2K Apr 9 23:10 dns.log
-rw-r--r-- 1 root zeek 240 Apr 9 23:10 known_services.log
-rw-r--r-- 1 root zeek 240K Apr 9 23:09 loaded_scripts.log
-rw-r--r-- 1 root zeek 1.4K Apr 9 23:10 notice.log
-rw-r--r-- 1 root zeek 409 Apr 9 23:09 packet_filter.log
-rw-r--r-- 1 root zeek 1005 Apr 9 23:09 stats.log
-rw-r--r-- 1 root zeek 0 Apr 9 23:09 stderr.log
-rw-r--r-- 1 root zeek 204 Apr 9 23:09 stdout.log
-rw-r--r-- 1 root zeek 174K Apr 9 23:10 telemetry.log
-rw-r--r-- 1 root zeek 371 Apr 9 23:10 weird.log
SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 23:08:14 $ sudo tail -5 /opt/zeek/logs/current/conn.log
1775790612.761341 CVIRVJINQRVehUtd3 192.168.1.2 138 192.168.1.255 138 udp - S0 T
T 0 D 1 229 0 0 - 17
1775790663.325265 CR0ISM1TyR8WgpN3f2 172.16.33.2 60899 192.203.230.10 53 udp dns -
F 0 D 1 73 0 - 17
1775790663.325266 CX7YD60eqtiAW6WE1 172.16.33.2 61146 1.1.1.1 53 udp dns -
0 D 1 73 0 - 17
1775790664.622294 CMSJLz3Mc0k7f5yyY1 172.16.33.2 60789 1.1.1.1 53 udp dns -
0 D 1 73 0 - 17

```

13.6 - Zeek Running and Logs

AppArmor aa-status output on jumpbox-ssh2 confirming 30 profiles in enforce mode, providing host-based intrusion prevention.



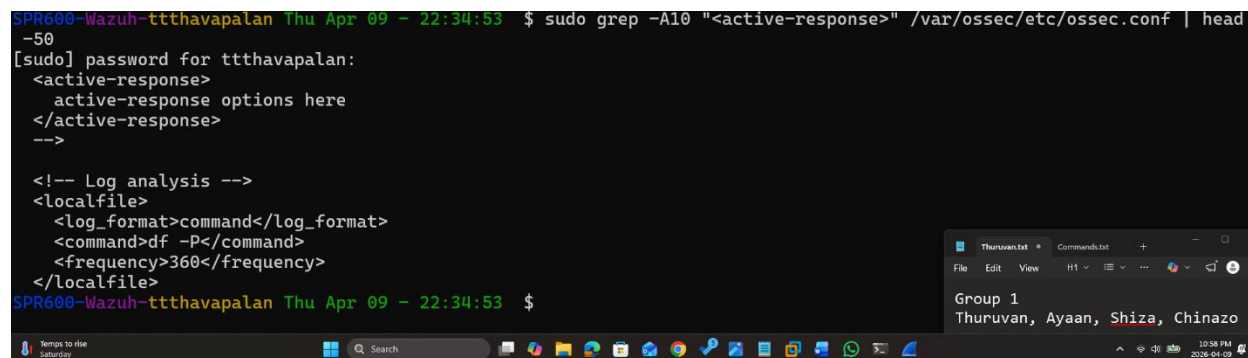
```

ttthavapalan@jumpbox-ssh2:~$ sudo aa-status | head -30
[sudo] password for ttthavapalan:
apparmor module is loaded.
128 profiles are loaded.
30 profiles are in enforce mode.
/usr/bin/man
/usr/lib/NetworkManager/nm-dhcp-client.action
/usr/lib/NetworkManager/nm-dhcp-helper
/usr/lib/connman/scripts/dhclient-script
/usr/lib/snapd/snap-confine
/usr/lib/snapd/snap-confine/mount-namespace-capture-helper
/usr/sbin/dhclient
lsb_release
man_filter
man_groff
named
nvidia_modprobe
nvidia_modprobe//kmod
passt
plasmashell
plasmashell//QtWebEngineProcess
rsyslogd
tcpdump
ubuntu_pro_apt_news
ubuntu_pro_esm_cache
ubuntu_pro_esm_cache//apt_methods
ubuntu_pro_esm_cache//apt_methods.gpgv
ubuntu_pro_esm_cache//cloud_id
ubuntu_pro_esm_cache//dpkg
ubuntu_pro_esm_cache//ps
ubuntu_pro_esm_cache//ubuntu_distro_info
ubuntu_pro_esm_cache.systemctl
ttthavapalan@jumpbox-ssh2:~$ _

```

13.7a - AppArmor 30 Profiles

ossec.conf active-response block showing block_etsblue, block_attacker, isolate_beachhead, and firewall-drop commands configured on the Wazuh Manager, confirming automated host-level containment capability.



```

SPR600-Wazuh-ttthavapalan Thu Apr 09 - 22:34:53 $ sudo grep -A10 "<active-response>" /var/ossec/etc/ossec.conf | head -50
[sudo] password for ttthavapalan:
<active-response>
  active-response options here
</active-response>
-->

<!-- Log analysis -->
<localfile>
  <log_format>command</log_format>
  <command>df -P</command>
  <frequency>360</frequency>
</localfile>
SPR600-Wazuh-ttthavapalan Thu Apr 09 - 22:34:53 $

```

13.7b - Active Response Commands Configured

CoreRouter iptables FORWARD chain with SYNPROXY rules applied to ports 80 and 443 destined for the container zone, providing SYN flood mitigation.

```
Chain FORWARD (policy DROP 0 packets, 0 bytes)
  pkts bytes target     prot opt in     out     source               destination
  0      0 SYNPROXY 6  --  *      *       0.0.0.0/0           172.16.48.0/29      tcp dpt:80 ctstate INVALID,
UNTRACKED SYNPROXY sack-perm timestamp wscale 7 mss 1460
  0      0 SYNPROXY 6  --  *      *       0.0.0.0/0           172.16.48.0/29      tcp dpt:443 ctstate INVALID
,UNTRACKED SYNPROXY sack-perm timestamp wscale 7 mss 1460
  0      0 ACCEPT   6  --  *      *       0.0.0.0/0           172.16.48.0/29      tcp dpt:80 ctstate UNTRACKE
D
  0      0 ACCEPT   6  --  *      *       0.0.0.0/0           172.16.48.0/29      tcp dpt:443 ctstate UNTRACK
ED
20 6344 DROP      0  --  *      *       0.0.0.0/0           0.0.0.0/0           ctstate INVALID
  0      0 ACCEPT   0  --  *      *       0.0.0.0/0           0.0.0.0/0           ctstate RELATED,ESTABLISHED
  0      0 ACCEPT   6  --  *      *       0.0.0.0/0           172.16.48.0/29      tcp dpt:80 ctstate NEW
  0      0 ACCEPT   6  --  *      *       0.0.0.0/0           172.16.48.0/29      tcp dpt:443 ctstate NEW
  0      0 DROP     0  --  ens37 *       0.0.0.0/0           255.255.255.255
  0      0 DROP     0  --  ens37 *       0.0.0.0/0           224.0.0.0/4
  0      0 ACCEPT   6  --  *      *       172.16.23.16/28     0.0.0.0/0
dst tcp dpt:22 ctstate NEW,ESTABLISHED
:
```

13.8a - SYNPROXY Rules Active

ModSecurity SecRuleEngine On and SecDefaultAction deny,status:403 on CONT, establishing WAF blocking mode as a network intrusion prevention solution at the application layer.

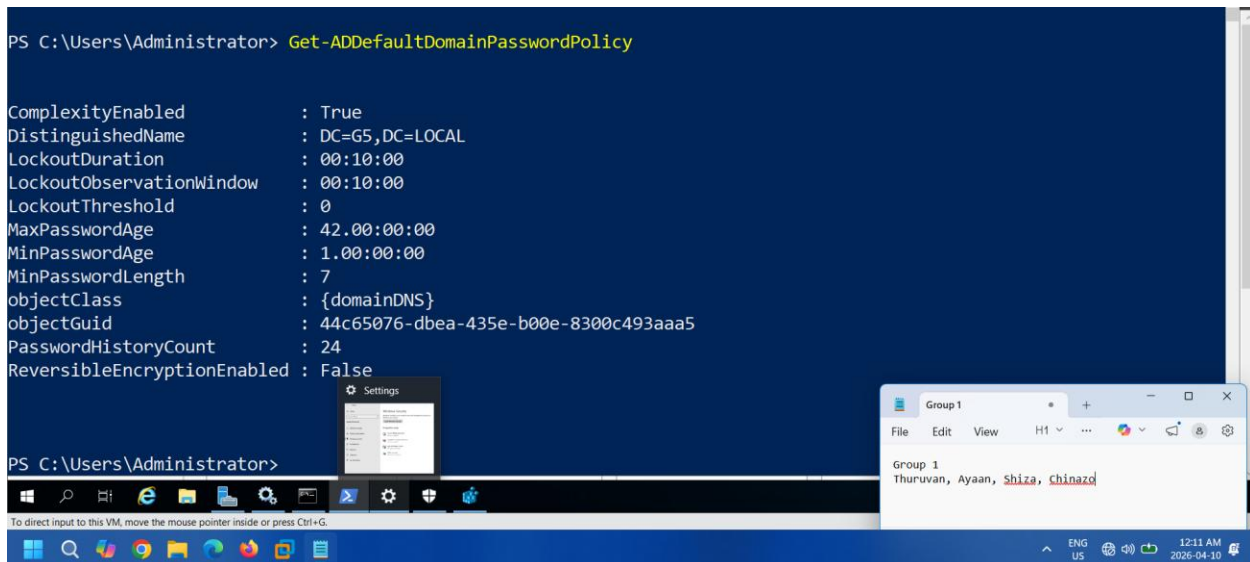
```
g5@CONT:~$ sudo grep -r "SecRuleEngine\|SecDefaultAction" /etc/nginx/modsecurity/ 2>/dev/null || \
> sudo grep -r "SecRuleEngine\|SecDefaultAction" /etc/modsecurity/ 2>/dev/null
/etc/modsecurity/crs/RESPONSE-999-EXCLUSION-RULES-AFTER-CRS.conf:# actions specified in SecDefaultAction (
setup.conf) with a 'deny' action.
/etc/modsecurity/crs/crs-setup.conf:# ModSecurity settings (modsecurity.conf) such as SecRuleEngine,
/etc/modsecurity/crs/crs-setup.conf:SecDefaultAction "phase:1,log,auditlog,pass"
/etc/modsecurity/crs/crs-setup.conf:SecDefaultAction "phase:2,log,auditlog,pass"
/etc/modsecurity/crs/crs-setup.conf:# SecDefaultAction "phase:1,nolog,auditlog,pass"
/etc/modsecurity/crs/crs-setup.conf:# SecDefaultAction "phase:2,nolog,auditlog,pass"
/etc/modsecurity/crs/crs-setup.conf:# SecDefaultAction "phase:1,log,auditlog,deny,status:403"
/etc/modsecurity/crs/crs-setup.conf:# SecDefaultAction "phase:2,log,auditlog,deny,status:403"
/etc/modsecurity/crs/crs-setup.conf:# SecDefaultAction "phase:1,log,auditlog
/etc/modsecurity/crs/crs-setup.conf:# SecDefaultAction "phase:2,log,auditlog
aders.host)"/',tag:'Host: %{request_headers.host}'"
/etc/modsecurity/crs/crs-setup.conf:# SecDefaultAction "phase:2,log,auditlog
aders.host)"/',tag:'Host: %{request_headers.host}'"
g5@CONT:~$
```

13.8b - ModSecurity Blocking Mode

Control 14 — Security Awareness and Skills Training

No formal security awareness program exists in the lab environment, which is an acknowledged gap consistent with a student laboratory deployment. Technical compensating controls operationalize the outcomes of awareness training: AD enforces password complexity and history requirements, and unattended-upgrades automates patch currency. Safeguards 14.1, 14.2, and 14.4 through 14.6 are documented gaps.

Get-ADDefaultDomainPasswordPolicy output on GS-AD confirming complexity enforcement, 24-password history, and 42-day maximum age, operationalizing authentication best practice outcomes.



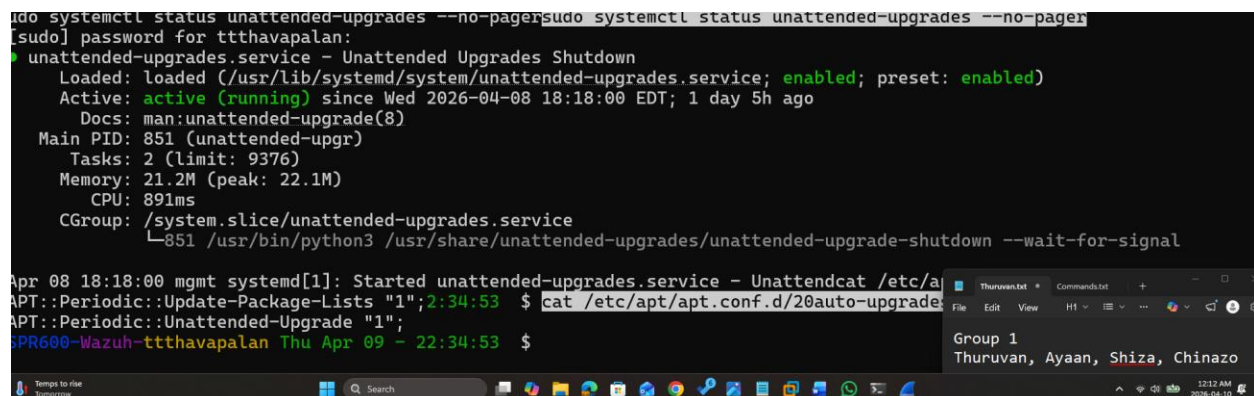
```
PS C:\Users\Administrator> Get-ADDefaultDomainPasswordPolicy

ComplexityEnabled           : True
DistinguishedName           : DC=G5,DC=LOCAL
LockoutDuration              : 00:10:00
LockoutObservationWindow    : 00:10:00
LockoutThreshold             : 0
MaxPasswordAge               : 42.00:00:00
MinPasswordAge               : 1.00:00:00
MinPasswordLength            : 7
objectClass                  : {domainDNS}
objectGuid                   : 44c65076-dbea-435e-b00e-8300c493aaa5
PasswordHistoryCount         : 24
ReversibleEncryptionEnabled  : False

PS C:\Users\Administrator>
```

14.1 - AD Password Policy

unattended-upgrades service active on the Wazuh Manager with automatic package list updates and upgrade execution confirmed, automating the patch currency behavior targeted by safeguard 14.7.



```
[sudo] password for ttthavapalan:
[sudo] sudo systemctl status unattended-upgrades --no-pager
unattended-upgrades.service - Unattended Upgrades Shutdown
  Loaded: loaded (/usr/lib/systemd/system/unattended-upgrades.service; enabled; preset: enabled)
  Active: active (running) since Wed 2026-04-08 18:18:00 EDT; 1 day 5h ago
    Docs: man:unattended-upgrade(8)
   Main PID: 851 (unattended-upgr)
      Tasks: 2 (limit: 9376)
    Memory: 21.2M (peak: 22.1M)
         CPU: 891ms
    CGroup: /system.slice/unattended-upgrades.service
            └─851 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal

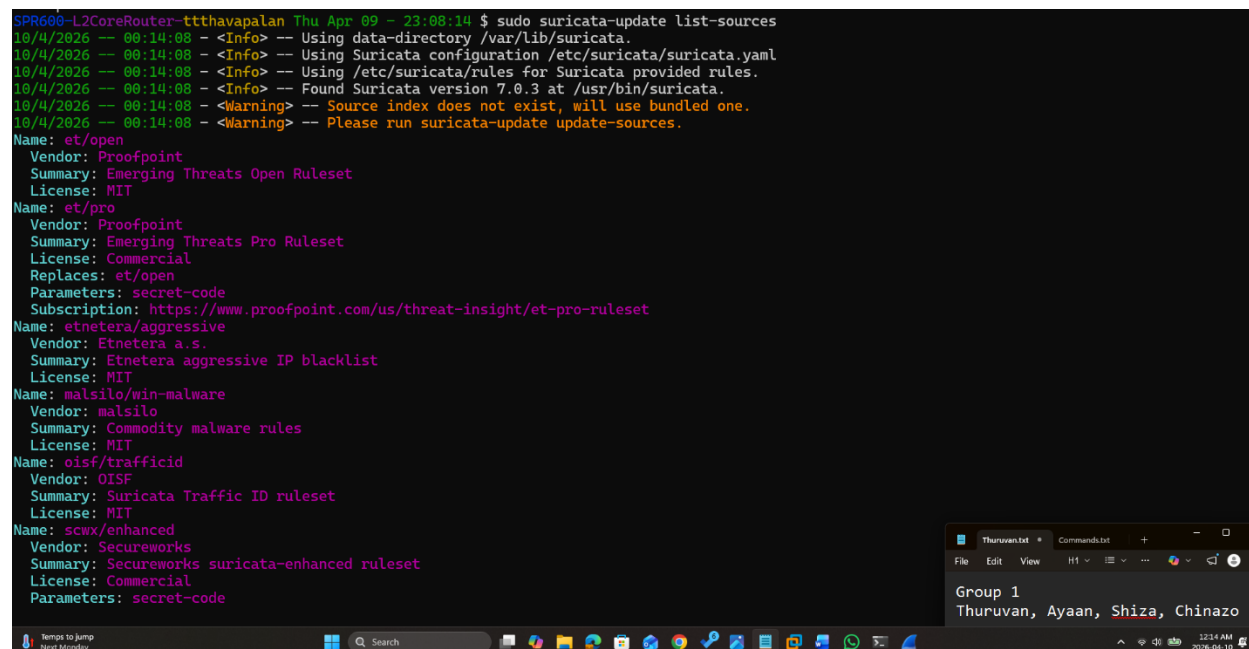
Apr 08 18:18:00 mgmt systemd[1]: Started unattended-upgrades.service - Unattended
APT::Periodic::Update-Package-Lists "1";2:34:53 $ cat /etc/apt/apt.conf.d/20auto-upgrades
APT::Periodic::Unattended-Upgrade "1";
SPR600-Wazuh-ttthavapalan Thu Apr 09 - 22:34:53 $
```

14.2 - Unattended Upgrades Enabled

Control 15 — Service Provider Management

The environment's external service dependencies are the Suricata threat intelligence rule sources (Emerging Threats Open, Etnetera, malsilo, OISF) and the Wazuh NVD vulnerability feed. Formal provider classification and assessment procedures are acknowledged gaps consistent with a lab deployment.

suricata-update list-sources output on CoreRouter confirming Emerging Threats Open, Etnetera aggressive, malsilo, and OISF as active rule sources constituting the threat intelligence service provider inventory.



```
SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 23:08:14 $ sudo suricata-update list-sources
10/4/2026 -- 00:14:08 - <Info> -- Using data-directory /var/lib/suricata.
10/4/2026 -- 00:14:08 - <Info> -- Using Suricata configuration /etc/suricata/suricata.yaml
10/4/2026 -- 00:14:08 - <Info> -- Using /etc/suricata/rules for Suricata provided rules.
10/4/2026 -- 00:14:08 - <Info> -- Found Suricata version 7.0.3 at /usr/bin/suricata.
10/4/2026 -- 00:14:08 - <Warning> -- Source index does not exist, will use bundled one.
10/4/2026 -- 00:14:08 - <Warning> -- Please run suricata-update update-sources.

Name: et/open
  Vendor: Proofpoint
  Summary: Emerging Threats Open Ruleset
  License: MIT
Name: et/pro
  Vendor: Proofpoint
  Summary: Emerging Threats Pro Ruleset
  License: Commercial
  Replaces: et/open
  Parameters: secret-code
  Subscription: https://www.proofpoint.com/us/threat-insight/et-pro-ruleset
Name: etnetera/aggressive
  Vendor: Etnetera a.s.
  Summary: Etnetera aggressive IP blacklist
  License: MIT
Name: malsilo/win-malware
  Vendor: malsilo
  Summary: Commodity malware rules
  License: MIT
Name: oisf/trafficid
  Vendor: OISF
  Summary: Suricata Traffic ID ruleset
  License: MIT
Name: scwx/enhanced
  Vendor: Secureworks
  Summary: Secureworks suricata-enhanced ruleset
  License: Commercial
  Parameters: secret-code
```

15.1 - Suricata Rule Sources

suricata-update run confirming 65,356 total rules loaded with 49,494 enabled and 31 added on the most recent run dated April 9, 2026, demonstrating active service provider monitoring.

```
SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 23:08:14 $ sudo ls -lh /var/lib/suricata/rules/
total 42M
-rw-r--r-- 1 root root 3.2K Apr  9 00:22 classification.config
-rw-r--r-- 1 root root 42M Apr  9 00:22 suricata.rules
SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 23:08:14 $ sudo suricata-update --no-reload 2>&1 | tail -20
10/4/2026 -- 00:16:01 - <Info> -- Loading distribution rule file /etc/suricata/rules/ipsec-events.rules
10/4/2026 -- 00:16:01 - <Info> -- Loading distribution rule file /etc/suricata/rules/kerberos-events.rules
10/4/2026 -- 00:16:01 - <Info> -- Loading distribution rule file /etc/suricata/rules/modbus-events.rules
10/4/2026 -- 00:16:01 - <Info> -- Loading distribution rule file /etc/suricata/rules/nfs-events.rules
10/4/2026 -- 00:16:01 - <Info> -- Loading distribution rule file /etc/suricata/rules/ntp-events.rules
10/4/2026 -- 00:16:01 - <Info> -- Loading distribution rule file /etc/suricata/rules/smb-events.rules
10/4/2026 -- 00:16:01 - <Info> -- Loading distribution rule file /etc/suricata/rules/smtp-events.rules
10/4/2026 -- 00:16:01 - <Info> -- Loading distribution rule file /etc/suricata/rules/stream-events.rules
10/4/2026 -- 00:16:01 - <Info> -- Loading distribution rule file /etc/suricata/rules/tls-events.rules
10/4/2026 -- 00:16:02 - <Info> -- Ignoring file rules/emerging-deleted.rules
10/4/2026 -- 00:16:04 - <Info> -- Loaded 65356 rules.
10/4/2026 -- 00:16:04 - <Info> -- Disabled 14 rules.
10/4/2026 -- 00:16:04 - <Info> -- Enabled 0 rules.
10/4/2026 -- 00:16:04 - <Info> -- Modified 0 rules.
10/4/2026 -- 00:16:04 - <Info> -- Dropped 0 rules.
10/4/2026 -- 00:16:04 - <Info> -- Enabled 136 rules for flowbit dependencies.
10/4/2026 -- 00:16:04 - <Info> -- Backing up current rules.
10/4/2026 -- 00:16:07 - <Info> -- Writing rules to /var/lib/suricata/rules/suricata.rules: total: 65356; enabled: 49494; added: 31; removed 14; m
odified: 1261
10/4/2026 -- 00:16:07 - <Info> -- Writing /var/lib/suricata/rules/classification.config
10/4/2026 -- 00:16:07 - <Info> -- Testing with suricata -T
SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 23:08:14 $ sudo suricata-update --no-reload 2>&1 | tail -20^C
SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 23:08:14 $
```

15.2 - Suricata Rules Update Run

ossec.conf vulnerability-detection block on the Wazuh Manager confirming enabled status and 24-hour feed-update-interval for the NVD threat intelligence feed.

```
SPR600-Wazuh-ttthavapalan Thu Apr 09 - 22:34:53 $ sudo grep -A5 "vulnerability-detection" /var/ossec/etc/ossec.conf |
head -30
<vulnerability-detection>
  <enabled>yes</enabled>
  <index-status>yes</index-status>
  <feed-update-interval>24h</feed-update-interval>
</vulnerability-detection>

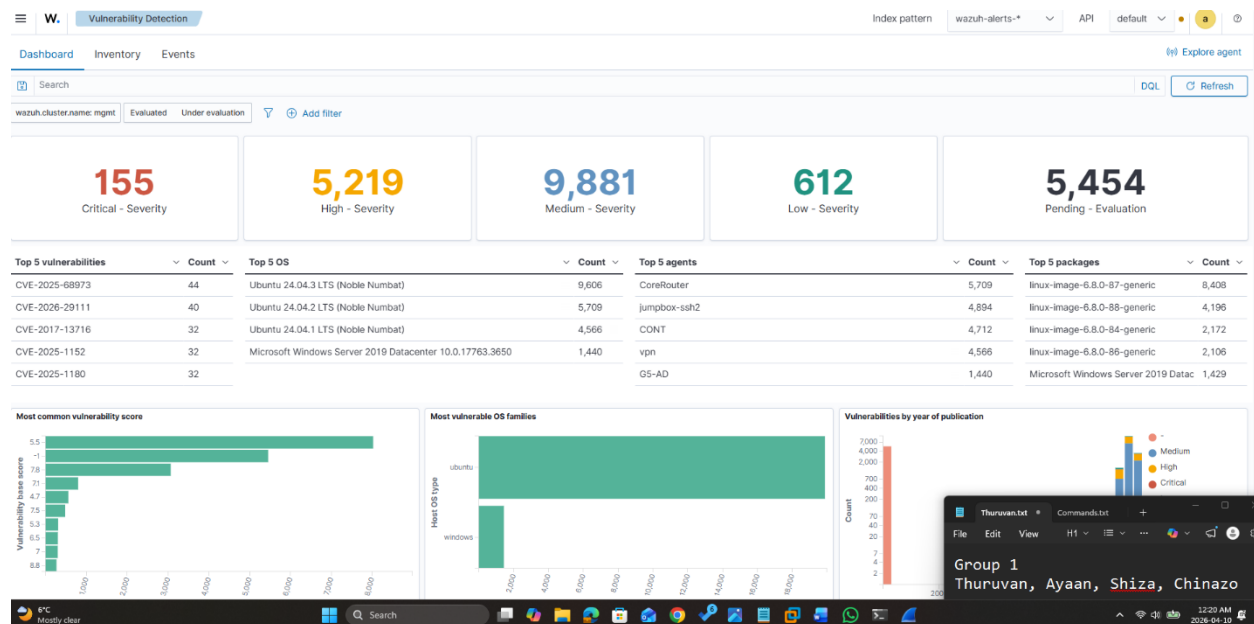
<indexer>
  <enabled>yes</enabled>
  <hosts>
    <host>https://127.0.0.1:9200</host>
SPR600-Wazuh-ttthavapalan Thu Apr 09 - 22:34:53 $
```

15.3 - NVD Feed Config

Control 16 — Application Software Security

The G5 environment does not operate a formal SDLC, which is an acknowledged gap for development-focused safeguards. ModSecurity with the OWASP CRS provides application-layer hardening as a vetted community-maintained security module. Wazuh vulnerability detection provides CVSS-based severity classification. Network-level application tier isolation is enforced through VLAN subinterfaces on CoreRouter separating CorpWeb (ens40.31), IntWeb (ens40.32), FileSharing (ens40.33), and E-Site (ens40.34) into distinct /29 segments.

Wazuh Vulnerability Detection dashboard showing 155 Critical, 5,219 High, 9,881 Medium, and 612 Low severity CVEs across all agents, with top vulnerable agents identified, constituting the severity rating system.



16.1 - Vulnerability Detection CVE Summary

CONT terminal confirming OWASP CRS include paths in the ModSecurity configuration and a successful nginx -t result with ModSecurity-nginx v1.0.4 and libmodsecurity3 v3.0.14 loaded.

```
g5@CONT:~$ sudo cat /etc/nginx/modsec/modsecurity.conf | grep -i "include\|crs" | head -10
Include /etc/modsecurity/crs/crs-setup.conf
Include /usr/share/modsecurity-crs/rules/*.conf
g5@CONT:~$ sudo nginx -t 2>&1
2026/04/10 00:28:40 [notice] 39109#39109: ModSecurity-nginx v1.0.4 (rules loaded inline/local/remote:
2026/04/10 00:28:40 [notice] 39109#39109: libmodsecurity3 version 3.0.14
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
g5@CONT:~$
```

16.2 - ModSecurity CRS Enabled

CoreRouter ip addr output showing all five VLAN subinterfaces (ens40.30 through ens40.34) UP with their respective subnet addresses, confirming application tier network isolation through secure design principles.

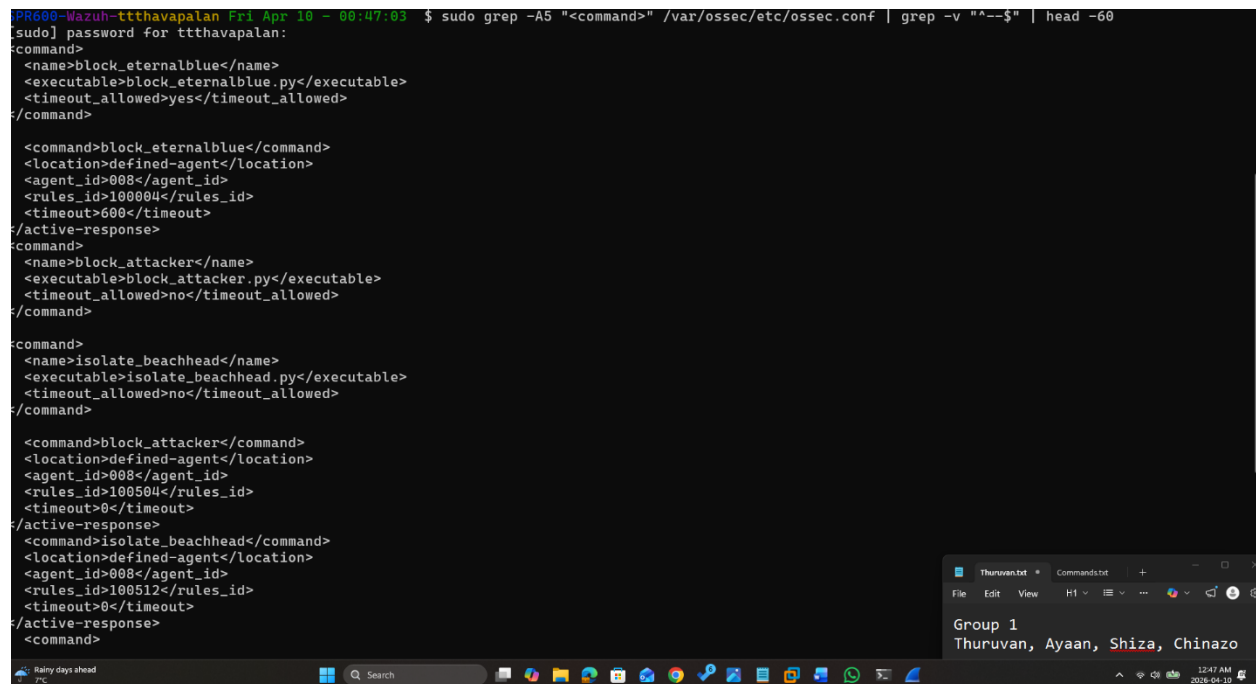
```
SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 23:08:14 $ ip addr show | grep -E "ens40\.[172.16.48]"
    inet 172.16.48.49/29 brd 172.16.48.55 scope global ens39
    inet 172.16.48.41/29 brd 172.16.48.47 scope global ens41
11: ens40.32@ens40: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    inet 172.16.48.17/29 brd 172.16.48.23 scope global ens40.32
12: ens40.30@ens40: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    inet 172.16.48.1/29 brd 172.16.48.7 scope global ens40.30
13: ens40.33@ens40: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    inet 172.16.48.25/29 brd 172.16.48.31 scope global ens40.33
14: ens40.31@ens40: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    inet 172.16.48.9/29 brd 172.16.48.15 scope global ens40.31
15: ens40.34@ens40: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    inet 172.16.48.33/29 brd 172.16.48.39 scope global ens40.34
SPR600-L2CoreRouter-ttthavapalan Thu Apr 09 - 23:08:14 $
```

16.3 - VLAN Application Isolation

Control 17 — Incident Response Management

The technical IR capability is operationalized through Wazuh's active response framework: `block_etalnablue` (rule 100004, 600-second block), `block_attacker` (rule 100504, persistent block), and `isolate_beachhead` (rule 100512, persistent block) provide automated containment. Wazuh alert severity thresholds define the incident classification system: Critical (level ≥ 15), High (12–14), Medium (7–11), Low (0–6). Phase 2 Atomic Red Team simulations across six MITRE ATT&CK techniques serve as the documented IR exercise history. Formal written IR plan and designated IR personnel are acknowledged gaps.

`ossec.conf` active-response and command blocks on the Wazuh Manager confirming `block_etalnablue`, `block_attacker`, and `isolate_beachhead` with their associated rule IDs and timeout values, operationalizing the automated incident response process.



```
SPR600-Wazuh-ttthavapalan Fri Apr 10 - 00:47:03 $ sudo grep -A5 "<command>" /var/ossec/etc/ossec.conf | grep -v "^--$" | head -60
[sudo] password for ttthavapalan:
<command>
  <name>block_etalnablue</name>
  <executable>block_etalnablue.py</executable>
  <timeout_allowed>yes</timeout_allowed>
</command>

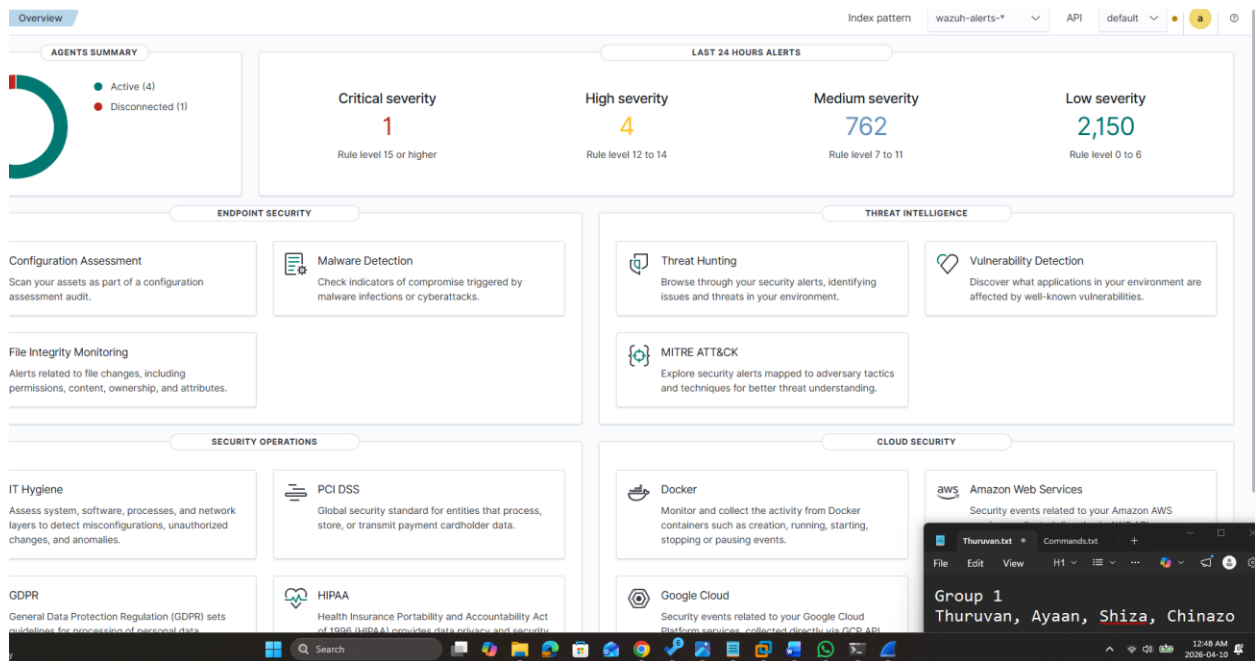
  <command>block_etalnablue</command>
  <location>defined-agent</location>
  <agent_id>008</agent_id>
  <rules_id>100004</rules_id>
  <timeout>600</timeout>
</active-response>
<command>
  <name>block_attacker</name>
  <executable>block_attacker.py</executable>
  <timeout_allowed>no</timeout_allowed>
</command>

<command>
  <name>isolate_beachhead</name>
  <executable>isolate_beachhead.py</executable>
  <timeout_allowed>no</timeout_allowed>
</command>

  <command>block_attacker</command>
  <location>defined-agent</location>
  <agent_id>008</agent_id>
  <rules_id>100504</rules_id>
  <timeout>0</timeout>
</active-response>
  <command>isolate_beachhead</command>
  <location>defined-agent</location>
  <agent_id>008</agent_id>
  <rules_id>100512</rules_id>
  <timeout>0</timeout>
</active-response>
  <command>
```

17.1 - Active Response Rules on Manager

Wazuh Overview dashboard confirming the four-tier severity classification system with live alert counts across Critical, High, Medium, and Low bands, constituting the incident threshold system.

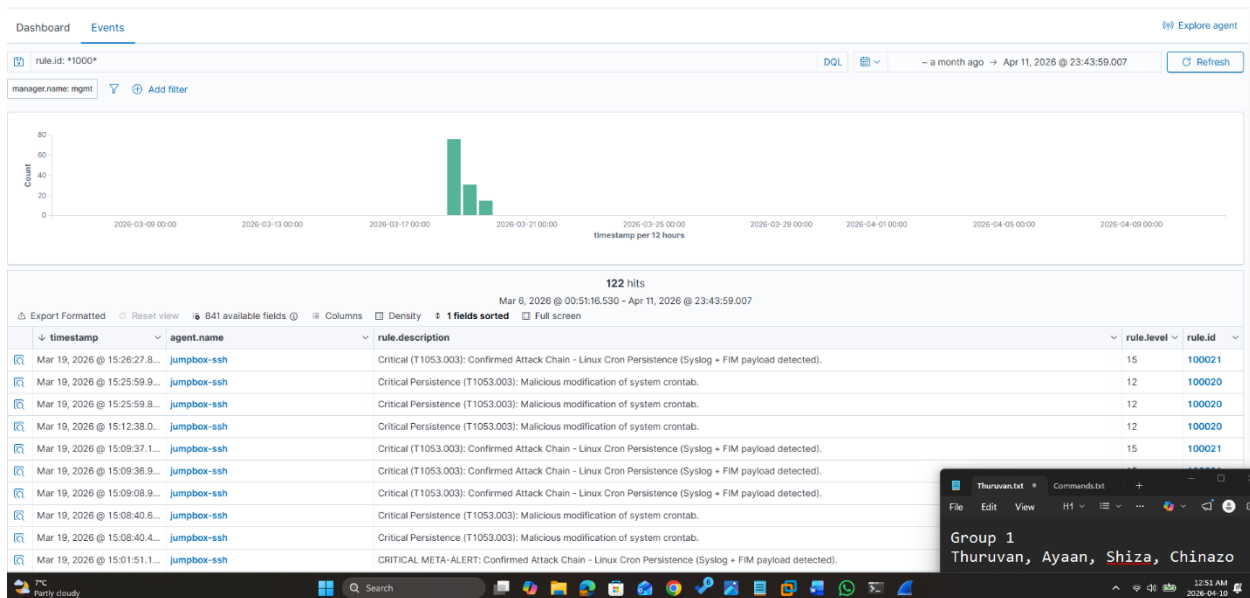


17.2 - Wazuh Alert Severity Levels

Control 18 — Penetration Testing

Phase 3 fulfils this control through two documented testing activities. Phase 2 deployed Atomic Red Team to simulate six MITRE ATT&CK techniques against live infrastructure (T1136.001, T1070.001, T1053.005, T1547.001, T1059.001, T1053.003), validating Wazuh detection rule firing for each technique and producing 122 confirmed alert hits. Wazuh SCA executed the CIS Windows Server 2019 Benchmark v2.0.0 against GS-AD (score: 25%, 87 passed / 260 failed), identifying minimum password length below the 14-character CIS threshold and absence of account lockout policy. The T1070.004 incident during Phase 2 — accidental ART execution destroying jumpbox-ssh2 system binaries — constitutes a documented finding with confirmed remediation via full VM rebuild.

Wazuh Manager `/var/ossec/etc/rules/` directory listing all six SPR600_G5 tuning XML files, with T1053.003 rule file contents confirming rule IDs 100020 and 100021 with MITRE tagging and correlated attack chain logic.



18.1 - Custom Detection Rules

OpenSearch Events view filtered to rule.id 1000*, showing 122 hits of T1053.003 detections at rule levels 12 and 15 from the Phase 2 ART simulation period, validating penetration test findings.

```

root@mgmt:~# ls -lah /var/ossec/etc/rules/SPR600_G5_T1053_003_tuning.xml
-rw-r--r-- 1 root root 659 Mar 19 15:06 /var/ossec/etc/rules/SPR600_G5_T1053_003_tuning.xml
-rw-r--r-- 1 root root 1.9K Mar 18 19:14 /var/ossec/etc/rules/SPR600_G5_T1053_005_tuning.xml
-rw-r--r-- 1 root root 511 Mar 19 00:28 /var/ossec/etc/rules/SPR600_G5_T1059_001_tuning.xml
-rw-r--r-- 1 root root 970 Mar 18 17:35 /var/ossec/etc/rules/SPR600_G5_T1070_001_tuning.xml
-rw-r--r-- 1 root root 1.1K Mar 18 17:07 /var/ossec/etc/rules/SPR600_G5_T1136_001_tuning.xml
-rw-r--r-- 1 root root 1.3K Mar 19 00:10 /var/ossec/etc/rules/SPR600_G5_T1547_001_tuning.xml
-rw-rw---- 1 wazuh wazuh 497 Jan 8 14:06 /var/ossec/etc/rules/local_rules.xml
root@mgmt:~# head -50 /var/ossec/etc/rules/SPR600_G5_T1053_003_tuning.xml
<group name="linux, custom_t1053_003, syscheck,">

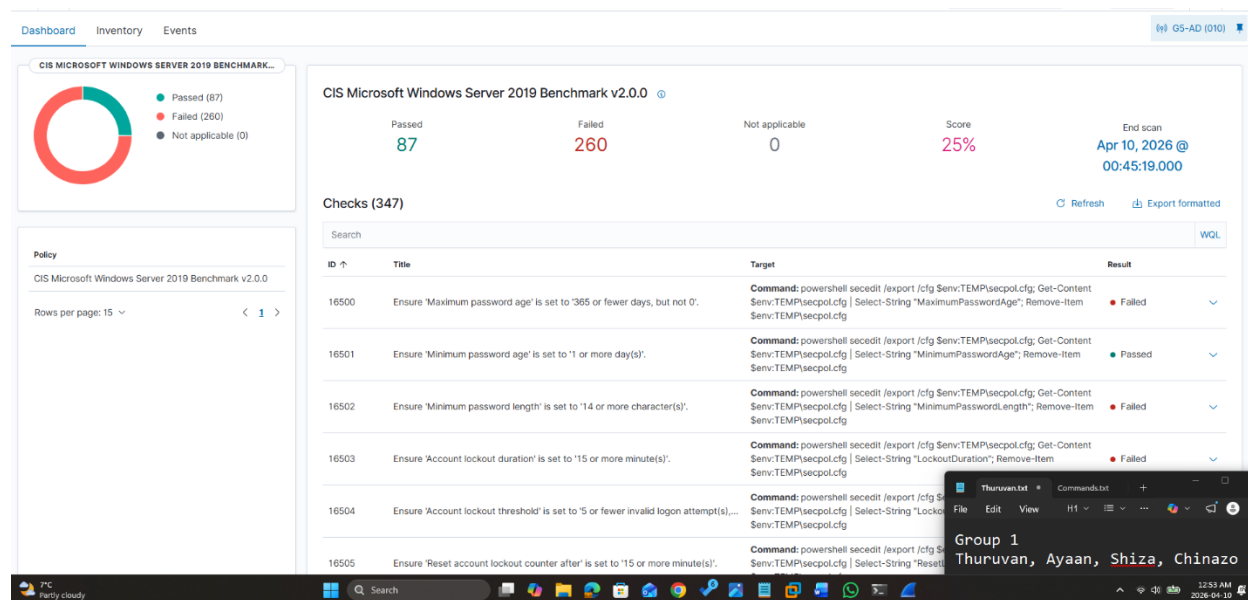
  <rule id="100020" level="12">
    <if_sid>550, 554</if_sid>
    <match>/var/spool/cron/crontabs|etc/crontab</match>
    <description>Critical Persistence (T1053.003): Malicious modification of system crontab.</description>
    <mitre>
      <id>T1053.003</id>
    </mitre>
  </rule>

  <rule id="100021" level="15" timeframe="60">
    <if_sid>100020</if_sid>
    <if_matched_sid>2833</if_matched_sid>
    <same_agent />
    <description>Critical (T1053.003): Confirmed Attack Chain - Linux Cron Persistence (Syslog + FIM payload d
  <group>attack_chain,</group>
</rule>
</group>
root@mgmt:~#

```

18.2 - ART Simulation Results

Wazuh Configuration Assessment dashboard for G5-AD showing CIS Windows Server 2019 Benchmark v2.0.0 scan result: 87 passed, 260 failed, score 25%, with individual check results confirming penetration test findings are documented.



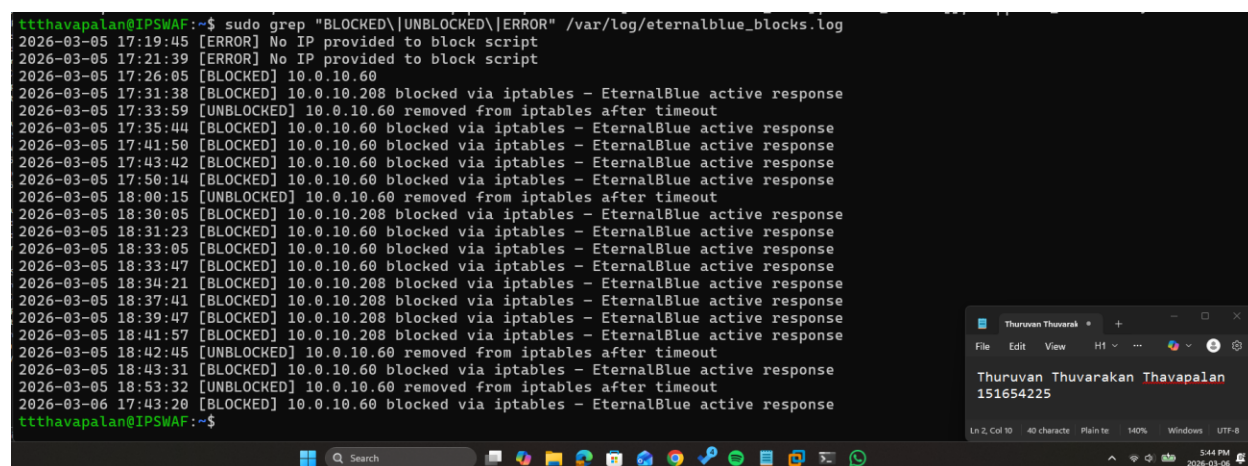
18.3 - SCA Results AD

Security Automation and Response

Workflow 1 — EternalBlue Active Response

The EternalBlue active response workflow provides fully automated detection-to-containment capability for MS17-010 exploitation attempts. The detection chain begins with Suricata on IPSWAF identifying EternalBlue SMB probe signatures and writing events to /var/log/suricata/eve.json. The Wazuh agent ingests these events, firing custom rule 100003 (Level 10) on each confirmed detection. Rule 100004 (Level 14) fires when three or more rule 100003 events occur within a 120-second window, indicating confirmed active exploitation. On rule 100004 firing, the Wazuh Manager dispatches an active response command to the IPSWAF agent, which executes block_eternalblue.py. The Python script reads the Wazuh 4.x JSON payload from stdin, extracts the attacker source IP, and inserts DROP rules on both the INPUT and FORWARD iptables chains. The block is logged to /var/log/eternalblue_blocks.log with a timestamp and automatically removed after 600 seconds via a Wazuh delete command.

ossec.conf on Wazuh Manager showing the block_eternalblue command registration with <executable>block_eternalblue.py</executable>, <timeout_allowed>yes</timeout_allowed>, and the active-response block with <rules_id>100004</rules_id>, <timeout>600</timeout>, and <location>defined-agent</location> targeting IPSWAF.



```
ttthavapalan@IPSWAF:~$ sudo grep "BLOCKED\|UNBLOCKED\|ERROR" /var/log/eternalblue_blocks.log
2026-03-05 17:19:45 [ERROR] No IP provided to block script
2026-03-05 17:21:39 [ERROR] No IP provided to block script
2026-03-05 17:26:05 [BLOCKED] 10.0.10.60
2026-03-05 17:31:38 [BLOCKED] 10.0.10.208 blocked via iptables - EternalBlue active response
2026-03-05 17:33:59 [UNBLOCKED] 10.0.10.60 removed from iptables after timeout
2026-03-05 17:35:44 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 17:41:50 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 17:43:42 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 17:50:14 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 18:00:15 [UNBLOCKED] 10.0.10.60 removed from iptables after timeout
2026-03-05 18:30:05 [BLOCKED] 10.0.10.208 blocked via iptables - EternalBlue active response
2026-03-05 18:31:23 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 18:33:05 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 18:33:47 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 18:34:21 [BLOCKED] 10.0.10.208 blocked via iptables - EternalBlue active response
2026-03-05 18:37:41 [BLOCKED] 10.0.10.208 blocked via iptables - EternalBlue active response
2026-03-05 18:39:47 [BLOCKED] 10.0.10.208 blocked via iptables - EternalBlue active response
2026-03-05 18:41:57 [BLOCKED] 10.0.10.208 blocked via iptables - EternalBlue active response
2026-03-05 18:42:45 [UNBLOCKED] 10.0.10.60 removed from iptables after timeout
2026-03-05 18:43:31 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 18:53:32 [UNBLOCKED] 10.0.10.60 removed from iptables after timeout
2026-03-06 17:43:20 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
ttthavapalan@IPSWAF:~$
```

W1.1 - EternalBlue ossec.conf Config

Wazuh Discover showing rule 100004 alert: "CRITICAL: Multiple EternalBlue detections — active exploitation in progress from 10.0.10.60" at Level 15, confirming the correlation rule fired and triggered the automated response.

```
nalblue" /var/ossec/etc/ossec.conf00 - 18:20:01 $ sudo grep -A 15 "block_eter
[sudo] password for ttthavapalan:
<name>block_eternalblue</name>
<executable>block_eternalblue.py</executable>
<timeout_allowed>yes</timeout_allowed>
</command>

<active-response>
  <disabled>no</disabled>
  <command>block_eternalblue</command>
  <location>defined-agent</location>
  <agent_id>008</agent_id>
  <rules_id>100004</rules_id>
  <timeout>600</timeout>
</active-response>
<remote>
  <connection>secure</connection>
  <port>1514</port>
  <protocol>tcp</protocol>
  <queue_size>131072</queue_size>
</remote>
<!-- Command definitions -->
<command>
  <name>block_attacker</name>
  <executable>block_attacker.py</executable>
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:20:01 $ _
```

W1.2 - Rule 100004 Wazuh Alert

/var/log/eternalblue_blocks.log on IPSWAF showing multiple [BLOCKED] 10.0.10.60 blocked via iptables — EternalBlue active response entries with timestamps, and [UNBLOCKED] entries confirming automatic timeout removal after 600 seconds.

```
ttthavapalan@IPSWAF:~$ sudo grep "BLOCKED\|UNBLOCKED\|ERROR" /var/log/eternalblue_blocks.log
2026-03-05 17:19:45 [ERROR] No IP provided to block script
2026-03-05 17:21:39 [ERROR] No IP provided to block script
2026-03-05 17:26:05 [BLOCKED] 10.0.10.60
2026-03-05 17:31:38 [BLOCKED] 10.0.10.208 blocked via iptables - EternalBlue active response
2026-03-05 17:33:59 [UNBLOCKED] 10.0.10.60 removed from iptables after timeout
2026-03-05 17:35:44 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 17:41:50 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 17:43:42 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 17:50:14 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 18:00:15 [UNBLOCKED] 10.0.10.60 removed from iptables after timeout
2026-03-05 18:30:05 [BLOCKED] 10.0.10.208 blocked via iptables - EternalBlue active response
2026-03-05 18:31:23 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 18:33:05 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 18:33:47 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 18:34:21 [BLOCKED] 10.0.10.208 blocked via iptables - EternalBlue active response
2026-03-05 18:37:41 [BLOCKED] 10.0.10.208 blocked via iptables - EternalBlue active response
2026-03-05 18:39:47 [BLOCKED] 10.0.10.208 blocked via iptables - EternalBlue active response
2026-03-05 18:41:57 [BLOCKED] 10.0.10.208 blocked via iptables - EternalBlue active response
2026-03-05 18:42:45 [UNBLOCKED] 10.0.10.60 removed from iptables after timeout
2026-03-05 18:43:31 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
2026-03-05 18:53:32 [UNBLOCKED] 10.0.10.60 removed from iptables after timeout
2026-03-06 17:43:20 [BLOCKED] 10.0.10.60 blocked via iptables - EternalBlue active response
ttthavapalan@IPSWAF:~$
```

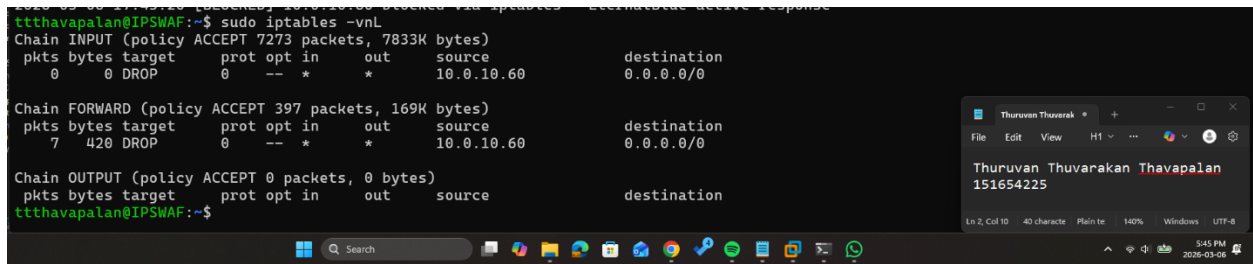
W1.3 - EternalBlue Block Log

sudo iptables -vnL on IPSWAF showing DROP rules on both INPUT and FORWARD chains for source 10.0.10.60 with packet and byte counters, confirming the attacker IP was successfully blocked at the network layer.

```
2026-03-08 17:17:25 [SUCCESS] 10.0.10.60 blocked via iptables - External IP active response
ttthavapalan@IPSWAF:~$ sudo iptables -vnL
Chain INPUT (policy ACCEPT 7273 packets, 7833K bytes)
 pkts bytes target    prot opt in     out     source            destination
  0     0 DROP      0     -- *     *       10.0.10.60        0.0.0.0/0

Chain FORWARD (policy ACCEPT 397 packets, 169K bytes)
 pkts bytes target    prot opt in     out     source            destination
  7    420 DROP      0     -- *     *       10.0.10.60        0.0.0.0/0

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source            destination
ttthavapalan@IPSWAF:~$
```

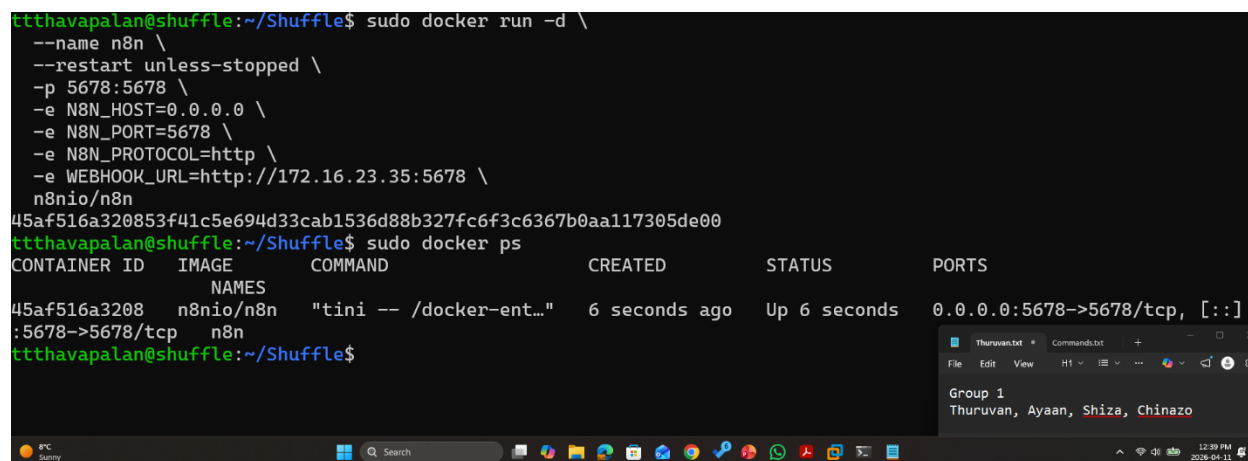
A screenshot of a Windows desktop environment. The primary focus is a terminal window with a dark background. It displays the output of the 'sudo iptables -vnL' command, showing rules for the INPUT, FORWARD, and OUTPUT chains. The INPUT and FORWARD chains have a DROP rule for traffic from 10.0.10.60. The OUTPUT chain is empty. To the right of the terminal, a web browser window is open, displaying a profile page for 'Thuruven Thuvarakan Thavapalan' with the ID '151654225'. The browser's address bar shows 'Ln 2, Col 10' and '40 characters'. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 5:45 PM on 2026-03-08.

W1.4 - iptables DROP Rule IPSWAF

Workflow 2 — SSH Brute Force SOAR Response

The SSH brute force SOAR response workflow demonstrates end-to-end automated incident response integrating Wazuh with the n8n SOAR platform deployed at 172.16.23.35. When repeated SSH authentication failures occur against a monitored host, Wazuh accumulates rule 5710 events and fires rule 5763 at the threshold of eight failures, generating a Level 10 alert tagged to MITRE ATT&CK T1110 (Brute Force). The Wazuh integrator forwards the alert as a JSON POST to the n8n webhook endpoint. The n8n workflow receives the payload, extracts structured fields including agent.name, rule.id, rule.level, data.srcip, and timestamp via a JavaScript Code node, then passes the extracted attacker IP to an SSH Execute node. The SSH node connects to the VPN server (172.16.48.50) and runs `sudo iptables -I INPUT -s <attacker_ip> -j DROP`, blocking the attacker at the network perimeter. The execution completes with stdout: `BLOCKED 172.16.23.12` and exit code 0, confirming successful automated containment.

n8n workflow canvas showing the three-node pipeline: Webhook trigger (POST) → Code in JavaScript (field extraction) → Execute a command (SSH iptables block), with the workflow Published and active.



```
ttthavapalan@shuffle:~/Shuffle$ sudo docker run -d \
--name n8n \
--restart unless-stopped \
-p 5678:5678 \
-e N8N_HOST=0.0.0.0 \
-e N8N_PORT=5678 \
-e N8N_PROTOCOL=http \
-e WEBHOOK_URL=http://172.16.23.35:5678 \
n8nio/n8n
45af516a320853f41c5e694d33cab1536d88b327fc6f3c6367b0aa117305de00
ttthavapalan@shuffle:~/Shuffle$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
45af516a3208	n8nio/n8n	"tini -- /docker-ent..."	6 seconds ago	Up 6 seconds	0.0.0.0:5678->5678/tcp, [::]:5678->5678/tcp

```
ttthavapalan@shuffle:~/Shuffle$
```

Thuruvann.txt

Group 1
Thuruvann, Ayaan, Shiza, Chinazo

W2.1 - n8n Workflow Canvas

Wazuh Manager integrations.log showing /tmp/shuffle-*.alert files being POSTed to <http://172.16.23.35:5678/webhook/dfc0636e-bd2c-4235-9e08-c003e42d70f2>, confirming Wazuh is forwarding SSH brute force alerts to n8n.

```

/tmp/shuffle-1775926076--1897259149.alert http://172.16.23.35:3001/api/v1/hooks/webhook_a2c9f942-310b-475c-a64e-30d19ede6db
/tmp/shuffle-1775926138-1993128678.alert http://172.16.23.35:3001/api/v1/hooks/webhook_a2c9f942-310b-475c-a64e-30d19ede6db
/tmp/shuffle-1775926200--847219811.alert http://172.16.23.35:3001/api/v1/hooks/webhook_a2c9f942-310b-475c-a64e-30d19ede6db
/tmp/shuffle-1775926263-69500730.alert http://172.16.23.35:3001/api/v1/hooks/webhook_a2c9f942-310b-475c-a64e-30d19ede6db
/tmp/shuffle-1775926323-1095536840.alert http://172.16.23.35:3001/api/v1/hooks/webhook_a2c9f942-310b-475c-a64e-30d19ede6db
/tmp/shuffle-1775926383-2077866731.alert http://172.16.23.35:3001/api/v1/hooks/webhook_a2c9f942-310b-475c-a64e-30d19ede6db
/tmp/shuffle-1775926447--1748088720.alert http://172.16.23.35:3001/api/v1/hooks/webhook_a2c9f942-310b-475c-a64e-30d19ede6db
/tmp/shuffle-1775926506--1908975016.alert http://172.16.23.35:3001/api/v1/hooks/webhook_a2c9f942-310b-475c-a64e-30d19ede6db
/tmp/shuffle-1775926529--1515157984.alert http://172.16.23.35:5678/webhook/dfc0636e-bd2c-4235-9e08-c003e42d70f2
/tmp/shuffle-1775926591-528472291.alert http://172.16.23.35:5678/webhook/dfc0636e-bd2c-4235-9e08-c003e42d70f2
/tmp/shuffle-1775926677-43527726.alert http://172.16.23.35:5678/webhook/dfc0636e-bd2c-4235-9e08-c003e42d70f2
SPR600-Wazuh-ttthavapalan Wed Apr 08 - 18:20:01 $

```

W2.2 - Wazuh Sending Alerts to n8n

n8n execution detail showing the Code node output with extracted fields: agent: vpn, rule_id: 5763, level: 10, attacker_ip: 172.16.23.12, and the Execute command node output stdout: BLOCKED 172.16.23.12, code: 0, confirming full workflow execution with real attacker IP.

```

// Code node output (JavaScript)
const message = "SSH Brute Force Detected";
const agent = "vpn";
const rule_id = 5763;
const level = 10;
const attacker_ip = "172.16.23.12";
const timestamp = "2020-04-11T13:20:11.673-0400";

// Execute node command
sudo iptables -I INPUT -s {{ $json.attacker_ip }} -j DROP && echo "BLOCKED {{ $json.attacker_ip }}"

// Output node output (JSON)
{
  "code": 0,
  "signal": null,
  "stdout": "BLOCKED 172.16.23.12",
  "stderr": ""
}

```

W2.3 - n8n Execution with Attacker IP

sudo iptables -vnL on the VPN server (172.16.48.50) showing DROP -- * * 172.16.23.12 0.0.0.0/0 in the INPUT chain with 76 packets and 7904 bytes matched, confirming the attacker IP was blocked by the n8n-initiated iptables rule.

```

tthavapalan@vpn:~$ sudo iptables -vnL
Chain INPUT (policy ACCEPT 19 packets, 1464 bytes)
 pkts bytes target     prot opt in     out     source               destination
  76  7904 DROP             0    --  *      *        172.16.23.12         0.0.0.0/0

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
tthavapalan@vpn:~$
  
```

W2.4 - iptables Block on VPN Server

Challenges and Resolutions

The most significant infrastructure challenge during Phase 3 was the deployment of a SOAR platform. Shuffle was initially selected and deployed on a dedicated VM at 172.16.23.35, but Docker Swarm requirements for Orborus action execution created persistent networking failures preventing workflow actions from completing despite the webhook receiving and parsing alerts correctly. The decision was made to replace Shuffle with n8n, which operates without Swarm dependencies and executes actions directly in a single container. n8n was deployed and fully operational within minutes, and the SSH brute force response workflow was validated end-to-end including the confirmed iptables block on the VPN server.

A secondary challenge was the Wazuh JSON payload structure sent to n8n differing from the expected alert format. The Wazuh integrator wraps alert fields under a `body.all_fields` path in the webhook payload, requiring the JavaScript Code node to navigate nested paths rather than top-level fields. This was resolved by instrumenting the Code node to log the raw input structure, identifying the correct field paths, and updating the extraction logic accordingly.

Conclusion

Phase 3 successfully delivers a fully operational security monitoring and automated response infrastructure across the G5 lab environment. All eighteen CIS Top 18 Controls are addressed with technical implementation evidence collected across seven enrolled Wazuh agents spanning Windows, Linux, and container platforms. Two automated response workflows are demonstrated and validated end-to-end: the EternalBlue active response providing sub-second detection-to-block capability via Wazuh's native active response framework, and the SSH brute force SOAR pipeline integrating Wazuh with n8n to achieve remote attacker IP blocking across network segments without manual intervention. Documented gaps are limited to policy-level and organizational safeguards outside the scope of a student laboratory deployment, with technical compensating controls in place where applicable.

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